Derivative Securities Notes

Derivative Securities (University of Melbourne)



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Week 1: Introduction to Derivatives

Futures Contracts

- A futures contract is an agreement to buy or sell a specified quantity of a specified asset at a specified time in the future for a specified price (delivery price)

  A futures contract is an agreement to buy or sell an asset at a certain time in the future for a certain price
- Buyer needs to pay only a small fraction (2-5%) of the delivery price as deposit
- By contrast in a spot contract there is an agreement to buy or sell the asset immediately (or within a very short period of time e.g. same business day) -> pay 'now
  - The difference between spot and future contract is: spot contract you pay right away and receive the asset you buy (or receive the money and delivery the asset if selling)
  - The spot price is for immediate, or almost immediate, delivery. The futures price is the price for delivery at some time in the future. The two are not usually equal.

The Nature of Derivatives: What are they?

- A derivative security is an instrument (or contract) whose payoff and, thus, value depends on the values (or, prices) of one (or, more) other variables (referred to as **underlying assets**)
  - Such security derives its value from the value of another (traded or not) asset
  - The value of the future contract (derivate asset) will **depend** on the price of the asset today (**the underlying asset**).

Type of Derivatives We Will Consider

- Futures and Forward Contracts
- Standard (or, Plain Vanilla) Options
- Primarily written on (i.e., whose underlying assets are): equities (individual and indices), currencies, commodities
- Traded either in organized exchanges or over-the-counter

Spot market: trading is immediate (buyers pay and assets received)

Future market: buyers buy but not pay until a certain future date

Organized exchanged: official exchange institutions e.g. New York Stock Exchange

Over-the-counter: parties agree to each other outside of the organized exchange

Derivatives Trading: Exchanges

- Products and trading terms are standardized

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- People can only buy the specific offering in a specific quantity
- The trading term is standardized.

- Increasingly this is being replaced by electronic trading where a computer matches buyers and sellers rather than trading on the floor.

Derivatives Trading: Over-the Counter (OTC) Markets

- Important alternative to exchanges
- It offers a more flexibility
- It is a telephone and computer-linked network of dealers who do not physically meet
- Trades are usually between *financial institutions, corporate treasurers, and fund managers*
  - Typically, small investors go to organized exchanges, and bigger investors go to OTC.
  - This means that they are always prepared to quote a bid price (at which they are prepared to take one side of a derivatives transaction) and an offer price (at which they are prepared to take the other side). When they start trading with each other, two market participants often sign an agreement covering all transactions they might enter into in the future.
- Contract flexibility (as opposed to standardized), private market, relatively unregulated
- Credit Risk
  - OTC involves more credit risk.
    - ⇨ When there is an organized exchange, the exchange itself, the organization will guarantee the execution of the trade (even when one party fails, the organization will continue the transaction).
    - ⇨ Whereas, in OTC if the counterparty betrays, then another party has to bear the loss and no compensation for the loss.
  - Prior to credit risk, the OTC markets were largely unregulated. Following the credit crisis and failure of Lehman Brothers, there was development of many new regulations affects the operation of OTC markets.
    - ⇨ The purpose of the regulations is to improve the transparency of OTC markets, improve market efficiency, and reduce systemic risk

Big Picture Questions

- WHY derivatives?
  1. Hedging the risk (eliminating the price risk)
  2. Speculation => to make profit for it

     Trading, in order to make an advantage of the price movements in the future
- WHO uses them - HOW to use them?
- How to PRICE them?

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- How to assess their riskiness - Are they "dangerous"?

LET'S START ANSWERING THE 'WHY' AND 'WHO'

Hedging Example

- A farmer with wheat in the field knows that the crop will be ready for sale in December
  1. Wait until December, then sell wheat in the spot market (sales price uncertain)
  2. Sell (go short) Wheat Futures for December delivery (sales price predetermined)
  - ⇨ Through the futures contracts the farmer locks in the price at which he delivers wheat
  - ⇨ Futures can be a form of insurance (or, hedging) -> Eliminate the price risk
- There is a fundamental difference between the use of forward contracts and options for hedging.
  - Forward contracts are designed to neutralize risk by fixing the price that the hedger will pay or receive for the underlying asset.
  - Option contracts, by contrast, provide insurance. They offer a way for investors to protect themselves against adverse price movements in the future while still allowing them to benefit from favorable price movements.
  - Unlike forwards, options involve the payment of an up-front fee.

Speculation Example

- Whereas hedgers want to avoid an exposure to adverse movements in the price of an asset, speculators wish to take a position in the market. Either they are betting that the price of the asset will go up or they are betting that it will go down.
- In July an investor with $20,000 feels that the price of wheat will increase over the next 2 months. The current spot price is $4.00 per bushel and the futures price for the September contract (i.e., expiring in 2 months) is $4.20
- What could the investor do?
  1. Buy Spot 5000 bushels of Wheat -> put the $20,000 down now
  2. Buy (Go Long) Wheat Futures -> no money down, except a 5% deposit (see later)
  - ⇨ Futures allow more leverage (no full amount of money need to down now)
- Trading, in order to make an advantage of the price movements in the future and hopefully can make profit.
- Futures and options are similar instruments for speculators in that they both provide a way in which a type of leverage can be obtained. However, there is an important difference between the two. When a speculator uses futures the *potential loss as well as the potential gain is very large*. When options are used, no matter how bad things get, the speculator's *loss is limited to the amount paid for the options*

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Arbitrage Example

- Arbitrage involves locking in a riskless profit by simultaneously entering into transactions in two or more markets
- A stock price is quoted as £100 in London and $182 in New York
- The current exchange rate is 1.8500 (London $185)
- What is the arbitrage opportunity (or, free lunch)?

  Buy low, sell high (the stock is cheaper in NY) => buy in NY, and sell in London $3 per share **without risk**

  The **difference** between speculation and arbitrage is: arbitrage allows people to make no-risk profit; whereas in speculation, the risk (uncertainty) is high.

Three Reasons for Trading Derivatives

- To hedge risks (=insurance)
- To speculate (=take a view on the future direction of a market value)
- To lock in an arbitrage profit (=free lunch)

Who Uses Derivatives? For What Purpose?

1. Treasury/Risk Management departments, consumers/investors hedging FX, interest rate, raw material price risk, firm-wide risk (hedging)
2. Proprietary trading groups/desks exploiting mispricing across related securities (arbitrage)
3. Speculators (Position Traders) engaging in leveraged bets (speculation)
4. Hedge funds trade derivatives for all three reasons (See Business Snapshot 1.3)

Derivatives Misuse

- When a trader has a mandate to use derivatives for hedging or arbitrage, but then switches to speculation, large losses can result. (*See Business Snapshot 1.4*)
  - When speculate, if price turns against you => not hedging! => will increase the loss even more
  - It varies very much, what and how you are using the derivatives for.
  - ⇨ For insurance purpose, ok
  - ⇨ If saying doing insurance, but actually speculating, then big problem

One of the risks faced by a company that trades derivatives is that an employee who has a mandate to hedge or to look for arbitrage opportunities may become a speculator.

Futures and Forward Markets

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Reading: chapter 2

Futures contracts are traded on an exchange and the contract terms are standardized by that exchange.

Forward contracts are traded in the over-the-counter market.

Futures

- Contract Specification: what, how much, where, how, when, price
- Available on a wide range of underlying assets
- **Exchange** traded ("anonymously") - If outside of the exchange, it is NOT a future contract.
- Settled daily - Every day, you have a cash inflow / outflow (as long as have a position outstanding).

Futures: Contract Specification

- The *asset* (what)
- The *contract size* (how many units of the underlying per contract) e.g. 100 shares of Telstra; $US 400
    - The contract size specifies the amount of the asset that has to be delivered under one contract.
    - The exchange size is determined by the exchange organization
- The *delivery arrangement* (where, how)
    - Where: the asset will be exchanged (in what location – the address)
    - The place where delivery will be made must be specified by the exchange. This is particularly important for commodities that involve significant transportation costs.
    - When alternative delivery locations are specified, the price received by the party with the *short position is sometimes adjusted according to the location chosen by that party*. The price tends to be higher for delivery locations that are relatively far from the main sources of the commodity.
- The *delivery month* (when)
    - The exchange must specify the precise period during the month when delivery can be made. For many futures contracts, the delivery period is the whole month.
- The delivery prices
    - The price of which the transaction will take place
- Position limits

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- The contract will also specify, the biggest position you can take. So, individual trades do not generate too much volatility
- Position limits are the maximum number of contracts that a speculator may hold.
- The purpose of these limits is to *prevent speculators from exercising undue influence on the market*.

- Price limits
  - Daily price movement limits are specified by the exchange.
  - If in a day the price moves down from the previous day's close by an amount equal to the daily price limit, the contract is said to be limit down.
  - If it moves up by the limit, it is said to be limit up.
  - A limit move is a move in either direction equal to the daily price limit.
  - Normally, trading ceases for the day once the contract is limit up or limit down.
  - However, in some instances the exchange has the authority to step in and change the limits.
  - The purpose of daily price limits is to prevent large price movements from occurring because of speculative excesses. **However, limits can become an artificial barrier to trading when the price of the underlying commodity is advancing or declining rapidly**. Whether price limits are, on balance, good for futures markets is controversial.

Futures: Underlying

- Agricultural and other commodities (CBOT, CME)
- Natural resources (NYMEX)
- Foreign currencies (PHLX)
- T-bills and Eurodollars (CME)
- T-bonds (CBOT)
- Equities (ASX 200, S&P 500 – CME)
  - Single stock futures

Futures: Exchange Trading

- The contract between the two parties is replaced with separate contracts with an intermediary (the clearinghouse). It guarantees the performance of the parties to each transaction.
  - For me, the effective counter party is NOT Roger, it is the Exchange. As long as I pay the money to Exchange, the Exchange will provide the asset.
  - Technically, two contracts: my contract with Exchange and Roger with Exchange.
- The clearinghouse is both long and short => offsetting position, the intermediary who guarantee the transaction takes place.

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- The clearinghouse has to monitor credit risk.
- Open outcry system (on the floor) or electronic platform

Opening Futures Positions

- To open a position, you call broker or enter into the contract via online trading account.
  - Contracts are referred to by their delivery month.
  - e.g. "long May WTI" (oil futures)

    You commit to purchase certain number of oil coming May
- No initial payment, except bid-ask spreads, commissions and margins (see later)
  - No down payment, but need to pay commissions, pay the deposit (**margins**), the bid-ask spreads
  - ⇨ Bid-ask spreads: at a given point in time, the price which you actually buy does not equal to the price you actually sell, because there is a bid-ask spread (transaction costs that incur)
- Price agreed upon today is the price at which transactions will take place in the future
  - Buyer pays the seller the futures price and seller deliver the asset. Or does he?

Closing Out Futures Positions

- Two options: *take delivery* or *reverse a position*
  - Take the delivery: until the maturity of the contract to pay the money and get the asset
  - Reverse a position: from long to short position or from short to long
- Most contracts don't lead to delivery (less than 2%).
  - Might be inconvenient (especially for speculators).
  - Can be expensive (storage costs, transport).
  - The reason is that most investors choose to close out their positions prior to the delivery period specified in the contract. Making or taking delivery under the terms of a futures contract is often inconvenient and in some instances quite expensive
- Reversing a position (especially, the reason for hedgers not hold until maturity)

  Closing out position: Submitting an order, which is exactly the same but with opposite sign.
  - Enter into a contract that is opposite of the original contract
  - Example: March 6: long (buy) 1 July corn, April 12: short(sell) 1 July corn
    - ⇨ Profit or loss is determined by the change in the futures price between March 6 & April 12

Delivery

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- If a futures contract is not closed out before maturity, it is settled by delivering the assets underlying the contract.
- When there are alternatives about what is delivered, where it is delivered, and when it is delivered, the party with the short (sell) position chooses.
- A few contracts (for example, those on stock indices) are necessarily (i.e., always) settled in cash => only money transfer (offsetting of the position) no physical assets transferred
  - This is because, it is inconvenient or impossible to deliver the underlying asset.
  - E.g. future contract on the S&P 500, delivering the underlying asset would involve delivering a portfolio of 500 stocks.
    - ⇨ Because, if going to deliver the SPX 500, need to buy the stock in specified quantities => lots of transaction costs
    - ⇨ Also because, the composition of the index may change over time => the underlying asset is different from original thought.
  - When a contract is settled in cash, all outstanding contracts are declared *closed on a predetermined day*. The final settlement price is set equal to the spot price of the underlying asset at either the open or close of trading on that day.

  98% of the time, settlement takes place in cash.

  For some assets (e.g. stock) the settlement 100% take places in cash.

Terminology: Long & Short

- Long: The party that has agreed to receive (buy) the underlying asset through the derivative contact acquires a long position on the underlying
- Short: The party that has agreed to deliver (sell) the underlying through the derivative contact acquires a short position on the underlying

Long – receive; Short – delivery (the underlying)

- Warning: the long/short concept looks plain and simple. Make sure you don't overlook its nuances!

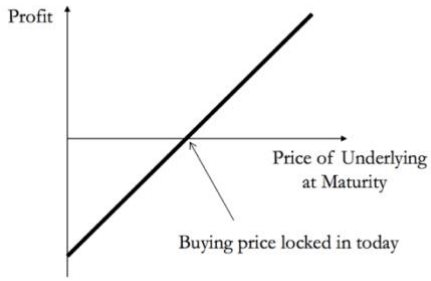
Payoffs to Long and Short Positions

- ***Long***
  - Generally used to describe the buyer of the underlying
  - Long positions agree to receive an asset at today's futures price
  - Longs profit when spot price increases
  - Payoff (when closing out) = Spot - Futures

Profit/Losses from a Long Futures Position

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⇨ For every dollar above the agreed upon price => a dollar in profit

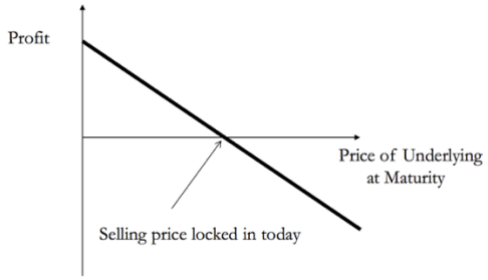


⇨ For every dollar below the agreed upon price => a dollar in loss

	If price rise	If price fall
Long (buying) future contract	Gain	Lose
Underlying spot (gold)	Loss	Gain
Net outcome	Approximately 0	Approximately 0

- ***Short***
  - Generally used to describe the seller of the underlying
  - Short positions agree to deliver an asset at today's futures price
  - Shorts profit when spot price decrease
  - Payoff (when closing out) = Futures - Spot

Profit/Losses from a Short Futures Position



⇨ Earn money when price decrease

⇨ Loss money when price increase

	If price rise	If price fall
Short (selling) future contract	Loss	Gain
Underlying spot (USD)	Gain	Lose
Net outcome	Approximately 0	Approximately 0

Margin Accounts

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- To reduce risk of default, futures exchanges require parties to post margin. Open a margin account, where you put some money just in case, to reduce the risk of default.
- In practice, a trader posts margin (cash or liquid securities) with a broker through whom they trade. The broker in turn places margin with the clearinghouse.

  Individual, margin account with broker => broker put margin with the clearinghouse
- Margin = leverage on the position as only about 5% of the value of a contract is required
- Initial margin (the amount that must be deposited at the time the contract is entered into)
- Maintenance margin: say 75% of initial margin
  - To ensure that the balance in the margin account never become negative, a maintenance margin, which is somewhat lower than the initial margin, is set.
  - If the balance in the margin account *falls below the maintenance margin*, the investor receives a margin call and is expected to *top up the margin account to the initial margin* level the next day.
- Minimum levels for the initial and maintenance margin are set by the exchange clearing house.
  - Individual brokers may require more margin from their clients than the minimum level specified by the exchange clearing house.
  - Minimum margin levels are determined by the variability of the price of the underlying asset and are revised when necessary. The higher the variability, the higher the margin levels. The maintenance margin is usually about 75% of the initial margin.

Margins and Marking-to-Market

- The payoff profiles we saw earlier are snapshots at the contract maturity.
- In reality, a cash exchange between parties takes place at the end of each trading day (daily settlement)
  - Margin accounts are adjusted daily to reflect gains and losses (**marking to market**)
  - This practice is referred to as daily settlement or marking to market.
- If the balance in the margin account dips below the maintenance margin, the holder of the account gets a "margin call"
  - Need to post additional margin or the position is closed out

Marking to Market

- The difference between today's and yesterday's settlement prices is determined.
- If the difference is positive, then the long position's account is credited
- If negative, short is credited from long's accounts

Example of Marking to Market

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An investor takes a long position in **2 December** (Number of contracts: 2) gold futures contracts on June 5

- contract size is 100 oz. => 2 contracts 200 oz.
- futures price is US$400 per oz.

  on June 5th, the agreed upon price is $400.
- margin requirement is US$2,000/contract (US$4,000 in total as 2 contracts)
- maintenance margin is *US$1,500*/contract (US$3,000 in total)

Example of Marking to Market

Day	Futures Price (US$)	Daily Gain (Loss) (US$)	Cumulative Gain (Loss) (US$)	Margin Account Balance (US$)	Margin Call (US$)
	400.00			4,000	
5-Jun	397.00	(600)	(600)	3,400	0
⋮	⋮	⋮	⋮	⋮	⋮
13-Jun	393.30	(420)	(1,340)	2,660	1,340
⋮	⋮	⋮	⋮	⋮	⋮
19-Jun	387.00	(1,140)	(2,600)	2,740	1,260
⋮	⋮	⋮	⋮	⋮	⋮
26-Jun	392.30	260	(1,540)	5,060	0

2,660 + 1,340 = 4,000

< 3,000

The price pre-determined is $400. At the end of the day, the price of gold decreased by $3 per oz.

⇨ Because have 200 oz. in total, the total loss is 200 x $3 = **$600**

Because your position is less valuable by $600, the Exchange will reduce the Margin Account balance.

It happens every day, the balance goes up and down
e.g. by 13 Jun, lost $1340 in total => by 13 Jun, the balance is $1340 lower than the initial

⇨ If the account goes to too low (13 Jun) => **margin call**

No enough money in the account, put some more. And how much more depends on the maintenance margin => whenever the margin account balance goes below the maintenance margin ($3000 in the case), need to put money in.

Thus, at Jun 13, there is $1340 more into the account and the balance would back to $4000. It is above the maintenance margin $3000, because need to replenish not up to $3k but **the initial margin**.

⇨ Settlement happens every day => **marking to market**

Futures Price Quotes

- Open interest: the total number of contracts outstanding (It is the value of all open positions)
  - Equal to number of long positions or number of short positions
  - Indicates liquidity of the contract

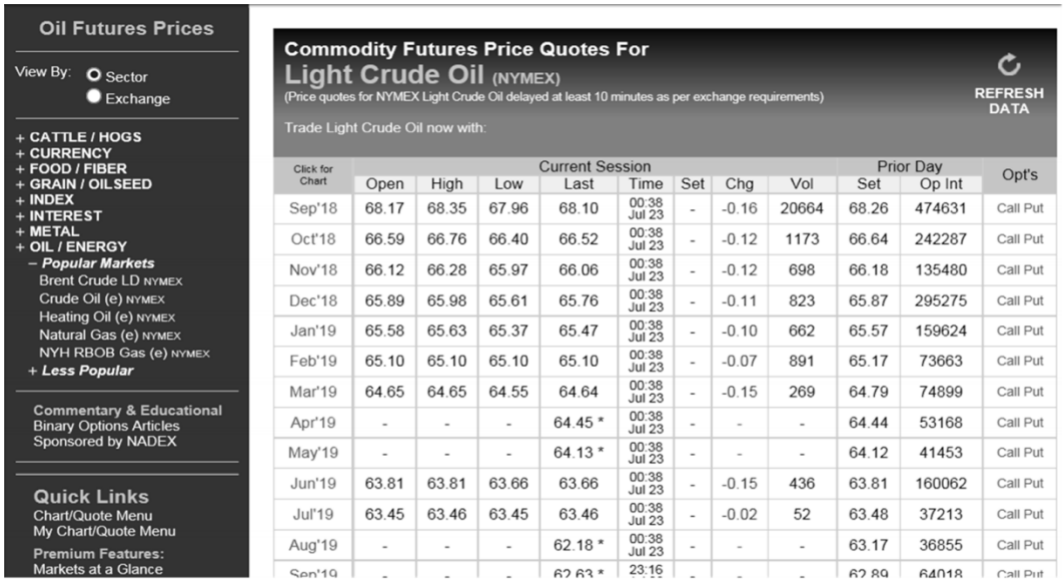
    ➔ This is the value of open contracts, which is the open interest
  - This is the contract being entered into and not closed out.

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- For every long, there is a short. But, the number of people with long position does NOT necessarily equal to the number of people with short positions.
  - ➔ The number of contracts on the long side = the number of contracts on the short side

- Settlement price: the price just before the final bell each day, used for marking the market
  - used for the daily settlement process
    - ➔ this is the one at which profit / loss are computed on a daily basis
- Volume of trading: the number of trades in 1 day



Oil Futures Prices

View By: Sector / Exchange

+ CATTLE / HOGS
+ CURRENCY
+ FOOD / FIBER
+ GRAIN / OILSEED
+ INDEX
+ INTEREST
+ METAL
+ OIL / ENERGY
– Popular Markets
Brent Crude LD NYMEX
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Click for Chart	Current Session								Prior Day		Opt's
	Open	High	Low	Last	Time	Set	Chg	Vol	Set	Op Int	
Sep'18	68.17	68.35	67.96	68.10	00:38 Jul 23	-	-0.16	20664	68.26	474631	Call Put
Oct'18	66.59	66.76	66.40	66.52	00:38 Jul 23	-	-0.12	1173	66.64	242287	Call Put
Nov'18	66.12	66.28	65.97	66.06	00:38 Jul 23	-	-0.12	698	66.18	135480	Call Put
Dec'18	65.89	65.98	65.61	65.76	00:38 Jul 23	-	-0.11	823	65.87	295275	Call Put
Jan'19	65.58	65.63	65.37	65.47	00:38 Jul 23	-	-0.10	662	65.57	159624	Call Put
Feb'19	65.10	65.10	65.10	65.10	00:38 Jul 23	-	-0.07	891	65.17	73663	Call Put
Mar'19	64.65	64.65	64.55	64.64	00:38 Jul 23	-	-0.15	269	64.79	74899	Call Put
Apr'19	-	-	-	64.45 *	00:38 Jul 23	-	-	-	64.44	53168	Call Put
May'19	-	-	-	64.13 *	00:38 Jul 23	-	-	-	64.12	41453	Call Put
Jun'19	63.81	63.81	63.66	63.66	00:38 Jul 23	-	-0.15	436	63.81	160062	Call Put
Jul'19	63.45	63.46	63.45	63.46	00:38 Jul 23	-	-0.02	52	63.48	37213	Call Put
Aug'19	-	-	-	62.18 *	00:38 Jul 23	-	-	-	63.17	36855	Call Put
Sep'19	-	-	-	62.63 *	23:16	-	-	-	62.89	64018	Call Put

  - Last: last trade of the day (for Sep contract) took place at $68.10.
    - ➔ They don't indicate the settlement price because when the market is closed, the settlement price is which the last trade took place.
  - Change: price change from settlement price of that day compared to the settlement price at the end of previous day.
    - ➔ Important: Oct contract lost 0.12 => if you have a long position at the end of that day, would lose 12 cents per oil.
    - ➔ Negative change: long's lose, short's gain

From the movie

- The price is determined by the demand and supply
- Because this is the derivative market, the demand and supply for futures contracts will depend on the expectation of the underlying asset.
- From spot market to form opinion how the price of futures market will evolve

Convergence of Futures Price to Spot Price

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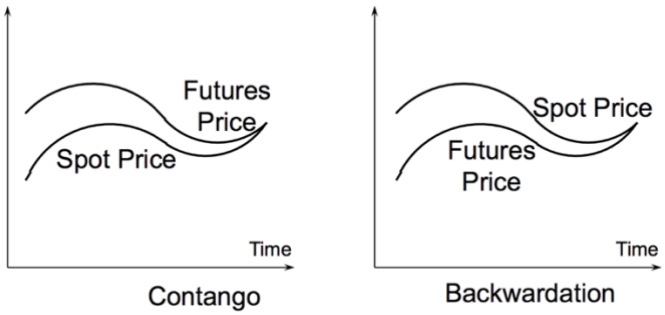
- As futures approaches expiration, futures price converges to the spot price. When the delivery period is reached, the futures price equals, or is very close to the spot price.
- Otherwise there's an arbitrage opportunity
- Example:
  - Assume futures is above the spot at maturity $F_T > S_T$
  - ⇨ sell overvalued (futures) security and buy undervalued (spot) one.
  - ⇨ Arbitrageurs sell (short) a futures contract, buy the asset, and make the delivery.
  - ⇨ Future price decreases & spot price increases
  - ⇨ Continue until prices are equal (subject to transaction costs)

    These steps are certain to lead to a profit equal to the amount by which the future price exceeds the spot price.

    As the traders exploit this arbitrage opportunity, the futures price will fall.

    At the expiration, two prices have to the same. Spot = Future

Convergence of Futures to Spot



Left: the future price is above the spot price prior to the delivery period.

Right: the futures price is below the spot price prior to the delivery period.

Forward Contracts

- A forward contract is similar to a futures contracts in that it is an agreement to buy or sell an asset at a certain time in the future for a certain price.
- But, whereas futures contracts are traded on exchanges, forward contracts trade in the over-the-counter market.
- A forward contract is an OTC agreement to buy or sell a certain quantity of a certain asset at a certain time in the future for a certain price
  - E.g. forward contract on oil, no intermediaries, deal with another party directly.
- Profit/losses profiles are the same as for futures (see earlier)
- There is no daily settlement. At the end of the life of the contract one party buys the asset for the agreed price from the other party

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- There is *no cash flow*, *until the expiration of the contract*.

  Whereas, for futures, there is cash flow transaction every day.

- Typically, in forward, there is a delivery
- Forward contracts are (more) popular on currencies and interest rates
  - Most large banks employ *both spot and forward foreign exchange traders*.
    - ➔ Spot traders are trading a foreign currency for almost immediate delivery.
    - ➔ Forward traders are trading for delivery at a future time.

Table 1.1 Spot and forward quotes for the USD/GBP exchange rate, June 22, 2012 (GBP = British pound; USD = U.S. dollar; quote is number of USD per GBP)

	Bid	*Offer*
Spot	1.5585	1.5589
1-month forward	1.5582	1.5587
3-month forward	1.5579	1.5585
6-month forward	1.5573	1.5580

The quote is for the number of USD per GBP. Bid: Buy; Offer: Sell

The first row indicates that the bank is prepared to buy GBP (also known as sterling) in the spot market (i.e., for virtually immediate delivery) at the rate of $1.5585 per GBP and sell sterling in the spot market at $1.5589 per GBP.

The second row indicates that the bank is prepared to buy sterling in one month at $1.5582 per GBP and sell sterling in one month at $1.5587 per GBP.

Forward Contracts vs Futures Contracts

Forward	**Future**
Private contract between two parties	Traded on an exchange
Not standardized (more tailored to the needs of customers)	Standardized
Usually one specified delivery date	Range of delivery dates
Settled at end of contract	Settled daily
Delivery or final settlement usual	Usually closed out prior to maturity
Some credit risk	Virtually no credit risk

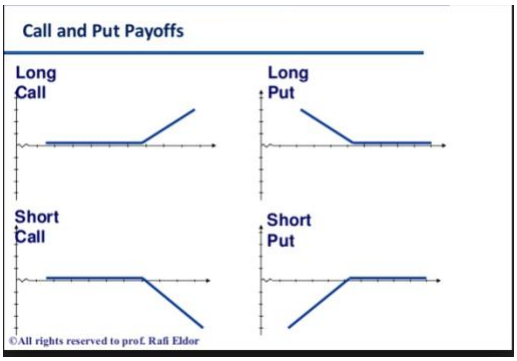
- Forward contracts traded over the counter and do not have to conform to the standards of a particular exchange.
- The contract delivery date can be any date mutually convenient to the two parties. Usually, in forward contracts a single delivery date is specified, whereas in futures contracts there is a range of possible delivery dates.

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- Unlike futures contracts, forward contracts are not settled daily. The two parties contract to settle up on the specified delivery date. Whereas most futures contracts are closed out prior to delivery, most forward contracts do lead to delivery of the physical asset or to final settlement in cash
- Under the forward contract, the whole gain or loss is realized at the end of the life of the contract.
- Under the futures contract, the gain or loss is realized day by day because of the daily settlement procedures.
- Futures prices are always quoted as the number of U.S. dollars per unit of the foreign currency or as the number of U.S. cents per unit of the foreign currency. Forward prices are always quoted in the same way as spot prices. The forward quotes show the number of U.S. dollars per unit of the foreign currency and are directly comparable with futures quotes (number of foreign dollar – the underlying asset per dollar)
    - Consider the Canadian dollar (CAD). A futures price quote of 0.9500 USD per CAD corresponds to a forward price quote of 1.0526 CAD per USD (1.0526 1⁄4 1/0.9500).

Option

Contract type	*Buyer or Holder*	*Seller or Writer*
Call option	*Right*, but no obligation, to *buy* the underlying security	*Obligation*, and not a right, to *sell* the underlying security if option exercised
Put option	*Right*, but no obligation, to *sell* underlying security	*Obligation*, and not a right, to *buy* the underlying security if option exercised



- Options are traded both on exchanges and in the over-the-counter markets.
- There are two types of option: calls and puts.
    - A call option gives the holder the right to buy an asset by a certain date for a certain price.
    - A put option gives the holder the right to sell an asset by a certain date for a certain price.
- The price in the contract is known as the exercise price or the strike price; the date in the contract is known as the expiration date or the maturity date.
- A European option can be exercised only on the maturity date; an American option can be exercised at any time during its life.
- The price of a call option decreases as the strike price increases; the price of a put option increases as the strike price increases.
- Both types of options tend to become more valuable as their time to maturity increases.

Lecture 2 – Hedging Using Futures and Forwards

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Hedging with Futures/Forwards

- In the context of futures/forwards, we will interpret hedging to mean minimizing or eliminating the variance of a portfolio (i.e., a set of positions) at a particular date in the future.
  - Buy insurance against price risk
  - Reduce the volatility
- Restated, hedging can be seen as a way to reduce or eliminate price risk of a position (or, portfolio)
- Examples
  - Pension/Mutual fund managers holding stock/bond portfolios.
  - Airlines purchasing fuel.
  - Nestle and Hershey buying cocoa for chocolate production
- Questions when deciding to hedge:
  1. Where is your exposure coming from?
  2. Should you hedge it – as eliminating risk means reduce expected return
  3. If so, how much should you reduce the risk of your portfolio?
  4. What futures contract to use?
  5. How many contracts to trade?
- Ch. 3 is about answering questions 2 through 5
- Today we will focus on 1,3, 4 and 5 (as we assume, we will hedge all the risk)

Consider a company that knows it will gain $10,000 for each 1 cent increase in the price of a commodity over the next three months and lose $10,000 for each 1 cent decrease in the price during the same period. To hedge, the company's treasurer should take a short futures position that is designed to offset this risk.

- The futures position should lead to a loss of $10,000 for each 1 cent increase in the price of the commodity over the three months and a gain of $10,000 for each 1 cent decrease in the price during this period.
- If the price of the commodity goes down, the gain on the futures position offsets the loss on the rest of the company's business.
- If the price goes up, the loss on the futures position is offset by the gain on the rest.

Hedges: Some Definitions

- A hedge is a trade used to reduce some preexisting risk exposure due to uncertainty about the evolution of asset prices.
- A static hedge is a hedge that once entered, is not rebalanced during its lifetime.
- A dynamic hedge is a hedge that must be rebalanced periodically to continue to reduce the pre-existing risk.
- For today and until otherwise noticed, we will *consider static hedges*

Short Hedges

- Involves a short position (commitment to deliver/sell the underlying) in the futures contract
  - E.g. sell 200 Apr contracts at $x
- It is appropriate when the hedger owns the underlying asset, expects to sell it in the future and wants to lock in the selling price

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- Natural Long Positions: owns/holds underlying now or expects to receive underlying in the future.
  - ⇨ Either way, needs to sell it at some time in the future => so, wants to lock the selling price => take a short hedge = natural long position in the spot
  - ⇨ E.g., US exporter and euros, farmer and wheat, portfolio manager and S&P500, oil producer
- Concerns about decreasing prices => eliminates the uncertainty about the price you are going be selling at.

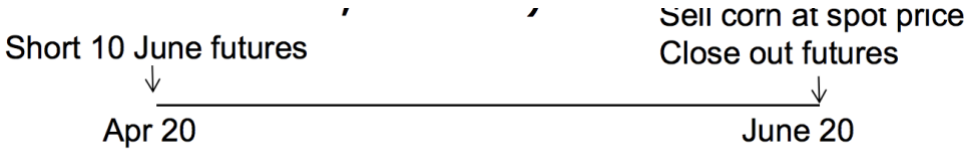
- Profit on futures will (fully or partially) offset losses in the spot asset (from seller's perspective)
  - In a spot market => natural long position (price goes down, loss money)
  - In a future market => short position (price goes down, make money)
- Hedging is if you lose in the spot market you can make it up in the futures market.

- Sell: if the spot price *below* the futures price (pre-determined price) -> the seller *gains* from the *short futures* position and lose from the spot sell => thus, offset each other.
- Sell: if the spot price *above* the futures price -> the sellers gains from the spot sell (deliver the asset) and lose on the short futures (close out the position) => offset

➔ At the end, the sell price should be approximately to pre-determined futures price

Short Hedge: Example

- April 20: Farmer negotiates to sell 50,000 bu of corn at the spot price on June 20 (naturally long corn)
- June 20 is futures maturity date
- Quotes are given as:
  - Spot price of corn: $3.50/bu
  - June corn futures price: $3.35/bu (each contract is for 5,000 bu)



Naturally long corn => take a short position in corn futures

- After futures gains & losses the price received by the farmer should be $3.35/bu.

- Scenario 1: Spot price June 20 $3.10/bu

  Spot: Farmer sells corn at: $3.10 x 50,000 = 155,000

  Futures: Gain ($3.35 – $3.10) x 50,000 = $12,500
  - ⇨ Short position in the future market
  - ⇨ Price going down => gain in futures
  - ⇨ When short at $3.35, will close out at $3.10 (spot price) because at expiration day, the spot price should equal future price

  Total received: $155,000 + $12,500 = $167,500

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Price / bu = $167,500 / 50,000 = $3.35

- Still selling the spot market and close out the position in futures market
- But in combination (money made in spot and money made in future), receive exactly the future price that you locked in

- Scenario 2: Spot price June 20 $3.70/bu

  Spot: Sell corn at: $3.70 x 50,000 = $185,000

  ⇨ Make more money in the spot

  Futures: Loss ($3.70 – $3.35 x (50,000) = $17,500

  ⇨ Lose money in the futures market

  Total received: $185,000 – $17,500 = $167,500

  Price/bu = $167,500 / 50,000 = $3.35
- Either way, price is locked in, even though there is no delivery in the futures market (as he delivered in the spot market)

  By combining the two market, the price will be the future price (the price that locked in)

  No matter what the spot price on Jun 20 is, will always receive $3.5 / bu.

Hedging is taking opposite position in two markets, so that they can offset by each other.

Long Hedges

- Involves a long position (commitment to receive/buy the underlying) in the futures contract
- It is appropriate when the hedger has (or, expects) to purchase the asset in the future and wants to lock in the buying price.
  - Natural Short Positions: needs underlying, does not have it yet.
    - ⇨ E.g., US importer and euros, cereal producer and wheat, Portfolio manager and S&P500
    - ⇨ If you are a cereal producer, you are naturally short wheat => as you need to buy wheat in the future
    - ⇨ Or, if you are a US importer, import from Europe, you are naturally short euros => as purchase euros
    - ⇨ To hedge from a naturally short position => long
  - Also appropriate for someone who is short the underlying (e.g. from a short-sale)
  - Concerns about increasing prices
- Profit on futures, if price increases, offset higher costs in the spot market

Notation

- S_0: spot price today
- S_t: spot price at time t
- S_T: spot price at expiration
- f_0: futures price today
- f_t: futures price at time t
- f_T: futures price at expiration
- t: some point prior to expiration

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- T: last day of hedge
- Π: profit / loss from strategy

Payoff (Profit/Loss) from Hedges

- Π (short hedge / naturally long) = $S_T - S_0 - (f_T - f_0)$
  - Selling the underlying at some point, hence short the futures.
- Π (long hedge) = $-S_T + S_0 + (f_T - f_0)$

  Π (short hedge) = - Π (long hedge)
- An expiring long futures contract is the same as the purchase of a spot asset ($S_T = f_T$).
  - Thus the payoff from the short hedge is simply $f_0 - S_0$
    - ⇨ Which is known at time 0 => no risk!
  - The essence of hedging is not making profit or avoiding a loss. The essence is eliminating uncertainty.
- If we *close a position prior to expiration*, say at time t, then a short hedge payoff is

  $$S_t - S_0 - (f_t - f_0)$$
  - The profit or loss is NOT a no-risk proposition
    - ⇨ If we take the future contract until expiration and close on expiration day => we have NO risk => the risk is **eliminated**
    - ⇨ If we close out PRIOR to expiration => still face some risks, even when you are hedging => the risk is **reduced**, not eliminated.

Perfect vs. Imperfect Hedges

- In the short hedge example we just saw that price risk is fully eliminated.
- Such a situation is defined as perfect hedge: no price risk (It is rare in reality)
- Perfect hedges occur if
  1. Derivative is written on the same assets being hedged (corn -> corn) (*no asset mismatch*)
  2. Date asset being hedged will be traded is known with certainty (Farmer knows it will be June 20)
  3. Expiration date of derivative matches with certainty date asset being hedged is traded (June 20) (*no maturity mismatch*)
     - ⇨ Expire on the same day, the future contract needs to expire on that day also => because this is not up to the seller, rather the organized *exchange*
- Forwards (tailored) can achieve perfect hedge if you find counterparty

- Hedging with futures contracts may not yield a perfect hedge because:
  1. There may be differences between the hedged asset and the asset underlying the futures contract => the asset you trading does not have a future contract
     - ⇨ i.e. airline purchase jet fuel (airlines are naturally short jet fuel), which means they need to take long position in futures.
     - ⇨ However, there is no futures contract written on jet fuel
     - ⇨ Thus, not likely to be a perfect hedge.
     - ⇨ And many airlines hedge jet fuel with long futures position, they take long position

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in gasoline futures

2. The date on which the asset will be purchased or sold may not be known with certainty
3. There may be a mismatch between futures contract expiration date and date when the asset is bought or sold

- Each of these issues is related to the concepts of basis and basis risk
  - Everybody deals with futures, unless they can do perfect hedge, deals with basis risk.

The Basis

It is the risk that still there even after the implementation of hedge

- Basis = Spot price of asset to be hedged – futures price of contract used to hedge
- If the hedged asset is the same as the asset underlying the futures contract, the basis equals 0 on the expiration day of the futures (otherwise -> arbitrage)
  - $S_T = f_T$
- Before the expiration day, the basis can be positive or negative
- As time passes, the spot price and a particular month's futures price do not necessarily change by the same amount and, as a result, the basis changes.
  - An *increase in the basis* is referred to as a strengthening of the basis and a *decrease* in the basis is referred to as a weakening of the basis

The Basis

- Initial basis: $b_0 = S_0 - f_0$
- Basis at t: $b_t = S_t - f_t$
- Basis at expiration: $b_T = S_T - f_T$ = 0 (if the assets are the same)
- The payoff from a short hedge closed out at time t is

$$S_t - f_t - (S_0 - f_0) \; OR \;\; b_t - b_0$$

  - The payoff from a hedge equals the change in the basis
- A hedge replaces the uncertainty about the spot price with the uncertainty about the basis
  - When hedge with futures, I replace the spot price risk (the uncertainty about the spot price) with the uncertainty about the basis.

The Basis: example

- Say the spot price today is \$2.50, the spot price at *t is \$2.00*. The current futures price is \$2.20 and the futures price at t is \$1.90.
- What is the basis at each point in time?

  $b_0 = S_0 - f_0$ = \$0.30 (2.5 – 2.2)

  $b_t = S_t - f_t$ = \$0.10

- The hedger owns the asset today (worth S_0) and will sell it for S_t at t.
- Today, he will also take a short position in the futures. Thus, he will
  - Receive S_t at t (from the spot market)

    Sell t and close out futures position (short)

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The day you trade your underlying, you close out the future position.

- Gain $f_0 - f_t$ on the futures position
- Total: $S_t + f_0 - f_t = f_0 + b_t$
  ⇨ 2.00+2.20-1.90 = 2.20 + 0.10
- Total: 2.30 = $f_0 + b_2$, since $b_2 = S_t - f_t$

- f_0 is known today, but b_t is unknown since S_t is unknown today.
  - Even after hedging, you still have uncertainty about b_t and this is why hedging is not perfect => face some risks after hedging.
  - The hedging risk is the uncertainty associated with Bt and is known as basis risk.
- A hedge replaces the uncertainty about the spot price with the uncertainty about the basis
- Since the change in the basis is less volatile than changes in the spot price, the hedged position is less risky than the unhedged one
  - Basis: The leftover after the hedge
  - Basis risk is much smaller than price risk.

Effective price = St + gain on future = initial future + basis

- Long position

  The price paid for the asset is S_t and the loss on the hedge is $f_0 - f_t$

  The effective price paid with hedging is therefore

$$S_t + f_0 - f_t = f_0 + b_t$$

  - It is the same as the short position.
  - The value of F0 is known at time 0, and the term Bt represents basis risk.

Basis and Basis Risk: Summary

- Basis risk arises because of the uncertainty about the basis when the hedge is closed outs
  - Face basis risk when using future contract because don't know what basis will be when close out the position.
  - Unless close at T.
- Perfect hedges may not be possible due to basis risk
- If hedges are held until expiration and there is no asset mismatch, basis and basis risk are reduced to zero
- However, most hedges are either rolled forward or are closed out prior to expiration, and/or are subject to asset mismatch
- Note that basis risk can lead to an improvement or a worsening of a hedger's position.
- Consider a company that uses a short hedge because it plans to sell the asset.
  - If the basis *strengthens* (i.e., increases) unexpectedly, the company's position *improves* because it will get a higher price for the asset after futures gains or losses are considered.
  - If the basis *weakens* (i.e., decreases) unexpectedly, the company's position *worsens*.

- For a company using a long hedge because it plans to buy the asset, the reverse holds.
  - If the basis *strengthens* unexpectedly, the company's position *worsens* because it will pay a higher price for the asset after futures gains and losses are considered
  - If the basis *weakens* unexpectedly, the company's position *improves* because the short the asset and they long the futures, when the spot decrease, they gain and when the futures increase gain also. **initial future + basis = effective price**

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Asset Mismatch & Cross Hedging

- Asset mismatch: sometimes a futures contract does not exist for the asset to be hedged
  - So, we going to use a future contract that is available, which is going to be written on another asset (e.g. try to hedge frozen orange, if we cannot find a future contract written on the vary asset that I am going to hedge, I am going to look for a future contract written on an asset looks like the asset I am trying to asset)
- A cross-hedge in constructed using a futures contract so that there is the highest possible correlation among the hedged asset and the asset underlying the futures contract
  - It occurs when the two assets are different.
  - Looking for a contract written on an asset whose price is highly correlated with the price of the asset that I am trying to hedge.
- A typical example of cross-hedging is *using gasoline futures* to *hedge jet fuel* price risk (done by airlines)
  - No exchange offered on jet fuel – airlines often use futures contract on gasoline to hedge jet fuel => cross hedge
- Cross hedging: the asset that gives rise to the company's exposure is sometimes different from the asset underlying the futures contract that is used for hedging.
  - It leads to an increase in basis risk.

Rolling the Hedge Forward – *maturity mismatch*

- What happens if the expiration date on the position to be hedged is later than the delivery dates for all (or, liquid) contracts available?
  1. The future contract may not be offered in that specific month or for such long maturity (e.g. 10 months) – as the exchange decide the expiration
  2. The liquidity issue – even for a very liquid futures market, only the closer maturities are traded, the trading volume was only for next month or up to 4 months. Contracts with more than 5-month expiration are not traded.
     - ⇨ When an asset is not traded, then no one wants to be the only one who trade the asset. Because the asset is considered as an *illiquid asset* (by definition), and illiquid asset has many problems, the most several one is you may not be able to *trade it at the price you see in the market* (no one on the other side of the trade) => you may suffer a much worse price as no one want to trade it with you.
  - The hedger can roll the hedge forward by:

    In order to trade the more liquid ones (with shorter maturity), hedgers open and close several positions over time.

    - ⇨ Closing out the initial futures position
    - ⇨ Taking the same position in the same contracts with a later delivery date
    - ⇨ Doing it many times, if necessary

    e.g. have a 12-month exposure, we going to use 3-month contract to hedge it; when those about to expire, we close out – open a new position for the next 3-month and keep going like it. It is done because

    - ⇨ Longer date future contracts do not exist
    - ⇨ Or because they are liquid.

    Clearly, rolling the hedge forward *create a maturity mismatch* because none of the positions you open expires the same time as the underlying asset (except the last one but need to be lucky enough)

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- ⇨ There is always a ***maturity mismatch*** when roll the hedge forward
- ⇨ Thus, there is basis risks.

Basis and Choice of Maturity

- Generally, basis risk *increases* as the time difference between delivery month (for futures) and date the underlying is traded *increases*
  - Basis increases typically when maturity mismatch increases
- Typically, one chooses contracts where such difference is as close as possible but delivery month for the futures is later
  - It assumes there is sufficient liquidity in all contracts to meet hedger's requirements. In reality, liquidity tends to be greatest in short maturity futures contracts.
  - Ex: Choose March contract for hedge expirations in Dec, Jan, Feb

    You have an exposure expiry in Dec, Jan, Feb; and the futures contracts that are available are Dec and Mar.
    - ⇨ Use the Mar contract to hedge the exposure for Jan and Feb.
    - ⇨ Choose the *Mar contract even for the Dec exposure*. Even you know you are trading in Dec, and there is one contract expires in Dec and the next one is expiring on Mar => use the expiry in Mar.
  - Futures prices may be "too" volatile during delivery month
    - ⇨ Try to match the maturities BUT we don't want it to happen in the same month unless they expiry on the same day.
    - ⇨ More trading when the contract approaches to maturity as this is when speculators and arbitrageurs come into play as well.

      Few days before maturity that is when the trading volume is highest.

      When there is lots of trading on a particular asset (specific for futures contract), **the more trading there is, the more prices move**.
      - ✧ If there is no trading, then the price will not move (stay as the previous day)
    - ⇨ Hedgers do not want hedges a contract whose price fluctuates a lot, because this create basis risk.

      Too much volatility (basis risks) in the delivery month (for hedgers only, not for speculators and arbitrageurs)

      The only exception, when there is no maturity mismatch (same day => no basis risk => perfect hedge)
  - Taking delivery can be expensive and/or inconvenient for, e.g., a cereal producer (long hedger on wheat)
- Liquidity considerations may alter the choice
  - It is a trade-off between volatility (basis risk) and liquidity
  - If the following contract is very liquid right now, then need to reconsider the choice.
    - ⇨ Look the market situation, try to avoid same month contract, but if the following month contract is too illiquid, you may need to revert to same month contract.

- At the time you trade your spot, you close out your future position.

Basis Risk and Size of Hedge

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- I have a Natural Long Position of 1 million barrels of crude oil: how many futures contracts should I go short => the size
- Restated, for every dollar ($) I'm exposed, what % of that $ should I hedge?
- If there is no basis risk
  - The size of the hedge should equal the size of the exposure
  - E.g.: Natural Long Position of 1 million barrels of crude oil should be hedged with short futures position on 1000 contract (each contract is for 1000 barrels per contract) => hedge is perfect

- If basis risk is present the size of the hedge that minimizes volatility does not need to equal the size of the exposure
  - As a hedger, still try to minimize the overall risk, but the risk minimization not necessarily achieved by hedging 1 for 1.
  - In the end, if you don't know at what price you will close out the position (either maturity or asset mismatch => basis risks), then hedging 1 for 1 does not necessarily do it.
- If there is basis risk, the best we can do is to minimize it, as we unable to eliminate it.

Optimal (i.e., Minimum Variance) Hedge Ratio

- Hedge ratio: the ratio of the *size of the position taken in the futures contract* to the *size of the exposure* (natural position)

  Hedge ratio: the size of the future position / the size of natural position
- Optimal hedge ratio is generally not 1, unless the hedge is perfect.
- Criterion for the optimal hedge is to minimize the risk (specifically, variance) of the hedger's position
- Remember that the hedger's position results from the combination of futures and spot positions

  Minimize the variance of my position (as a hedger) => a portfolio of *two positions,* my natural and my future position
  - Minimize the variance of 2 assets portfolio.

Notation

- The minimum variance hedge ratio depends on the relationship between changes in spot price and changes in the future price.
- ΔS: change is spot price during life of hedge
- ΔF: change in futures price during life of the hedge
- σ_s standard deviation of ΔS
- σ_F: standard deviation of ΔF
- ρ: correlation between ΔS & Δ F
- h: hedge ratio
- h^*: optimal (minimum variance) hedge ratio

Deriving h^*

- H* is the slope of the best-fit line from a linear regression of ΔS against ΔF.
- We would expect h* to be the ratio of the *average change in S for a particular change in F*.

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- Assume that the hedger is *short the futures* and *long the asset* => naturally long, so short the futures
  - Then, the change in the value of the position is ΔS - hΔF

    ΔS is the actual change in the value of my spot position (because I am naturally long)

    Because I am short futures, I need to put a - ΔF (put a negative sign in the front as to show my position of short futures – to correctly measure the dollar change in my position)

    If *short in the future* market, *price decrease is a good* thing for you.

    ⇨ ΔF => price goes up in the future market, which is bad for short futures.
  - If hedge 1 for 1, the change in your portfolio position would be ΔS - ΔF
    - ⇨ But, I don't have 1 unit of future exposure for every unit of naturally exposure.
    - ⇨ I have *h* units of future exposure for every unit of spot (natural) exposure.
  - The variance of the changes in the value of the hedge position is then

$$= Var(\Delta S)+h^2Var(\Delta F)-2h\rho\sigma_S\sigma_F$$
$$= \sigma_S^2+h^2\,\sigma_F^2-2h\rho\sigma_S\sigma_F$$

ΔF is the uncertain part.
h is the constant.

- To minimize the variance of the hedger's position, choose h such that

$$\frac{\partial\sigma_F^2}{\partial h} = 2h\sigma_F^2 - 2\rho\sigma_S\sigma_F$$

Inputs are based on estimations.
Price change (change in future, change in spot price) => run the regression

- Set the first derivative equal to zero and solve for h:

$$h^* = \rho\frac{\sigma_S}{\sigma_F}$$

This is the optimal hedge ratio

- Computationally, it is the slope coefficient of a linear regression of ΔS (dependent / y variable) on ΔF (independent / x variable)
- If ρ = 1 and $\sigma_F = \sigma_s$, the hedge ratio h* = 1. This result is to be expected, because in this case the *futures price mirrors the spot price perfectly*.
- If ρ = 1 and $\sigma_F = 2\,\sigma_s$ the hedge ratio h is 0.5. This result is also as expected, because in this case the futures price always *changes by twice as much as the spot price*.
- The hedge effectiveness can be defined as the proportion of the variance that is eliminated by hedging
  - This is the $R^2 from\ the\ regression\ and = \rho^2$

Optimal Hedge Ratio: Example

- A company will buy 1 million gallons of jet fuel in 3 months
- ρ(jet fuel, gasoline) = 0.80
- σ_S= 0.032
- σ_F = 0.040
- What is the optimal hedge ratio?

$$h^* = \rho\frac{\sigma_S}{\sigma_F} = 0.80\left(\frac{0.032}{0.04}\right) = 0.64$$

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- 0.64 tells you: how many units in the futures market you need to trade for each unit of your natural position
- The company needs 640,000 of gasoline to hedge their jet fuel needs. Since each gasoline contract covers 42,000 gallons, they need 15 or 16 contracts
  - ⇨ Not hedging 1 for 1; for every gallon of jet fuel (natural position) that we short, we going to long 0.64 gallons of gasoline => short 1M jet fuel, long 640,000 gasoline.
  - ⇨ In terms of how many contracts you are going go long

- *Optimal number of contracts*
  - Q_A: *size of position being hedged (units)*
  - Q_F: *size of one futures contracts (units)*
  - N^*: *optimal number of futures contracts for hedging*

$$N^* = \frac{h^* Q_A}{Q_F}$$

Hedging with Stock Index Futures

- A stock index tracks changes in the value of a hypothetical portfolio of stocks.
- Stock index futures can be used to hedge a well-diversified equity portfolio.
- We can used stock index futures to hedge an existing equity portfolio
  - Pension funds, mutual funds
- Define:
  - V_A = Current value of the portfolio
  - V_F = Current value of stocks underlying one futures contract (future price x contract size)

Hedging with Stock Index Futures

- If the portfolio exactly mimics the index, if the maturity of available futures matches the hedging horizon and if daily settlements can be ignored (see next time if not), then the optimal hedge ratio should be 1

  If your portfolio is a replicator of the index => no asset mismatch => perfect hedge (if your portfolio is an index bond)
- In such case, the optimal number of contracts to be shorted is

$$N^* = \frac{V_A}{V_F}$$

  - ⇨ Take the dollar size of your natural position (stock portfolio) / dollar size of 1 future contract => how many futures contracts you are going to trade

Stock Index Futures Hedging Example

- Suppose you have a portfolio worth $10 million (natural position) which *mirrors the S&P 500 index*. The current index value is 1254 and each futures is $250 times the index value (this is the index multiplier). How many contracts should be shorted?
  - Index multiplier, which is determined by the exchange. This is what you need to compute the dollar value of 1 future contract on S&P.

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- ⇨ $V_F =$ Index value x multiplier
- ⇨ The multiplier is not the dollars, it is just a number => to translate the number into a dollar figure, the exchange gives us a *dollar multiplier*

$$N^* = \frac{V_A}{V_F} = 10 \text{ mill} / (1254*250) = 31 \text{ or } 32 \text{ K}$$ should be shorted

- ⇨ If your portfolio is more risky than overall market in order to achieve the same level => more than 31 contracts
- ⇨ If your portfolio is less risky than the overall market, to achieve the same level of protection, need to buy less insurance. The key is the β, as whether your portfolio is more or less risky than the market

Stock Index Futures Hedging

- When the portfolio does not exactly mirror the index, we can use the parameter beta from the capital asset pricing model to determine the appropriate number of contracts to short.
- We used the β from the CAPM to "weight" the appropriate hedge ratio:

$$N^* = \beta(\frac{V_A}{V_F})$$

  - $\beta > 1$: need to buy more insurance to protect => because it is twice as sensitive to index movements as a portfolio with a beta of 1. It is therefore necessary to use twice as many contracts to hedge the portfolio.
  - $\beta < 1$: need to buy less insurance to protect

- Note 1: the formula assumes (again) no maturity mismatch (or very small mismatch such as a few days)
- Note 2: if forwards are used instead of futures, V_F is the $ value of the assets (index) underlying the forward

Stock Index Futures Hedging: Example

- S&P500 Spot = 1000
- Hedging horizon = 3 months
- S&P500 Futures 3 months = 1010
- Contract size = $250 times index
- Portfolio Value $5,050,000
- Portfolio beta = 1.5
- Risk-free rate = 4% p.a.
- Dividend yield on S&P500 = 1%
  - ➔ N* = 1.5 x 5,050,000 / (1010 x 250) = 30
  - ➔ Go short (this is the example where the hedger not holding passive index)

- Assume S&P500 spot in 3 months ends up at 900 and futures ends up at 902
- Need to include dividends when calculation **profits/losses on spot position**
  - ➔ Index Return Spot = -10% (from $1k to $900) + 1%/4 = -9.75%

    1%: dividend yield over 1 year => 3 months horizon => divided by 4

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- Need to use CAPM to compute expected portfolio return
  - ➔ 1% + 1.5*(-9.75%-1%) = -15.125%
  - ➔ For a given market return, I expect the market goes down by 9.75%
    The CAPM shows my portfolio expects to lose 15.125%

  Shorting S&P at 1010; 3 months later close out at 920 (sell at 1010; buy back at 920)
  The losses in the spot are at least partially offset by gains in the futures.
  - But, not perfectly offset, as if perfectly offset, then would be a perfect hedge.
  - But, it is not a perfect hedge because asset mismatch => the asset underlying the future contract is S&P, your natural position is not S&P.

- Need to compute value of overall (spot + futures position)
- Repeat for different values of S&P spot and futures in 3 months => end up with similar numbers
- Table 3.4 shows that the overnall value of the hedger's position does not vary (much) as the S&P does. Actually, the hedger makes the risk-free rate.
  - Compute the difference between what you make in the spot and what you make in the futures, end up with similar numbers.
  - In fact, no matter what happen to the S&P, future and spot => end up with something close to risk-free rate.

- **Why hedging then?** To earn the risk-free interest rate, the hedger can simply sell the portfolio and invest the proceeds in risk-free instruments such as Treasury bills.
- One answer to this question is that hedging can be justified if the hedger feels that the stocks in the portfolio have been chosen well. In these circumstances, the hedger might be *very uncertain about the performance of the market as a whole, but confident that the stocks in the portfolio will outperform the market* (after appropriate adjustments have been made for the beta of the portfolio).
  A hedge using index futures removes the *risk arising from market moves and leaves the hedger exposed only to the performance of the portfolio relative to the market.*
- Another reason for hedging may be that the hedger is planning to *hold a portfolio for a long period of time and requires short-term protection in an uncertain market situation.* The alternative strategy of selling the portfolio and buying it back later might involve unacceptably high transaction costs.
  - If you hedge properly, the rate of return should be very close to risk-free rate => which is the purpose of hedging – eliminating risk = risk-free rate
  - Unable to create risk-free portfolio that has a return higher than risk-free rate.
    If there is, there is the arbitrage opportunity.
  - The better you hedge, the more you reduce your risk, the more your overall rate of return close to risk-free rate.
  - The reason to hedge is: if the beta > 1; when market do well, you do better; but when market is poor, hedging can reduce the downside risk – to achieve the risk-free rate is much better than losing 10%.

Not for hedging over long-term; rather short-term positioning in the futures to still take advantage of stock picking ability over the longer term

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Stock Index Futures: Changing Betas

- Suppose instead you want to use futures not to reduce betas to zero, but to some other value (larger or smaller than the current portfolio beta)
- If the current beta is greater (less) than the desired beta: a short (long) position in N* contracts is required

$$N^* = (\beta - \beta^*)\left(\frac{V_A}{V_F}\right) - short$$

$$N^* = (\beta^* - \beta)\left(\frac{V_A}{V_F}\right) - long$$

  - To change the beta of the portfolio without changing stocks in the portfolio.
  - If I can use futures contract to reduce my risk: long position – short futures
    To increase risk: long position – long futures
  - When N* negative, it suggests to do the opposite when it is positive

Futures contracts can be used to change the level of risk to a desired level (whether to reduce or increase it)

Locking the benefits of stock pricing

- Good at picking stocks that will outperform the market.
- You do not know how well the market will perform over the next few months, but you are confident that your portfolio will do better than the market. What should you do?

  - You should short $\beta\left(\frac{V_A}{V_F}\right)$ index futures contracts, where β is the β of your portfolio, V_A is the total value of the portfolio, and V_F is the current value of one index futures contracts.
  - If your portfolio performs better than a well-diversified portfolio with the same beta, you will then make money.
  - See exercise book example

Lecture 3 - Determination of Forward and Futures Prices (and their relation with Spot Prices)

Pricing Forwards & Futures

- We will determine forward/futures price in relation to the spot price by using a no-arbitrage argument

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- We start by looking at underlying assets that *do not provide interim cash payments* (dividends or interest payments) and require *no storage costs*
- Extensions (conceptually small) to cash payments, yields, storage costs will follow
- We will first consider *forward prices* (i.e., prices of forward contracts), as they are a bit simpler => forward contract: only one cash flow at the maturity of the contract
- Then, we will make the extension to *futures prices* => daily cash flow

Assumptions for Pricing

- No or negligible transaction costs
- Same tax rates for all market participants
- Market participants can borrow or lend at risk-free rate (Libor rate)
- Investors rationally take advantage of arbitrage opportunities
  - If there is arbitrage opportunities, the trader will act fast enough to make them disappear very quickly.
  - Hence, the price will move in response to the action of arbitrager
- All above assumptions need not apply to all investors: only a few large players suffice
  - The above assumptions will be sufficient if there are few large players => then the assumptions hold.

- It is the trading activities of these key market participants and their eagerness to take advantage of arbitrage opportunities as they occur that determine the relationship between forward and spot prices.

Arbitrage

- Arbitrage - arises if an investor can construct a zero-risk-investment portfolio with a sure (i.e., no risk) profit.
  - Since no investment is required, an investor can create large positions to secure large levels of profit.
  - In efficient markets, profitable arbitrage opportunities will quickly disappear.
  - One of the most important concepts in finance (and in this subject!)

    Key determinant of many prices in financial market

Notation

- T: Time to maturity
- S_0: Initial spot price
- S_T: Spot price at maturity of the contract
- F_0: Initial Forward or Futures Price
- r: risk free rate continuously compounded p.a.
- q: continuous p.a. (yield)
  - yield: the percentage of the price that is paid in either dividends or interest

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- I: Discrete payments (dividends, income)
  - The dollar value
- U: Discrete storage costs (gold, commodities)
  - Storage costs will generally apply to commodity – in dollar form
- u: continuous p.a. storage costs (like a negative dividend yield)
  - storage costs in percentage term

No Arbitrage Argument

Action	Cash Flow at t=0	Cash Flow at T
Buy 1 unit of asset at t=0	$-S_0$ (pay the price – cash outflow)	S_T (sell the underlying asset)
Short 1 forward at t=0	0 (no cash flow involved at day 0)	$F_0 - S_T$ (close out the forward contract and this is the payoff)
Borrow $\$S_0$ at rate r at t=0	S_0 (in order to cover the cost of the underlying asset, borrow money to cover the cost)	$-S_0\, e^{rT}$ (paying off the loan and the interest incurred – *continuous* compounding of interests)
Overall Position	0	$S_T + F_0 - S_T - S_0 e^{rT}$ $= F_0 - S_0 e^{rT}$

- At t = 0, open a position in the underlying asset; a position in the forward market; a position in the borrowing market => overall, *0 net cash flow*
- At t = T, sell (deliver) the unit of the underlying asset and close the forward position (because I am short, the payoff will be $F_0 - S_T$); paying off the loan and the interest incurred.
  - $F_0 - S_0 e^{rT} > 0$: everyone will want to implement this strategy as it costs nothing but can provide you with a positive cash flow => riskless opportunity of profit = arbitrage opportunity
    - ⇨ Because the nature of arbitrage opportunity, the difference of greater than 0 cannot exist for a long time

No Arbitrage Argument

- If it does not cost anything to enter into the position at time t, then riskless profits should not be earned at time T
- Therefore, the forward price at t=0 must be:

$$F_0 = S_0\, e^{rT}$$

  - Then no longer a profit to be made.
  - The forward price has to be the function of spot price, interest rate and maturity.

Example 1

- A forward contract is written on a stock. The maturity of the contract is 6 months. The stock price is \$50 today and the risk free rate is 10% per year. What is the no-arbitrage forward price?
- $F_0 = S_0\, e^{rT} = 50\, e^{0.1 * 0.5} = \$52.56$
- What is F_0 is not \$52.56?

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Arbitrage

- Suppose F_0=\$53 (overprice), and all other values are those on previous slide
- The forward is the overpriced asset and the spot is the underpriced asset => the stock is too cheap, and forward is too expensive.
  - you can earn a riskless profit by selling the forward.
  - At t = 0 (no cash outlay):
    - ⇨ Borrow \$50 at the 10% p.a. rate for 6 months to buy the stock (spot) => because the stock is the underpriced asset.
    - ⇨ Buy 1 share of stock at \$50
    - ⇨ Take a short position in the forward expiring at T for delivery at \$53:

      Selling the overpriced asset
  - At t = T
    - ⇨ Sell asset for 53 (this is the price we agreed on, deliver the stock at \$53 to the buyer)
    - ⇨ Pay back loan $50\ e^{0.1 * 0.5} = 52.56$
    - ⇨ Realize a profit of \$(53-52.56) = \$0.44
  - Enough people are buying the stock, price (S – spot) will go up; enough people are shorting the forward (f) will go down.

Table 5.2 Arbitrage opportunities when forward price is out of line with spot price for asset providing no income (asset price = \$40; interest rate = 5%; maturity of forward contract = 3 months)

Forward Price = \$43	***Forward Price = \$39***
Action now:	*Action now*:
Borrow \$40 at 5% for 3 months	Short 1 unit of asset to realize \$40
Buy one unit of asset	Invest \$40 at 5% for 3 months
Enter into forward contract to sell asset in 3 months for \$43	Enter into a forward contract to buy asset in 3 months for \$39
Action in 3 months:	*Action in 3 months*:
Sell asset for \$43	Buy asset for \$39
Use \$40.50 to repay loan with interest	Close short position
	Receive \$40.50 from investment
Profit realized = \$2.50	Profit realized = \$1.50

If $F_0 > S_0\ e^{rT}$, arbitrageurs **buy** the asset spot and **short** forward contracts
If $F_0 < S_0\ e^{rT}$, arbitrageurs **short** the asset spot and buy (**long**) forward contracts.

What Happens if the Security Pays Income?

- Dividend paying stocks with a known (discrete) dividend
  - Typically, dividend remain constant as if changes = bad signal
  - For futures or forward pricing, dividend price in 2 years are not that matter because forward (especially futures) don't go that long.
- Coupon bonds

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- Let 'I' be present value of all known cash income
- Still determine today's forward price using no-arbitrage argument

No Arbitrage…with Income

Action	Cash Flow at t=0	Cash Flow at T
Buy 1 unit of asset at t=0	$-S_0 + I$	S_T
Short 1 forward at t=0	0	$F_0 - S_T$
Borrow \$$S_0$ at rate r at t=0	$S_0 - I$	$-(S_0 - I)\, e^{rT}$
Overall Position	0	$S_T + F_0 - S_T - (S_0 - I)\, e^{rT} = 0$

Pricing a Forward with Income

- Thus, the price of a forward contract at time t=0 is:

$$F_0 = (S_0 - I)\, e^{rT}$$

  - Where $S_0 - I$ is the dividend adjusted underlying asset price and arbitrage activities force the relation to hold.

Known Yield

- For some securities, like stock indices, it is easier and more common to denote with a yield the income they pay.
- Assume that the yield is paid at an annual rate of q, continuously compounded
- Thus, we can use the same no arbitrage arguments to find the forward price.

No Arbitrage… with Known Yield

- Instead of subtracting the income from S_0, we reduce S_0 by adjusting it for the dividend yield

$$S_0\, e^{-qT}$$

  - ⇨ Providing q is positive, the outcome of the equation will < S_0
  - ⇨ Adjusting the spot price down because the present of dividend yield
  - If there is no arbitrage on assets paying a dividend yield of q

$$F_0 = S_0\, e^{-qT} \times e^{rT}(interest) = S_0\, e^{(r-q)T}$$

  - ⇨ Adjusting the spot price with dividends (with the present value of dividends => e^{-qT})

Stock Index Futures Pricing

- Treat the stock index as a dividend-paying stock
  - Dividend is paid continuously with a dividend yield rate of q
- Then (again)

$$F_0 = S_0\, e^{(r-q)T}$$

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Index Arbitrage

- Look for mispricing between the index spot and index future
- If $F_0 > S_0\, e^{(r-q)T}$ how can you profit?
  - Go **long** the index and **short** the futures
- If $F_0 < S_0\, e^{(r-q)T}$ how can you profit?
  - **Short** (sell) the index and go **long** the futures contract
    - ⇨ Done, e.g., by pension fund managers since they already own the index
- Implementation: program trading
  - When the index and futures prices get too out of line, a *computer program* is automatically enacted to trade so as to eliminate the arbitrage opportunity

Currency Forwards Pricing

- Notation:
  - S_0: price in dollars of one unit of foreign currency
  - r: domestic interest rate with continuous compounding
  - r_f: foreign interest rate with continuous compounding
    - A foreign currency is analogous to a stock paying a known dividend yield
    - The "yield" is r_f => able to get yield from foreign currency by investing them at what is in (e.g. US) risk-free rate -> the Libor
      - There is no qualitative difference between a stock paying 3% yield and US dollar that earns 3% interest rate on a Libor deposit
    - Interest earned on a foreign currency holding is dominated in a foreign currency
      - Close to Libor rate, and almost risk-free

Currency Forwards Pricing

- Recall that

$$F_0 > S_0\, e^{(r-q)T}$$

- Replacing q by r_f, we can obtain the forward / futures price on a foreign currency

$$F_0 = S_0\, e^{(r-r_f)T}$$

- *Also called Covered Interest Parity (CIP) in International Finance*
- *When* $r - r_f < 0$
  - *Futures price is less than the spot price,* S_0
- *When* $r - r_f > 0$
  - *Futures price is greater than the spot price,* S_0

Commodity Forward/Futures

- Unlike financial instruments which are designed for investment, commodities can be held for either immediate consumption or investment.
  - Examples: pork bellies, gold, sugar, crude oil, beef, wheat, corn, etc.

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- ⇨ Holds for production propose – consumption
- ⇨ Holds for speculation / hedge purpose – investment

- The price relationships we saw between the *spot and forward price* will change a little bit depends on whether it is consumption or investment commodities.

Gold Forward/Futures (and, more generally, on investment commodities)

- All commodities will have one common feature: require storage costs if you have a long position in them.
- If there are no storage costs (SC), then gold hold for investment purpose could be seen as an asset paying no income.

Thus, the futures price is $F_0 = S_0\, e^{r\,T}$

- If there are storage costs, they can be regarded as negative income.
  - If U is the PV(SC) incurred during the life of the contract and we replace I by **–U** in the previous expressions, then:

$$F_0 = (S_0 + U)\, e^{r\,T}$$

(Investment) Commodity Futures Example

- A futures contract is written on gold. The maturity of the contract is one year and the *storage cost is $2/oz per year*. The payment will be made at the end of the year. The spot price is $450 and the risk free rate is 7% per annum.
- First we need to find U
  - $U = 2e^{-0.07\ (1)} = 1.865$
  - Then the futures price is:

    $F_0 = (S_0 + U)\, e^{r\,T} = (450+1.865)\, e^{0.07\ (1)}$
    $F_0 = 484.63$

Gold Forward/Futures (and, more generally, on investment commodities)

- Storage costs can also be seen as a negative dividend yield
- Denote storage costs with 'u' (as it is in the percentage form)
- Conceptually, we can replace q with –u
- Then

$$F_0 = S_0\, e^{(r+u)\,T}$$

Short selling

- This trade, usually simply referred to as "shorting," involves selling an asset that is not owned
- It is something that is possible for some, but not all, investment assets

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Table 5.1 Cash flows from short sale and purchase of shares

Purchase of shares	
April: Purchase 500 shares for $120	−$60,000
May: Receive dividend	+$500
July: Sell 500 shares for $100 per share	+$50,000
	Net profit = −$9,500
Short sale of shares	
April: Borrow 500 shares and sell them for $120	+$60,000
May: Pay dividend	−$500
July: Buy 500 shares for $100 per share	−$50,000
Replace borrowed shares to close short position	
	Net profit = +$9,500

- The margin account should not represent a cost to the investor. This is because interest is usually paid on the balance in margin accounts and, if the interest rate offered is unacceptable, marketable securities such as Treasury bills can be used to meet margin requirements.

Lecture 4 main points

- We can value an open (i.e. pre-existing) forward contract today by involving no-arbitrage:
  - We have a long position in a forward contract entered previously at a price of K and maturity of T
  - We now (t = 0) short a forward contract with price of F_0 and maturity T
  - Our total payoff is $(S_T - K) + (F_0 - S_T) = F_0 - K$
    - ⇨ The payoff does not depend on prices at maturity T, all needed variables are known at t = o => risk-free
  - Given the total payoff is risk-free, its value at t = 0 must be $F_0 - K$ discounted back at the risk-free rate
  - $f = (F_0 - K)\, e^{-rT}$
  - Since $F_0 = S_0 e^{rt}$

$$\boldsymbol{f = S_0 - K\, e^{-rT}}$$

Week 4 - Forward and Futures Prices and their relation with Spot Prices & Options Markets: Overview and Basic Features

Futures/Forwards on Consumption Commodities

- If $F_0 > (S_0 + U)e^{rT}$
  - U = e^{-rT}
  - Borrow $S_0 + U$, buy commodity spot, short futures -> enough people buy spot (S) increases; enough people short the futures (F goes down) -> Equality
- If $F_0 < (S_0 + U)\, e^{rT}$

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- Forward is underpriced, and spot is overpriced.
- Sell commodity spot (to save storage cost), invest at risk-free rate, long futures
- But market participants holding commodity for consumption are reluctant to sell
  - ⇨ Because there is a premium (specialty) associate with the physical commodity position -> as producer need the physical asset to continue the business, they cannot just sell it and buy futures.
  - ⇨ Even there was an arbitrage opportunity, people will not take an advantage of it, because they would *give up too much*.

- $F_0 \leq (S_0 + U)\, e^{rT}$
  - Limited arbitrage
  - Thus, typically, for consumption commodity, we usually observe futures below spot => inequality
  - Forward price will always be few percentage less than the expression => because people attribute a premium to hold the physical commodity => the premium (2% or 3% difference) is the arbitrage opportunity they give up => convenience yield

Convenience Yield

- The previous concept is related to the convenience yield: i.e., the benefit from holding the physical asset
  - Convenience yield makes the difference between equality and inequality
- In formulas, c.y. is defined as the rate y such that

$$F_0 e^{yT} = (S_0 + U)\, e^{rT}$$

$$F_0 e^{yT} = S_0 e^{(r+u)T}$$

  - Once plugged the convenience yield (c.y) in, it compensates the inequality
  - y is unobservable => by observing others, able to work out convenience yield
- The greater the expectation of shortage the higher y (check level of inventories)
  - The convenience yield reflects the market's expectations concerning the future availability of the commodity
- y=0 for investment assets

Cost of Carry

- We can simplify and summarize our discussion using the cost of carry concept
- Cost of carry is equal to:
  - The storage cost
  - Plus, any interest paid to finance the asset => for all assets
  - Less any income earned on the asset
- For a non-dividend paying stock, c = r (financing cost / opportunity cost)
- For a dividend-paying stock, c = r (financing cost) – q (income that asset generated)
- For a currency, c = r – rf
- For a commodity, c = r + u (storage cost)

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- Thus, for any investment asset, the forward price is:

$$F_0 = S_0\, e^{cT}$$

- For any consumption asset

$$F_0 = S_0\, e^{(c-y)T}$$

Time 0 and Time t

- All previous arbitrage examples assumed that actions were taken at time t=0
- But everything conceptually holds at any time t during the life of the contract
- All previous pricing formulas hold at each time t between contact inception (t=0) and contract expiration (t=T)
- General formula

$$F_t = S_t\, e^{(c-y)(T-t)}$$

  - It can be used for any assets, including consumption commodity at any time during the life of the contract
  - F and S are simultaneously determined, as long as both are equally liquid, the price move simultaneously
  - In order for F > S
    - ⇨ $e^{(c-y)(T-t)} > 1$
    - ⇨ $(c-y)(T-t) > 0$ => when (c – y) > 0 (as T – t always > 0).
      - ✧ y only appears to consumption commodities, otherwise y = 0.
      - ✧ All contracts except consumption assets, for F > S, depends on c
    - ⇨ If c > 0 => F > S
  - Why people are willing to pay more in futures than spot market?
    - ⇨ C > 0 => as c is the cost of carry of the spot asset, and it costs your money.
      - ✧ It is an opportunity cost.
    - ⇨ If c is positive, meaning the cost of financing the purchase of the spot asset is higher than the yield.
    - ⇨ That is why people willing to pay more for futures position than spot position because not incur the cost of carry.

If cost of carry is positive, F > S.

  - For investment assets, if the financing costs < dividend yield => net cost is negative.
    - ⇨ Spot > futures
    - ⇨ e^ +: F > S; e^ -: F < S

Forward vs. Futures Prices

- With futures gains earn interest and losses need to be financed
- However, it turns out that forward and futures price may not differ by any appreciable amount

Relation between Forwards & Futures Prices

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- In fact, when interest rates are constant or change in a deterministic way (i.e., they are uncorrelated with spot / forward / futures prices) forward and futures prices are the same => interest rates and underlying price are uncorrelated – gains or losses can offset each other.
- At the intuitive level, accrued interest (on gains) and financing costs (on losses) tend to offset each other
  - The more they offset each other, the more a futures looks like a forward
- As interest rates change over time in an unpredictable (random, stochastic) way, there may be a divergence between forward and futures prices
- Future prices, especially short intervals are hard to predict
  - Under the assumption that price follows *random walk* (the best price to predict tomorrow's price is today's price), the *probability of making or losing money is equally likely* => gain and losses would offset each other
  - Then people in the futures position are the same as people in the forward position (as futures' daily gain or loss is offset by each other)
    - ⇨ Meaning people would accept that the futures prices = forward prices
  - If the futures prices follow random walk (as 50% of gain, 50% of loss), then what if they able to *reinvest the gains with a higher rate* than when financed the loss
    - ⇨ If this is the case, people with futures would be better off than people with forward contracts
    - ⇨ Meaning, willing to accept a *higher futures price* than forward price
  - But the situation can be vise versus
    - ⇨ Then people will want a lower price for futures price than forward

Understanding how interest rate are related to spot and futures price.

- If the underlying asset price is *positively* correlated with interest rates, futures prices tend to be higher than forward prices (long position is 'better off' with futures).
  - Positive correlation between prices and interest rate -> long in futures better off than long in forwards.
  - Because interests are higher when the asset price is higher => gains can be invested at a higher rate => better off for longs in futures
  - Then willing to pay more for futures contracts => futures price will be higher than forwards

- If the underlying asset price is *negatively* correlated with interest rates, futures prices tend to be lower than forward prices (long position is better off with forwards).

- If the underlying asset price is uncorrelated with interest rates (like, e.g., when interest rates are constant) futures and forward prices tend to be very close
- If the maturity is only a few months, the *difference between the two prices is very small* even if interest rates move randomly and are correlated with spot.

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- Other factors such as taxes, transaction costs and default risk may create a divergence between forward and futures prices.
- In general, we will assume that forward and futures prices are, in most situations, equal or almost so.

Value of a (Pre-existing) Forward Contract: Example

- 20 days ago, you went long on a forward contract with a forward price of $100. At that time the contract had 45 days to maturity. Today the risk-free rate is 10% and the underlying spot is at $102. What is the value of the forward contract to you today?
  - You care the price because you may want to unload your position to someone else => to establish a fair price.
  - Or, it might be disclosure issue by regulations.

Value of a Forward Contract

- Suppose a forward contract has already been entered into at a previous time with a forward price K (i.e., K is the delivery price in the contract – forward price was when open the position). Also suppose that the forward price today (=time 0) is F_0.
- Alternatively, if you have a forward position and want out of the contract at time t without closing it out, how much should you get (or, pay) to transfer your position to another investor?
- Conceptually, it is the same situation we saw for futures contracts
- Assessing the value of forward positions is relevant for understanding company's assets/liabilities
  - Assessing the value of the position
  - A very painful issue in the past credit crisis

    From a regulator's point of view, need to know the value of the position at any time

Value of a Forward Contract

- We will determine the value by no arbitrage
- Consider a portfolio of 2 positions:
  1. Long a forward contract entered into previously with delivery price K and maturing at T.
  2. Short a forward contract entered into at t = 0 with delivery price F_0 and maturing at T.
- Payoffs are $S_T - K$ from long and $F_0 - S_T$ from short
  - Total payoff at T is $F_0 - K$, known with certainty at t = 0
  - Since the portfolio is risk-free, its value at t = 0 must be

$$f = (F_0 - K)\, e^{-rT}$$

Value of a Forward Contract

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- Intuitively, the equation says that the gain/loss on a **long** forward contract is given by the discounted (as the cash flows occur at T) change in the forward price

  - **Since $F_0 = S_0 e^{(rT)} \rightarrow f = S_0 - Ke^{-(rT)}$**
  - **Similarly, the value of a short forward contract is**
    $$f = (K - F_0)\, e^{-rT}$$
  - **For assets paying known income, value of long forward is**
    $$f = S_0 - I - K\, e^{-rT}$$
  - **For assets paying known yield, value of long forward is**
    $$f = S_0\, e^{-qT} - K\, e^{-rT}$$

Value of a Forward Contract: Example

- 20 days ago you went long on a forward contract with a forward price of $100. At that time the contract had 45 days to maturity. Today the risk free rate is 10% and the underlying spot is at $102. What is the value of the forward contract to you today?

$$f_t = F_0 - Ke^{-rT} = 102 - 100e^{-0.1\frac{25}{365}} = \$2.68$$

Value of a Forward Contract

- The value of a forward contract is **not** the forward price.
  - F: forward price – it cannot be negative; f: value of a position, not a price – it can be negative.
- It is more intuitive to look at it as the value of an open forward position (conceptually, like the value of a futures position)
- The value of a forward contract can be negative
  - Negative f: losing money – if wants to give the contract to someone else, need to pay someone else.
- The value of the forward at maturity equals its payoff (S – K for long position…)
- The value of a forward contract at inception is zero

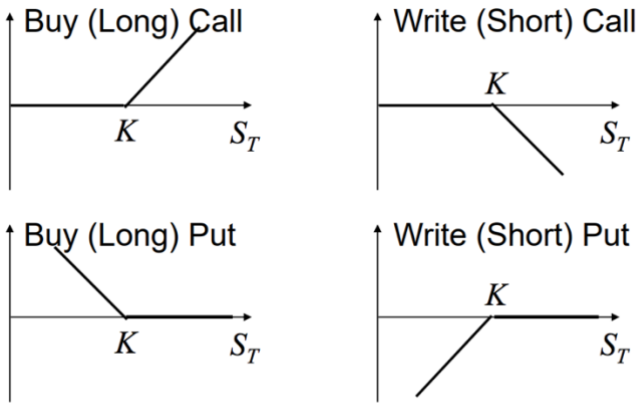
Lecture 4 main points

- Options
  - Call option – right to buy the underlying asset at a pre-specified (strike) price upon the option's exercise
  - Put option – right to sell the underlying asset at a pre-specified (strike) price upon the option's exercise
  - American – can be exercised any time during the option's life
  - European – can only be exercised at maturity
  - Naked – only the option position

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- Covered – option position + offsetting position directly in the underlying asset



- When you sell an option, you are required to post a margin, because you face a huge downside
- When a naked call option is written, the margin is the greater of:
  - 100% of sale proceeds + (20% of underlying share price – amount option is out of the money)
  - 100% of sale proceeds + 10% of underlying share price / exercise price (for puts)

- Intrinsic value = value of option of expiring immediately
  - For call option, c = max {S – K, 0}
    - ⇨ If S < K, call option has no intrinsic value and is out of the money
  - For put option, p = max {K – S,0}
    - ⇨ If S > K, the put option has no intrinsic value and is out of the money

Options Markets

Chapter 9: 9.1-9.4, 9.7, 9.8

Call and Put Options

- A call option gives the holder the right to buy a certain asset by a certain date for a certain price (the strike price – the delivery price)
- A put option gives the holder the right to sell a certain asset by a certain date for a certain price (the strike price)
- The date specified in the contract is known as the expiration date or the maturity date.
- The price specified in the contract is known as the exercise price or the strike price.

Options vs. Futures/Forwards

- A futures/forward contract gives the holder the obligation to buy or sell at a certain price

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- An option gives the holder the right to buy or sell at a certain price
  - Option contract is more flexible than the futures contract
  - Thus, need to pay for the options – more costly than futures / forward

American vs. European Options

- An American option can be exercised *at any time* during its life
- A European option can be exercised *only at maturity*

Exchange-Traded Options

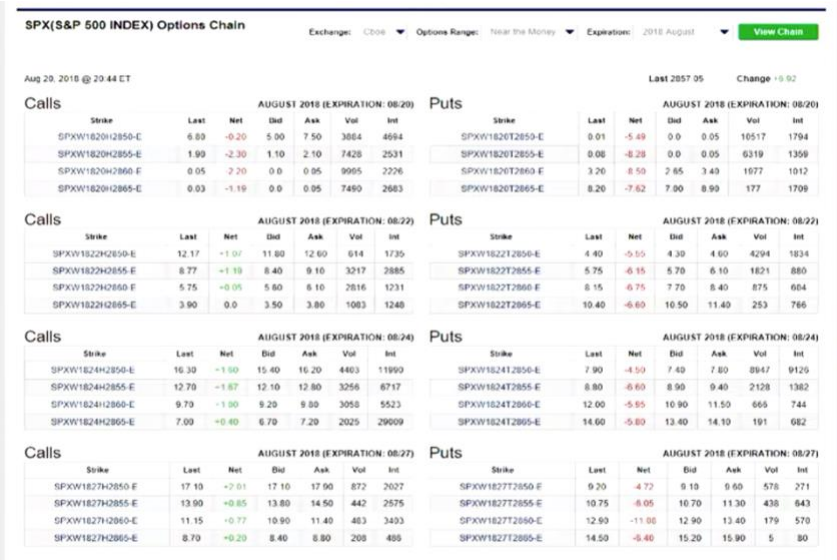
- Underlying Assets
  - Stocks
  - Foreign Currency
  - Stock Indices
  - Futures
- Specifications (specify by the clearinghouse)
  - Expiration date
  - Strike price
  - European or American: by and large, American
  - Call or Put
  - Contract size (in the US, stock options are for 100 shares)
  - Position Limits (max # of contracts on one side of the mkt)

- Option Clearing Corporation (OCC): much the same as clearinghouse for futures exchanges
- It guarantees that options writers will fulfill their obligations under the terms of options contracts and keeps a record of all long and short positions.
- If a broker is not itself a member of an exchange's OCC, it must arrange to clear its trades with a member. Members are required to have a certain minimum amount of capital and to contribute to a special fund that can be used if any member defaults on an option obligation.

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- Go to http://www.cboe.com/delayedquote/quotetable.aspx



- The underlying asset is S&P 500
  - Expiration is almost every trading day – higher frequency than futures and forward (every week)
  - For the current month and next month, expiration every week; after that only a few every month

Calls — NOVEMBER 2018 (EXPIRATION: 11/16)

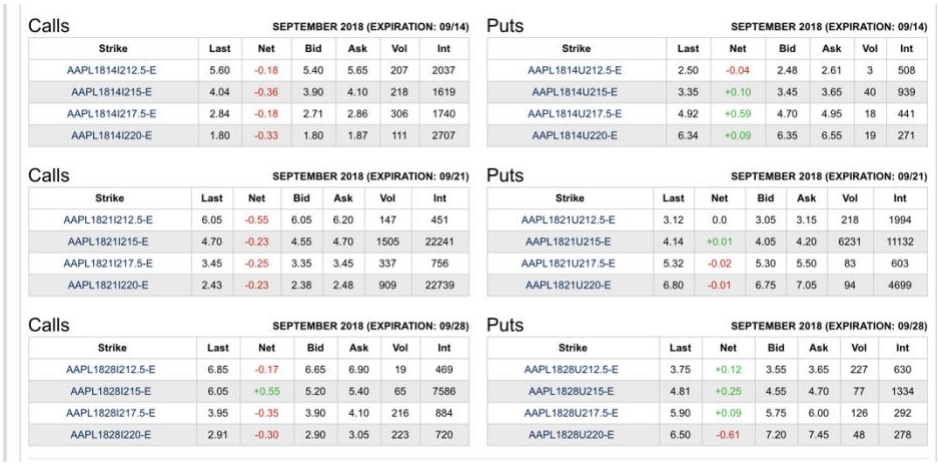
Strike	Last	Net	Bid	Ask	Vol	Int
SPX1816K2855-E	62.82	0.0	65.90	67.00	0	34
SPXW1816K2855-E	64.89	0.0	66.20	67.10	0	121
SPX1816K2860-E	66.20	+4.10	62.50	63.70	5	1207
SPXW1816K2860-E	64.36	+2.76	62.90	63.80	2	21
SPX1816K2865-E	66.20	+7.72	59.40	60.40	4	181
SPXW1816K2865-E	63.15	+14.40	59.80	60.60	3	1
SPX1816K2870-E	57.20	+2.89	56.30	57.30	80	207
SPXW1816K2870-E	62.78	+5.94	56.60	57.50	30	86

Puts — NOVEMBER 2018 (EXPIRATION: 11/16)

Strike	Last	Net	Bid	Ask	Vol	Int
SPX1816W2855-E	55.63	-4.47	57.60	58.70	12	38
SPXW1816W2855-E	0.0	0.0	58.00	58.90	0	0
SPX1816W2860-E	56.48	-3.87	59.30	60.40	26	166
SPXW1816W2860-E	56.40	-4.70	59.80	60.60	6	3
SPX1816W2865-E	60.80	-1.82	61.00	62.30	134	335
SPXW1816W2865-E	63.50	0.0	61.40	62.40	0	26
SPX1816W2870-E	57.80	-6.40	62.90	64.00	3	8
SPXW1816W2870-E	58.07	-21.43	63.30	64.30	10	22

- Strike price is following the letter 'K'
  - The first call option is with a strike price of $28.55
    - ⇨ If bought this option, the holder would have the right to acquire this S&P index for $28.55 on Nov 16.

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Calls — SEPTEMBER 2018 (EXPIRATION: 09/14)

Strike	Last	Net	Bid	Ask	Vol	Int
AAPL1814I212.5-E	5.60	-0.18	5.40	5.65	207	2037
AAPL1814I215-E	4.04	-0.36	3.90	4.10	218	1619
AAPL1814I217.5-E	2.84	-0.18	2.71	2.86	306	1740
AAPL1814I220-E	1.80	-0.33	1.80	1.87	111	2707

Puts — SEPTEMBER 2018 (EXPIRATION: 09/14)

Strike	Last	Net	Bid	Ask	Vol	Int
AAPL1814U212.5-E	2.50	-0.04	2.48	2.61	3	508
AAPL1814U215-E	3.35	+0.10	3.45	3.65	40	939
AAPL1814U217.5-E	4.92	+0.59	4.70	4.95	18	441
AAPL1814U220-E	6.34	+0.09	6.35	6.55	19	271

Calls — SEPTEMBER 2018 (EXPIRATION: 09/21)

Strike	Last	Net	Bid	Ask	Vol	Int
AAPL1821I212.5-E	6.05	-0.55	6.05	6.20	147	451
AAPL1821I215-E	4.70	-0.23	4.55	4.70	1505	22241
AAPL1821I217.5-E	3.45	-0.25	3.35	3.45	337	756
AAPL1821I220-E	2.43	-0.23	2.38	2.48	909	22739

Puts — SEPTEMBER 2018 (EXPIRATION: 09/21)

Strike	Last	Net	Bid	Ask	Vol	Int
AAPL1821U212.5-E	3.12	0.0	3.05	3.15	218	1994
AAPL1821U215-E	4.14	+0.01	4.05	4.20	6231	11132
AAPL1821U217.5-E	5.32	-0.02	5.30	5.50	83	603
AAPL1821U220-E	6.80	-0.01	6.75	7.05	94	4699

Calls — SEPTEMBER 2018 (EXPIRATION: 09/28)

Strike	Last	Net	Bid	Ask	Vol	Int
AAPL1828I212.5-E	6.85	-0.17	6.65	6.90	19	469
AAPL1828I215-E	6.05	+0.55	5.20	5.40	65	7586
AAPL1828I217.5-E	3.95	-0.35	3.90	4.10	216	884
AAPL1828I220-E	2.91	-0.30	2.90	3.05	223	720

Puts — SEPTEMBER 2018 (EXPIRATION: 09/28)

Strike	Last	Net	Bid	Ask	Vol	Int
AAPL1828U212.5-E	3.75	+0.12	3.55	3.65	227	630
AAPL1828U215-E	4.81	+0.25	4.55	4.70	77	1334
AAPL1828U217.5-E	5.90	+0.09	5.75	6.00	126	292
AAPL1828U220-E	6.50	-0.61	7.20	7.45	48	278

- The underlying asset is Apple's stocks (09 / 28)
  - Apple call option: acquire the right to buy the stock at $212.50 (strike price)
    - ⇨ The last price paid for this option is $6.85 => option price, also called **option premium**.
    - ⇨ Pay $6.85 par share in order to acquire the right to purchase Apple shares at $212.50. Each contract contains 100 shares, so one contract would cost you $685

- When you buying the right to treat the underlying, pay a **premium** to that right and the premium that you pay has to do with the
  - Spot price of the underlying
  - As well as the expectation of spot price to be in the future.

Exercising an option

- When an investor notifies a broker to exercise an option, the broker in turn notifies the OCC member that clears its trades. This member then places an exercise order with the OCC. The OCC randomly selects a member with an outstanding short position in the same option. The member, using a procedure established in advance, selects a particular investor who has written the option. If the option is a call, this investor is required to sell stock at the strike price. If it is a put, the investor is required to buy stock at the strike price. The investor is said to be assigned. When an option is exercised, the open interest goes down by one.
- At the expiration of the option, all in-the-money options should be exercised unless the transactions costs are so high as to wipe out the payoff from the option

Options Terminology

- Option class (calls or puts)
- Option series (same T, same K, within class: e.g., AAPL 107 September 9 calls)

Option Positions

- Long call
- Long put
- Short call
- Short put

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- Naked: option position is not combined with an offsetting position in the underlying asset
- Covered: option position is combined with offsetting position in the underlying asset

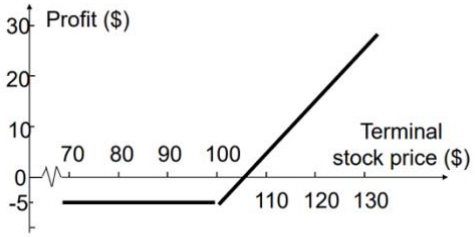
Contract type	*Buyer or Holder*	*Seller or Writer*
Call option	*Right*, but no obligation, to *buy* the underlying security	*Obligation*, and not a right, to *sell* the underlying security if option exercised
Put option	*Right*, but no obligation, to *sell* underlying security	*Obligation*, and not a right, to *buy* the underlying security if option exercised

Writer: sells an option that he does not have -> take a short position

- E.g. who goes short the stock, does not have the stock so goes with a short position

Long Call (at maturity) (profit & losses profile)

- Profit from buying one European call option: option price = $5, strike price = $100.

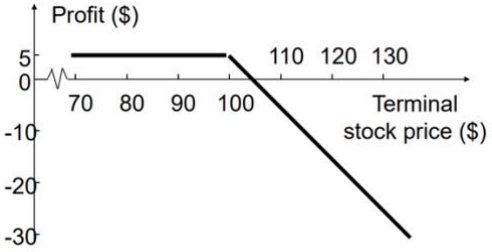


- Strike price is $100
- Option premium is $5
- Gives you the right to buy one unit of underlying for $100.

  - ⇨ If at maturity, the underlying < $100 => not exercise the option: loss of $5 (the premium)
  - ⇨ If at maturity, the underlying > $100 => exercise
  - ⇨ The break-even point is $105
- The higher the price of the underlying, the more money you can make
  - The graph is linear, with 45 degree
    - ⇨ Meaning, for every extra dollar the underlying goes up, you make an extra dollar for profit.
- Long call: unlimited upside potential, and limited downside risks.

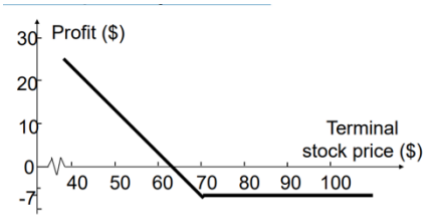
Short Call

- Profit from writing one European call option: option price = $5, strike price = $100
- The writer of an option receives cash up front, but has potential liabilities later.
- Short call: limited gains, unlimited losses



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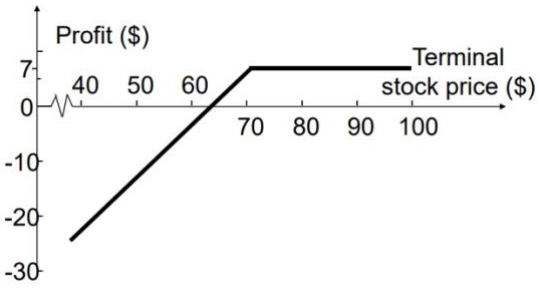
Long Put



- Profit from buying a European put option: option price = $7, strike price = $70
- As the holder of the put, you are acquiring by paying $7 to have the right to sell the underlying for $70.
  - If the underlying > $70 => not exercise.
- The lower the underlying prices, the more profits you make.
  - E.g. if the underlying is $50, the option gives you the right to sell at $70 -> buy the underlying in the spot market for $50 and exercises the option -> sell it at $70.
- The brake even here is $63.
- Long put: limited losses and limited gains as well (as the line cannot pass the y-axis -> underlying asset price cannot be negative).

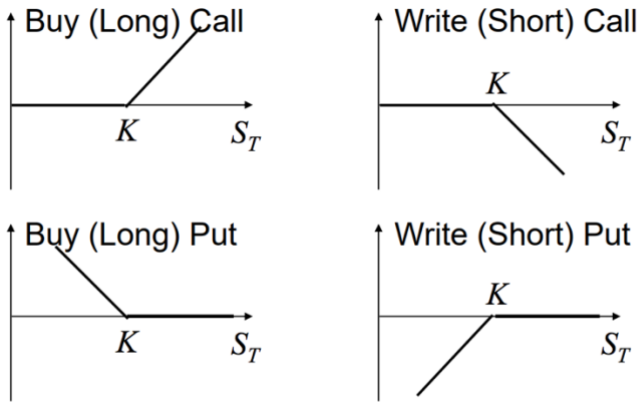
Short Put

- Profit from writing a European put option: option price = $7, strike price = $70



Payoffs from Options: Summary

- Note: cost of option non-included



Payoff vs. Profit Diagrams

- One can construct profit diagrams which subtract the initial cost of the position from the future payoffs.
- But, for most of the concepts we'll see during the course, the two are interchangeable

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Option Trading: Margins

- Margins are required when options are sold
  - When options are written – you are required to set up a margin if you write the option.
- The initial margin is usually 50% of the value of the shares, and the maintenance margin is usually 25% of the value of the shares.
- A trader who writes options is required to maintain funds in a margin account.
- For example, when a naked call option is **written** the margin is the greater of the two:
  1. A total of 100% of the proceeds of the sale plus 20% of the underlying share price less the amount (if any) by which the option is out of the money
     - ⇨ Out of the money: price of the underlying is such that the holder will not exercise.

       The writer is in the better position -> if the writer is in the better position, then he will put a smaller margin.

       The more the option is out of the money, the smaller the initial margin will be.
  2. A total of 100% of the proceeds of the sale plus 10% of the underlying share price
- Option holders, no need for an option margin because already paid the premium

Options: Notation

- K: strike price
- S_t: underlying asset price at time t
- c_t: value (premium) at time t of European call on 1 unit of underlying (C for American)
- p_t: value (premium) at time t of European put on 1 unit of underlying (P for American)

Option Trading: Margins Example

- Trader writes *4 naked calls* on stock XYZ. C=$5, K=$40, S=$38
  - Each contract is for 100 shares.
- Initial Margin requirement?
  1. 400 x ($5 + 20% x $38 - $2) = $4240
  2. 400 x ($5 + 10% x $38) = $3520
     - ➢ Initial Margin = $4240
- Same calculations are done every day as a function of the current stock price (marking to market, margin calls, etc.)

Options: Moneyness

- At the money (ATM): S = K for Call and Puts
- In the money (ITM): Calls S > K, Puts S < K
  - If for a call option, S > K => cheaper to buy the underlying by exercising the option than in the spot market
- Out of the money (OTM): Calls S < K, Puts S > K
  - For call option: S < K -> not worth to exercise, because can purchase the asset cheaper in the spot market than by exercising the option.

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Expiration date

- The last day on which options trade is the third Friday of the expiration month.
- If the expiration date for the current month has not yet been reached, options trade with expiration dates in the current month, the following month, and the next two months in the cycle. If the expiration date of the current month has passed, options trade with expiration dates in the next month, the next-but-one month, and the next two months of the expiration cycle.

    - IBM is on a January cycle. At the beginning of January, options are traded with expiration dates in January, February, April, and July; at the end of January, they are traded with expiration dates in February, March, April, and July

Strike prices

- If the stock price moves outside the range defined by the highest and lowest strike price, trading is usually introduced in an option with a new strike price. To illustrate these rules, suppose that the stock price is $84 when trading begins in the October options. Call and put options would probably first be offered with strike prices of $80, $85, and $90. If the stock price rose above $90, it is likely that a strike price of $95 would be offered; if it fell below $80, it is likely that a strike price of $75 would be offered; and so on.

Week 5 - Options Valuation: Basics

Readings: Chapter 10: 10.1 – 10.4

Options: Intrinsic value

- Value option would have if there were no time to maturity and, hence, exercise decision had to be made immediately. Value of option of expiring immediately
    - If exercise -> the option has intrinsic value; if not exercise -> no intrinsic value
        - ⇨ Zero for OTM and ATM options
        - ⇨ Intrinsic value is either 0 or positive.
    - Intrinsic value = difference between the asset price and the strike price (in the appropriate order…) for ITM options
- For calls the intrinsic value: max (S - K, 0)
    - If the difference is negative, then intrinsic value = 0.
    - If the difference is positive, there is intrinsic value.
- For puts: max (K - S, 0)
- Part of the option price will come from intrinsic value, and the IV can be 0.

Options: Time value

- For an American option, it may be optimal not to exercise and, instead, wait
- To get the intuition, think of an option written on a paying dividend stock: should you exercise as soon as the option is in the money?
    - If exercise, I pay the strike (money out); but I can dividends (money in) – there is a tradeoff.
    - But, if the option is out of money -> not exercise
    - In the money, it might not be optimal to exercise -> and whenever is NOT optimal to exercise means the investors attribute time value for this option.

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- The additional value induced by waiting is defined as the time value of the option
- We'll consider the details of optimal exercise of American option later in the term

- But European options have time value as well (see next slide, check http://www.cboe.com/delayedquote/quote-table).

SPX(S&P 500 INDEX) Options Chain

Exchange: CBOE | Options Range: Near the Money | Expiration: 2017 April | View Chain

Mar 27, 2017 @ 20:57 ET — Last 2341.59 — Change -2.39

Calls — APRIL 2017 (EXPIRATION: 04/03)

Strike	Last	Net	Bid	Ask	Vol	Int
SPXW1703D2335-E	18.07	-6.93	16.30	17.30	369	79
SPXW1703D2340-E	14.20	-6.51	13.10	14.00	385	68
SPXW1703D2345-E	11.40	-6.19	10.30	11.10	254	98
SPXW1703D2350-E	8.90	-4.55	7.80	8.60	201	1894

Puts — APRIL 2017 (EXPIRATION: 04/03)

Strike	Last	Net	Bid	Ask	Vol	Int
SPXW1703P2335-E	10.00	-2.60	9.50	10.10	139	449
SPXW1703P2340-E	11.80	-3.55	11.30	12.00	973	2002
SPXW1703P2345-E	13.70	-2.30	13.40	14.20	172	370
SPXW1703P2350-E	16.05	-1.95	15.80	16.60	182	1662

Calls — APRIL 2017 (EXPIRATION: 04/05)

Strike	Last	Net	Bid	Ask	Vol	Int
SPXW1705D2335-E	19.50	-7.07	18.40	19.50	145	265
SPXW1705D2340-E	16.20	-5.80	15.60	16.20	303	282
SPXW1705D2345-E	13.40	-4.70	12.70	13.30	87	181

Puts — APRIL 2017 (EXPIRATION: 04/05)

Strike	Last	Net	Bid	Ask	Vol	Int
SPXW1705P2335-E	12.15	-2.95	12.00	12.60	285	492
SPXW1705P2340-E	13.95	-2.85	13.90	14.50	139	921
SPXW1705P2345-E	15.15	-4.45	16.00	16.70	182	311

- The strike is $2340, and the S&P spot market is $2341; intrinsic value $1, yet someone paying $14 for this option.
  - This is a European option, with the intrinsic value of $1.5 and actually premium of $14 -> more than $12 in time value
  - Price > intrinsic value => there is time value

Options: Total Value

- Option total value: how much is the right to buy/sell worth?
- Total value is referred to as the option premium or option price (determined in the market place)
- Total Value = Intrinsic Value + Time Value

Option Valuation: Basic Questions

- How much should you charge when you write an option?
- How much would you pay to hold (buy) an option?
- More generally, what is a "fair" price for an option?
- What variables affect option value (prices)?
- Traders (speculators, arbitrageurs), brokers (market makers), hedgers, regulators need to answer these questions

Option Valuation: Basic Assumptions

- No transactions costs
- All trading profits (net of losses) subject to same tax rate
- Borrowing/lending at risk-free rate is possible
- Arbitrage opportunities vanish quickly -> we will price options by no-arbitrage

Optimal Exercise of European Options

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- Call
  - Exercise when S_T > K, $c_T = S_T - K$, in-the-money
    ⇨ Buy at a lower price than spot
  - Do nothing when S_T < K, c_T = 0, out-of-the-money

- Put
  - Exercise when K > S_T , p_T = K − S_T , in-the-money
    ⇨ Sell at a higher price than spot
  - Do nothing when K < S_T, p_T = 0, out-of-the-money

Call value(s) at expiration

$$C_T = c_T = Max\{0, S_T - k\}$$

- At the expiration, the payoff is either 0 or $S_T - k$
- Proof:
  - At expiration the call is either exercised, in which case the cash flow CF = S_T – K, or it is left to expire worthless, in which case, CF = 0.

Put value(s) at expiration

$$P_T = p_T = Max\{0, k - S_T\}$$

- Proof:
  - At expiration the put is either exercised, in which case CF = K - S_T, or it is left to expire worthless (out of money), in which case CF = 0

American vs. European Options

- American options are worth at least as much as the corresponding European options
  - Before expiration, American is always worth more than a corresponding European option.

$$c(S_0, T, K) \leq C(S_0, T, K)$$
$$p(S_0, T, K) \leq P(S_0, T, K)$$

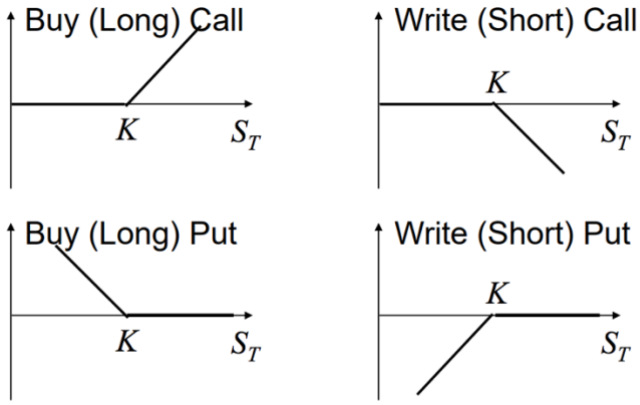
- Intuitively, an American option gives rights to the same things the corresponding European option does, and more…

Factors Affecting Option Prices

- Current stock price: S_0
- Strike price: K
- Time to maturity: T
- Risk-free rate: r
- Dividends: D
- Volatility of stock price: σ

Before expiration, there is time value and intrinsic value; at the expiration, there is only intrinsic value.

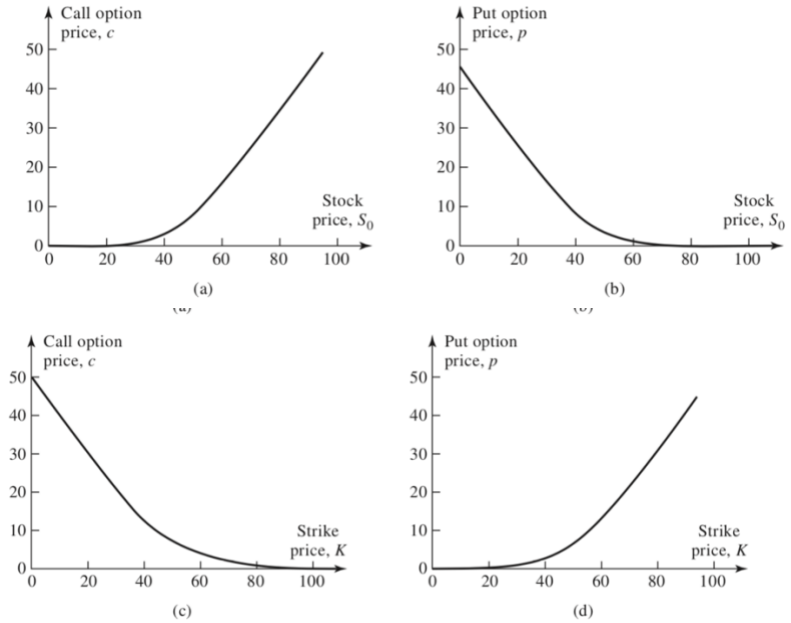
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Stock Price & Strike Price

- Call = S – K: options are *more valuable*:
  - When Stock Price increases
    - ⇨ As the underlying asset goes up, the payoff from the option goes up, and hence the price that people are willing to pay for the option increase.
  - The lower the Strike Price is, the higher the payoff, people are more willing to pay more.

- Put = K -S: options are *more valuable*:
  - When Stock Price decreases
  - The higher the Strike Price is

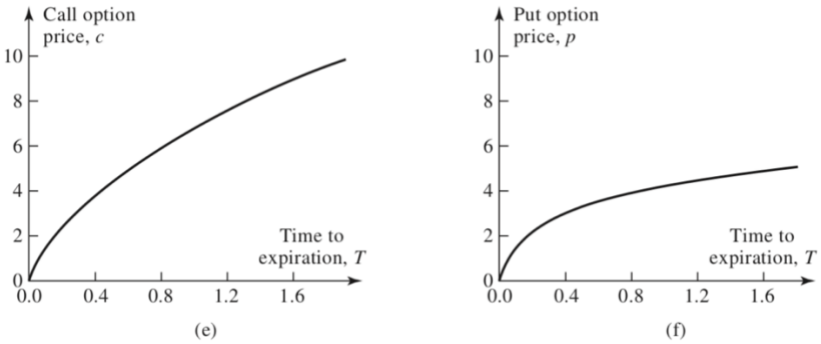


Time to Expiration

- American calls and puts: options become more valuable as the time to maturity increases
  - The longer the time, the more valuable the options
  - Consider two options that differ only in the time to maturity. The owner of the *long-lived option can take part in all of the exercise opportunities as the short-lived option, plus more*
    - ⇨ **More time till expiration = more valuable**; as for the (e.g. first two months, they can do exact the same thing) but there is an extra month for the longer holder.

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- European calls and puts: may go either way.
  - Longer option -> forgo dividend -> option value may decrease
    - ⇨ If the underlying asset is a stock. The stock pays a significant dividend – if the holder thinks the dividend is *good enough* will exercise the option (worth paying the strike price for it)
      - ✧ The shorter holder can exercise the option because it expires before the dividend is paid out.
      - ✧ The longer cannot do what holder shorter-lived option can do because the nature of European.
    - ⇨ It depends on whether the underlying asset pays a large enough income during the life of the short-lived option *but not during* the life of longer-lived option.



(Risk-Free) Interest Rate

- Two effects with opposite impacts on option value
- As interest rates in the economy increase, the *expected return required by investors from the stock tends to increase*. In addition, the *present value of any future cash flow* received by the holder of the option *decreases*.

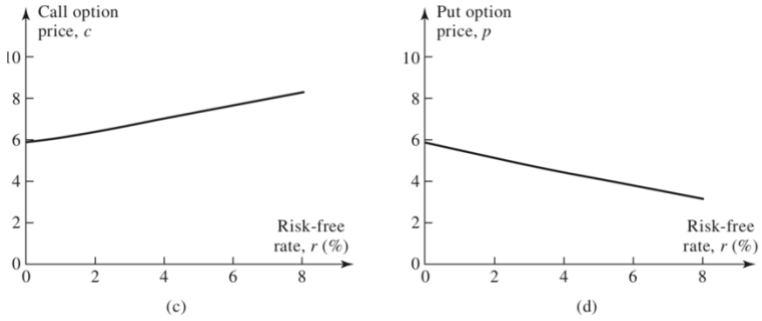
  - As *r increases*, the difference between strike price (saved) and option premium (spent) can be invested at a higher rate by the call holder.
    - ⇨ The money you saved by not exercising can be invested at a higher rate (if r increases)
  - But, *as r increases*, the present value of future cash flows received by option holder decreases

- For calls, the first effect (empirically) dominates
  - Thus, the value of a call increases as r increases
  - The buyer of a call option can defer paying for the shares. The higher the interest rate the more valuable is this right to defer

- For puts, the second effect is the only relevant one
  - Because When you exercise, you receive the strike

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- Thus, the value of a put decreases as r increases



- In practice, when interest rates rise (fall), stock prices tend to fall (rise).
  - The combined effect of an interest rate increase and the accompanying stock price decrease can be to *decrease the value of a call* option *and increase the value of a put* option.
  - Interest rate decrease & stock price increase: increase the value of the call and decrease the value of a put.

Dividends

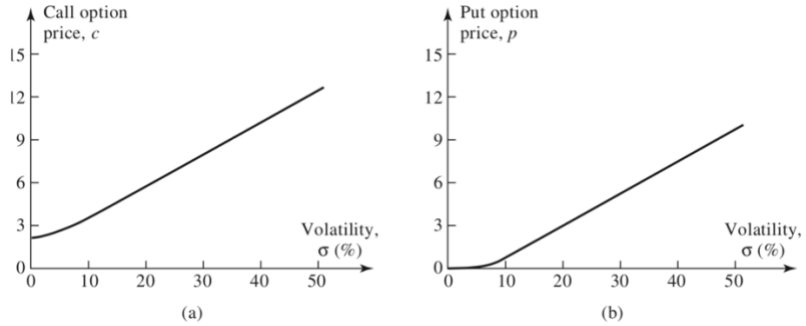
- Dividend distributions lead to a decline in the stock price, all other things equal
  - Bad news for call owners, good news for put owners
  - The value of a call decreases with dividends
  - The value of a put increases with dividends
- When dividends are paid, the value of the underlying falls.
  - The value falls at and after ex-dividend date
  - If the dividend is paid during the life of the option, the stock price will fall
    - ⇨ Bad for holder of call, as S goes down
    - ⇨ Good for holder of put

Volatility

- The most important variable, as it is the most difficult one to estimate (i.e., forecast during the life of the option). It is a measure of how uncertain we are about future stock price movements.
  - The other five are observable
  - This refers to how volatile the stock (price) return will be over the remaining life of the option.
- If volatility increases, it increases the probability that the stock price will move up or down
  - The greater the volatility, the greater the value of the option because greater chance of the price to move up and down.
    - ⇨ The upside and downside do not cancel out each other because option has an asymmetric payoff.
    - ⇨ The option allows you to keep the upward and limits the downside => volatility is valuable for option holders
  - The owner of the call will benefit from price increases, but has limited downside risk
    - ⇨ Value of a call increases when volatility increases
  - The owner of a put option benefits from price decreases, but again has limited downside risk (put premium)
    - ⇨ The value of a put option increases as volatility increases

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- Volatility modeling and forecasting is a big deal for option traders/markets (later in the term)
  - When volatility expects to increase, the option premium will increase.
  - Thus, predicting volatility of the underlying asset is the key



Effects of Variables on Option Pricing: Comparative Statics

Factors Affecting Option Prices

- $C = f(S, X, T, \sigma, r, D)$ *and* $P = f(S, X, T, \sigma, r, D)$

	Results in the value of a…	
An increase in…	*Call option*	*Put option*
1. Spot price (S)	+	–
2. Exercise price (X)	–	+
3. Time to expiration (T)	+	+
4. Price volatility (σ)	+	+
5. Riskfree interest rate (r)	+	–
6. Expected dividends (D)	–	+

- Time and volatility: Increase the probability that the option will become (more) profitable to exercise before it expires

Variable	**c**(S_0,T,K)	**p**(S_0,T,K)	**C**(S_0,T,K)	**P**(S_0,T,K)
S_0	+	-	+	-
K	-	+	-	+
T	?	?	+	+
σ	+	+	+	+
r	+	-	+	-
Dividend	-	+	-	+

+: there is a positive relation between the variable and the option price

Option Valuation: No-Arbitrage Upper Bounds

If an option price is above the upper bound or below the lower bound, there are profitable opportunities for arbitrageurs

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Upper bond: what is the maximum price that option can sell for

1. American and European Calls
   - Option can never be worth more than the stock.
   - If the option is exercised, its value is equal to S – K (this is the payoff we receive)
   - What is the maximum value of the call?
     - ⇨ When K = 0 -> max = S
       Because the payoff cannot be greater than S, hence you will never pay more than S for the option.
     - ⇨ So, call value cannot be higher than the **stock** price.

$$c(S_0, T, K) \leq S_0$$
$$C(S_0, T, K) \leq S_0$$

     - ⇨ The upper bound for a call is the stock price
       The value of the call option will never be above the value of the underlying asset. Otherwise, there is arbitrage opportunity – buying the stock and selling the call option.

2. American Puts:
   - The option can never be worth more than K.
   - If the option is exercised, its value is equal to K – S
     This is the payoff from the puts (K – S)
   - What is the maximum value of the put?
     - ⇨ When S = 0 -> max = K.
     - ⇨ So, the value of a put cannot be higher than the **exercise (strike)** price.

$$P(S_0, T, K) \leq K$$

3. European Puts:
   - p (S_0,T,K) ≤ Ke^{-rT}
   - The upper bound for a put is **PV of the strike** price.
     - ⇨ Because, we are NOT receiving k now => we are going to receive k => even with certainty to receive the k, there is a time delay => *the PV of the strike*.
   - Otherwise, riskless profit by writing the option and investing the proceeds if the sale at the risk-free interest rate.

No-Arbitrage Lower Bounds: American Calls
This is the minimum we expect to pay.

- At any time t, prior to expiration,

$$C_t \geq Max\{0, S_t - k\}$$

  - Prior to expiration, the option still has the time value.
  - So, the value of the option will always be no less the instinct value.
    - ⇨ Because there is time value, the total value (C) will > intrinsic value.

- Proof: Assume, to the contrary, that

$$C_t < Max\{0, S_t - k\}$$

  - Then, buy the call and simultaneously exercise it for an arbitrage profit of

$$S_t - k - C_t > 0$$

    - ⇨ Arbitrageurs will raise C_t and eliminate the opportunity

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- At any time, before the maturity, the option price can NOT be lower than the intrinsic value as otherwise there is arbitrage opportunity.

No-Arbitrage Lower Bounds: American Puts

- At any time t, prior to expiration,

$$P_t \geq Max\{0, K - S_t\}$$

  - At any time, before maturity, the price of an American put will never be lower than its intrinsic value.
    - ⇨ There is always time value.
- Proof: Assume to the contrary that

$$P_t < Max\{0, K - S_t\}$$

  - Then, buy the put and simultaneously exercise it for an arbitrage profit of

$$K - S_t - P_t > 0$$

No-Arbitrage Lower Bounds: European Calls & Puts on non-Dividend Paying (NDP) stocks

- Calls: $c_t \geq Max\{0, S_t - Ke^{-rT}\}$
- Puts: $P_t \geq Max\{0, Ke^{-rT} - S_t\}$
  - The expression is the PV of the payoff at T.
- Proofs: can be shown with usual no arbitrage argument
  - If the call price is less than the theoretical minimum price ($S_t - Ke^{-rT}$), then an arbitrageur can short the stock and buy the call to make riskless profit.
- Note: we are also showing formally that option values cannot be negative

Put-Call Parity: European options, no dividends

- We can use puts, stock and risk-free bonds to replicate the payoff of a call option. Then,

$$P(S_0, T, K) + S_0 = c(S_0, T, K) + Ke^{-rT}$$

  - This is the relation between the put price and call price
    - ⇨ Put price + stock (underlying) price = call price + strike (exercise) price

  - C = P + S – K(PV)
  - P = C – S + K(PV)
    - ⇨ A European long call C can be replicated => by going long a European put, long one unit of underlying and short the PV of K (borrowing the PV of k at risk-free rate)
    - ⇨ A European short put (-P = S – C – K) can be replicated by => long the underlying, short the call and borrow the PV of the strike.
    - ⇨ A risk-less position can also be created (K (PV) = P + S – C) => long the European put, long the underlying, short the call -> it is as if I was investing at risk-free rate.

- Proof by No-Arbitrage: Construct 2 Portfolios
  - A: One European Call + a $ amount = Ke^{-rT} in Cash to be invested at r
  - C: One European Put + One share of Underlying Stock
  - Next table shows that A and C have same payoffs at T regardless of S_T -> must have the same value (cost) at any time t prior to T
    - ⇨ They have the same payoff ta maturity regardless of what happens to the underlying.
    - ⇨ The costs for them are the same.

- Both of them are option holders, their payoffs depend on the underlying asset price.

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- The two portfolios are perfectly equivalent, people will pay the same price for each of them.

STRATEGY	Cash Flow @t	AT EXPIRATION (T)	
		$S_T < K$	$S_T > K$
BUY CALL	$-c_t$	0	$S_T - K$
LEND NPV(K)	$-Ke^{-r(T-t)}$	K	K
TOTAL	$-(c_t + Ke^{-r(T-t)})$	K	S_T

- A'a portfolio – buys the call

STRATEGY	Cash Flow @t	AT EXPIRATION (T)	
		$S_T < K$	$S_T > K$
BUY STOCK	$-S_t$	S_T	S_T
BUY PUT	$-p_t$	$K - S_T$	0
TOTAL	$-(S_t + p_t)$	K	S_T

- C: buys the stock
  - If S < K (strike) => exercise the put option, get a payoff of K - S
    - ⇨ And then sells the stock
  - If S > K => option will not be exercised, get a payoff of 0.
    - ⇨ But, she will still have the stock which worth St.
  - She will have a portfolio worth either K (if the option is ITM) or ST (if the option is OTM)

- Since the portfolios have identical values at time T, they must have identical values today; otherwise arbitrageur opportunity.

Put-Call Parity: American options, no dividends

- Put-call parity holds only for European options.
- However, it is possible to derive some results for American options when there are no dividends.

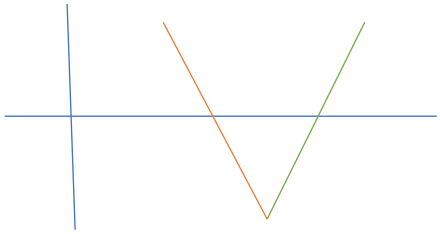
$$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$$

Week 6

- If the underlying going down significantly (you don't own the stock): shorting the stock / longing(holder) the put

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- If the stock price going up: long the stock / long the call
- If you know there is a great volatility in the stock, but you unsure about whether the price is moving up or down
  - Establish something that will make profit when price move either way and loss when price does not change.
  - Negative profit is what I am giving up if my expectation is wrong (price does not move a lot).
- To combine several options to change the profit loss profile, so that it can correspond to my view (forecast) to the underlying.
  - If it can go either up or down a lot, then short the stock or long the put may not be ok.
  - The possible strategy is long a put and long a call - ***Straddle***



Trading Strategies Involving Options

Chapter 11: 11.2 – 11.5

Option Trading Strategies

- The main point is that by combining option positions (or, options + underlying), investors can generate a variety of payoff functions (i.e., profit/losses profiles with many different shapes)
- The chosen strategy (and, thus, payoff profile) reflects views ('bets') on the evolution of the underlying, either directionally or on its volatility (either up or down)
- Most strategies we (and Ch. 11) consider focus on payoff profiles at options expiration

Option Trading Strategies

- Note: All Figures (payoff profiles) ignore time value of money – without discounting
- i.e., they simply subtract the initial cost from the final payoff

Three Types of Options Trading Strategies

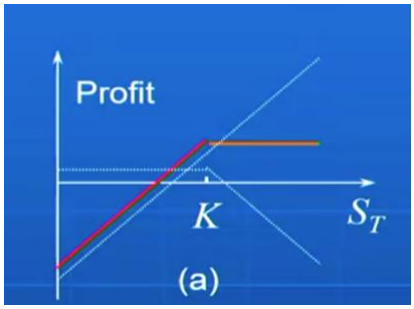
- Take a position in the option and the underlying
- Take a position in 2 or more options of the same type, all calls or all puts (a spread strategy)
- Take a position in a mixture of calls & puts (a combination strategy)

Covered Calls

- A covered call combines a position in an option with a position on the underlying (naked position is an option position without position in the underlying)
  a. Writing covered calls: combines a long stock (underlying) position with a short call

  By writing a covered call seeks protection against declines in the underlying spot price

  You are the option writer (receive the premium), short the call and take a long position in the underlying.
  - Potential losses from the shorted calls are offset by the long positions in the stock
  - ⇨ The long stock position 'covers' or protects the investor from the payoff on the short call that becomes necessary if there is a sharp rise in the stock price.
  - Aggregate payoff looks like a short put (put-call parity…)



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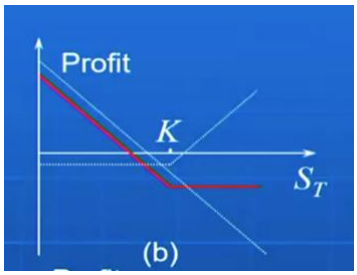
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- The straight line (45-degree line) – long stock position with a position of short call => covered call writing
- Limited upside potential and limited downside risk.

⇨ Whether the 45-degree line is better than the covered call or not fully depends on the characteristic of investors.

Covered call writing: less risky less return => able to protect fall in the underlying asset

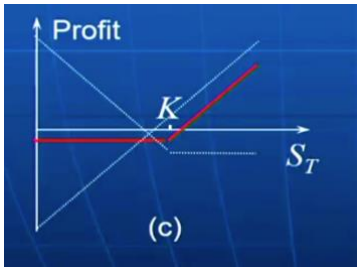
b. Buying (holding) covered Calls: the opposite strategy
   - Short stock and long call.
   - Aggregate payoff looks like a long put (put-call parity…)



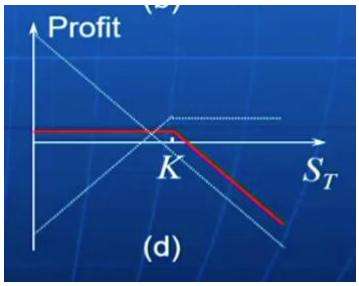
Protective Put – seeks protection against increase in the spot price

- A protective put is a combination of a long stock and a long put
  - The long position in the put protects the investor from a sharp decline in stock prices
  - Insurance on stock prices since downside risk is limited, but can take advantage of upside potential when stock prices increase
  - It is equivalent to a long European call position + a certain amount ($= Ke^{-rT} +$ $D(the\ PV\ of\ the\ dividneds\ anticipated\ during\ the\ life\ of\ the\ option)$ of cash.

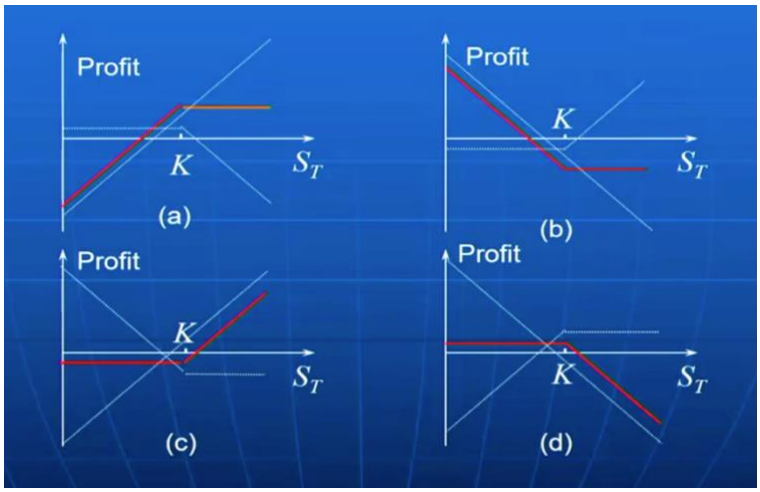
    ⇨ This explains why the profit pattern is similar to the profit pattern from a long call position.



- One could also take the opposite position: short position in a put combined with short position in a stock
  - Aggregate payoff looks like a short call



Positions in an Option & the Underlying



- If the shape is similar to the long / short call / put, why would we do this?
  - Despite the shape is similar, but not the same level.

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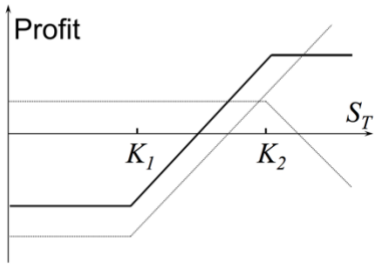
- By combining different positions in option -> can create payoff profile that match your expectations.

Spreads
- Combinations of two or more options of the same variety (all calls / all puts)
- Make bets on movement of prices in a given direction
  - Limited upside potential and downside risk
  - But, low cost strategy -> because it has a limited upside potential

Spread: Types
- Vertical: Same maturities, different strikes
  - With same underlying, same maturity, the one with lower strike is a more valuable call option
  - Debit: when set up, requires an outflow
- Horizontal (or calendar): Same strikes, Different maturities
- Diagonal: Different Strikes, different maturities

Bull Spreads
- Investors expect stock prices to increase
- Two methods:
  - Buy a call with a low strike price, sell a call with a high strike price (same T) or
    - ⇨ Call decrease with increase in K.
  - Buy a put with a low strike price, sell a put with a high strike price (same T)
- Limits upside potential and downside loss

1. ***Bull Spread Using Calls***
- Requires inotial outflow -> debit spread



  - It is a vertical strategy – same T, same asset, different strike
  - Because a call price always decreases as the strike price increases, the value of the option sold is always less than the value of the option bought. A bull spread, when created from calls, therefore requires an initial investment.
  - The profit from this strategy is: if the underlying goes up enough; you will make profit.
  - Limited downside loss and limited upside gain
    - ⇨ The maximum loss is given as the cost to set up the strategy C2 – C1
  - you are bullish on the underlying, this could be one reasonable strategy.
    - ⇨ Because it makes money for you, once the underlying goes up.

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Table 11.1 Payoff from a bull spread created using calls

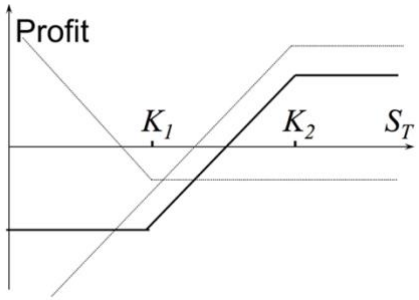
Stock price range	*Payoff from long call option*	*Payoff from short call option*	*Total payoff*
$S_T \leqslant K_1$	0	0	0
$K_1 < S_T < K_2$	$S_T - K_1$	0	$S_T - K_1$
$S_T \geqslant K_2$	$S_T - K_1$	$K_2 - S_T$	$K_2 - K_1$

(纸上写的是 profit / loss)

- The strategy can be described by saying that the investor has a call option with a strike price equal to K_1 and has chosen to give up some upside potential by selling a call option with strike price K_2 ($K_2 > K_1$).
- In return for giving up the upside potential, the investor gets the price of the option with strike price K_2

2. ***Bull Spread Using Puts***

- Buy low strike price (K) put and write high strike price put.
- Generates initial inflow -> credit spread



- Higher the strike, higher the value
- Even the shape is the same as the bull calls, the total cash flow is not at the same level. There is no guarantee that bull spread put is better than calls even at the beginning puts generate cash and calls generate outflows.

Spread calls and puts NEVER at the same level, they only have the same shape.

An investor who enters into a bull spread is hoping that the stock price will increase.

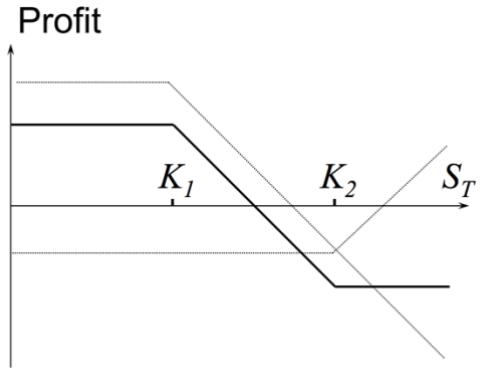
Bear Spreads

- Investors expects stock price to decrease
- Two methods:
  - Buy a call with a high strike price and sell a call with a low strike price or
  - Buy a put with a high strike price and sell a put with a low strike price
    - ⇨ Both options must have the same expiration
    - ⇨ The strike price of the option purchased is **greater** than the strike price of the option sold, the lower the strike price, the better
      - ✧ which is exactly opposite to a bull spread (where the strike price of the option purchased is always less than the strike price of the option sold).

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- Limits upside potential and downside loss

Bear Spread Using Calls



Bear Spread Using Puts

- Buy a put with a high strike price and sell a put with a low strike price
- A bear spread created from puts involves an initial cash outflow because the price of the put sold is less than the price of the put purchased.
  - ***Because with puts, higher the strike, higher the price***
  - In essence, the investor has bought a put with a certain strike price and chosen to give up some of the profit potential by selling a put with a lower strike price. In return for the profit given up, the investor gets the price of the option sold.

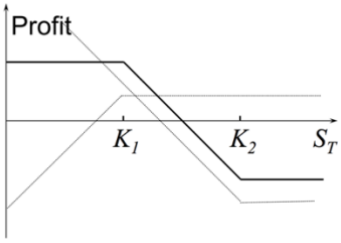


Table 11.2 Payoff from a bear spread created with put options

Stock price range	*Payoff from long put option*	*Payoff from short put option*	*Total payoff*
$S_T \leqslant K_1$	$K_2 - S_T$	$S_T - K_1$	$K_2 - K_1$
$K_1 < S_T < K_2$	$K_2 - S_T$	0	$K_2 - S_T$
$S_T \geqslant K_2$	0	0	0

- Bear spreads limit both the upside profit potential and the down- side risk.

Box Spread

- A combination of a bull call spread (with strikes at K_1 and K_2) and a bear put spread (same strikes)

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Table 11.3 Payoff from a box spread

Stock price range	*Payoff from bull call spread*	*Payoff from bear put spread*	*Total payoff*
$S_T \leqslant K_1$	0	$K_2 - K_1$	$K_2 - K_1$
$K_1 < S_T < K_2$	$S_T - K_1$	$K_2 - S_T$	$K_2 - K_1$
$S_T \geqslant K_2$	$K_2 - K_1$	0	$K_2 - K_1$

- If all options are **European** a box spread yields a constant payoff = $K_2 - K_1$
  - The value of a box spread is therefore always the PV of this payoff
    $(K_2 - K_1)\, e^{-rT}$

  - If the market price of the box spread is too low, it is profitable to buy the box.
    ⇨ Buying a call with strike price K_1, buying a put with strike price K_2, selling a call with strike price K_2, and selling a put with strike price K_1.

  - If the market price of the box spread is too high, it is profitable to sell the box.
    ⇨ Buying a call with strike price K_2, buying a put with strike price K_1, selling a call with strike price K_1, and selling a put with strike price K_2.

- If they are **American** this is not necessarily so (see Business Snapshot 11.1)
  - If, it is American, the payoff no longer constant.

- By combing these two => ends up with a straight line.
  - No matter what happens, you guarantee a payoff of K2- K1.
  - Meaning, you create a risk-free security

- Why buying box if we are going to generate risk-free rate?
  - It is an arbitrage condition
  - It involves 4 options => the price of the options match put-call parity of the box spread price

Butterfly Spreads

- They are volatility bets, NOT directional
- Can be made with either puts or calls
  - Buy lowest & highest strike calls (puts) and sell 2 middle strike calls (puts)

- Appropriate for investors who feel that large price moves are unlikely
  - Strike of middle calls typically close to current spot => meaning K2, is typically at-the-money call.
  - If St moves too much down or too much => loss.

- A butterfly spread leads to a profit if the stock price stays close to K_2 but gives rise to a small loss if there is a significant stock price move in either direction.

Thus, it is appropriate for people who expect small changes in the price.

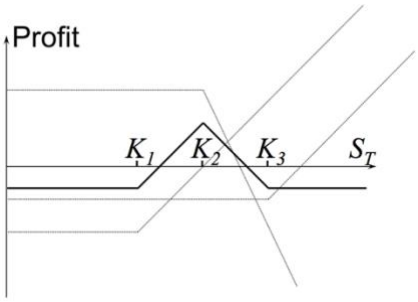
- Low cost option strategy

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- But requires a small investment initially.

- Reverse butterflies
  - Expect stock price to move "a lot", but don't know which direction
    ⇨ Still low-cost strategy

Butterfly Spread Using Calls



Butterfly Spread Using Puts

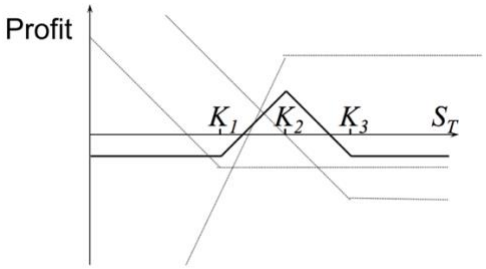


Table 11.4 Payoff from a butterfly spread

Stock price range	*Payoff from first long call*	*Payoff from second long call*	*Payoff from short calls*	*Total payoff**
$S_T \leqslant K_1$	0	0	0	0
$K_1 < S_T \leqslant K_2$	$S_T - K_1$	0	0	$S_T - K_1$
$K_2 < S_T < K_3$	$S_T - K_1$	0	$-2(S_T - K_2)$	$K_3 - S_T$
$S_T \geqslant K_3$	$S_T - K_1$	$S_T - K_3$	$-2(S_T - K_2)$	0

* These payoffs are calculated using the relationship $K_2 = 0.5(K_1 + K_3)$.

- Put–call parity can be used to show that the initial investment is the same in both cases (using puts and calls).

Calendar Spread

Strike is the same, but T different.

- Time (or, horizontal) spread involves the purchase of one option at a strike = K and $T = T_1$ and the sale of another with a strike = K and T = T_2, where $T_1 > T_2$
  - Typically sell longer option when shorter option expires
  - As T2 expires, you then close your position of T1.

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⇨ Short-maturity option expires on the assumption that the long-maturity option position is closed out at that time.
⇨ Longest holding period possible is T_2
⇨ More complicated than other spreads

- A calendar spread can be created by selling a European call option with a certain strike price and buying a longer-maturity European call option with the same strike price.
  - The longer the maturity of an option, the more expensive it usually is.
  - A calendar spread therefore usually requires an initial investment.

- Since both have the same K, both have the same intrinsic value
  - Profitability is determined solely by the difference in time values
  - This does not mean that you should always purchase the longer-term option and short the shorter-term option
    ⇨ Depends on investor's outlook for the stock

- Much like for butterfly and bull/bear spreads, volatility is a major factor in the performance of a calendar spread
  - Profits are highest for longs if low volatility
  - If prices move too far away from the exercise price, the time value becomes low for both options and profits of longs are eroded, as if the option is deep ITM or OTM, there is a little chance the price of the stock will increase or decrease even further. And thus, time value worth highest when the option is ATM because this is the time where we most uncertain about whether the option will exercise in or OTM.
  - To profit from this strategy, if the underlying does not move much away from strike.
  - Intrinsic value in this case, does not matter, because they offset each other => same underling, same strike = same intrinsic value.

    ⇨ This strategy assumes that investors expect low volatility in the price => because if there is high volatility, will be either deep in-the-money or deep out-the-money. And they have very low time value.

- The investor makes a profit if the stock price at the expiration of the short-maturity option is close to the strike price of the short-maturity option.
- However, a loss is incurred when the stock price is significantly above or significantly below this strike price.
- Stock price is very low when the short-maturity option expires.
  - The short- maturity option is worthless and the value of the long-maturity option is close to zero because it has no intrinsic value and very little time value.
  - The investor therefore incurs a loss that is close to the cost of setting up the spread initially.
- If the stock price, S_T , is very high when the short-maturity option expires.
  - The short-maturity option costs the investor S_T - K, and the long-maturity option is worth a little more than S_T - K, where K is the strike price of the options.
  - Again, the investor makes a net loss that is close to the cost of setting up the spread initially.
- If S_T is close to K, the short-maturity option costs the investor either a small amount or nothing at all.
  - However, the long-maturity option is still quite valuable. In this case a significant net profit is made.

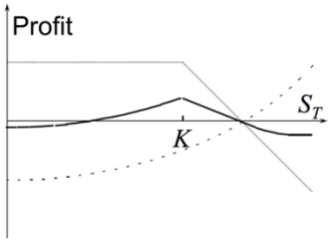
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- In a neutral calendar spread, a strike price close to the current stock price is chosen.
- A bullish calendar spread involves a higher strike price
- A bearish calendar spread involves a lower strike price.

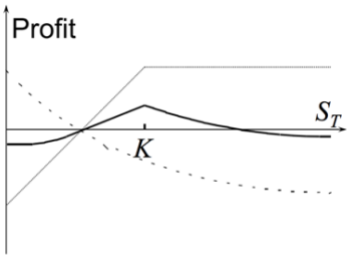
  - If volatility expectations are high, then enact a reverse calendar spread
    - ⇨ Time value worth highest when the option is at-the-money.
    - ⇨ Time value worth a little when either deep in the money or deep out the money, this is because there is a little change it will increase or increase even further.
    - ⇨ To profit from this strategy, if the underlying does not move much away from strike.
    - ⇨ Intrinsic value in this case, does not matter, because they offset each other => same underling, same strike = same intrinsic value.
      - ✧ Because, I am short one and long another, intrinsic value cancel out.
      - ✧ And because intrinsic value cancelled out, the only value left is time value of the first option.

Calendar makes money out of time value.

Calendar Spread Using Calls



Calendar Spread Using Puts



The investor buys a long-maturity put option

And sells a short-maturity put option

- A ***reverse calendar spread*** is the opposite to above figures.
  - The investor buys a short-maturity option and sells a long-maturity option.
  - A small profit arises if the stock price at the expiration of the short-maturity option is well above or well below the strike price of the short-maturity option.
  - However, a significant loss results if it is close to the strike price.

Combinations: Straddles, Strangles, Strips, and Straps

- Combination is an option trading strategy that involves taking a position in both calls and puts on the same stock.
- With any of the these strategies, the investor expects extremely large price movements (high volatility), but doesn't know in which direction the price will move
- Shorts of these strategies expect small price movement, long – large movements

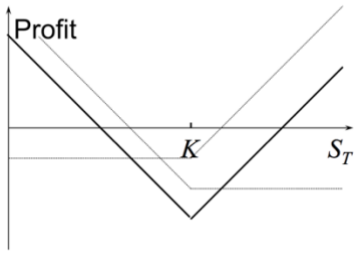
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- Typically, more expensive than spreads, typically have unlimited upside.

Straddle

- Straddle: buy a call and a put with the same strike, same T => high volatility bet
  - Maximum loss: is the lowest point on the graph, which is c + p (the sum of the premium pay for buying the call and put).
  - Short straddle: gives you more money upfront but more risky.



- If the stock price is close to this strike price at expiration of the options, the straddle leads to a loss. However, if there is a sufficiently large move in either direction, a significant profit will result.
- Appropriate when an investor is expecting a large move in stock price and does not in which direction the move will be.

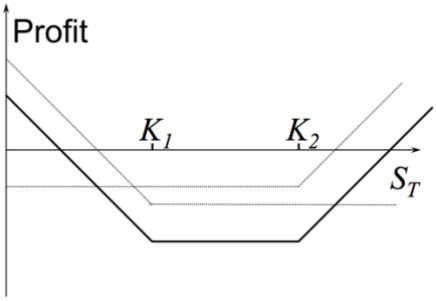
Table 11.5 Payoff from a straddle

Range of stock price	*Payoff from call*	*Payoff from put*	*Total payoff*
$S_T \leqslant K$	0	$K - S_T$	$K - S_T$
$S_T > K$	$S_T - K$	0	$S_T - K$

- The above figure is referred to as bottom straddle or straddle purchase.
- A top straddle or straddle write is the reverse position. ‘
  - It is created by selling a call and a put with the same exercise price and expiration date.
  - It is a highly risky strategy.
  - If the stock price on the expiration date is close to the strike price, a significant profit results. However, the loss arising from a large move is unlimited

Strangle

- Strangle: buy one put with low strike, one call with high strike, same T



  - The investor is betting that there will be a large price move, but is uncertain whether it will be an increase or a decrease.
  - Lower cost than others, but requires larger movement in price to profit
    - ⇨ The cost is lower because if the call is the same as straddle, the put has a lower strike => less costly.
    - ⇨ But, takes more movement to make money.
    - ⇨ Stock price has to move farther in a strangle than in a straddle for the investor to make a profit.
      - ✧ However, the downside risk if the stock price ends up at a central value is less with a strangle.

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⇨ The profit pattern obtained with a strangle depends on how close together the strike prices are. The farther they are apart, the less the downside risk and the further the stock price has to move for a profit to be realized

Table 11.6 Payoff from a strangle

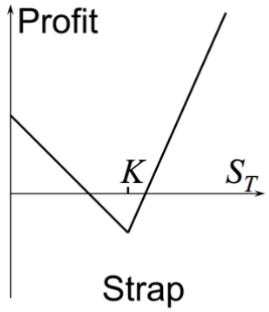
Range of stock price	*Payoff from call*	*Payoff from put*	*Total payoff*
$S_T \leq K_1$	0	$K_1 - S_T$	$K_1 - S_T$
$K_1 < S_T < K_2$	0	0	0
$S_T \geq K_2$	$S_T - K_2$	0	$S_T - K_2$

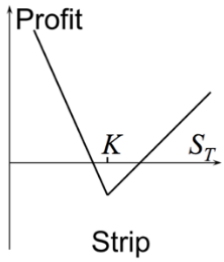
Strangle and straddle are pure volatility biased; spreads (bull and bear) pure directional biased

Stripes and straps

- Straps: buy one put and more than one call with same K, T
  - Volatility biased + some directional biased
  - More bullish strategy – believe up movements more likely than downward movement => make right leg steeper
  - In a strap the investor is betting that there will be a big stock price move – an increase in the stock price is considered to be more likely than a decrease.

- Strips: buy one call and more than one put with same K, T
  - More bearish strategy – believe downside movements more likely
  - In a strip the investor is betting that there will be a big stock price move and considers a decrease in the stock price to be more likely than an increase.





A Note of Caution

- Big move in stock price is expected given corporate events
- Straddle appears to be conceptually profitable strategy
- But what if rest of the market is also predicting these big movements => which means already in the price of the option.
  - Need to be good and to be different from the market, otherwise if what you expect is what market expect, then cannot make money.

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Week 9: Optimal Exercise (Ch. 10)
Extensions of Put-Call Parity (Ch. 10)

Readings: Chapter 10: 10.5 – 10.7

Optimal Exercise: American Options

- If the option is OTM before maturity -> no exercise
  - Even has the expectation of becoming ITM in the future, still don't exercise as you will lose money straight away.
  - If the option is ITM before maturity -> it depends on
    - ⇨ put vs. call
    - ⇨ NDP (non-dividend pay) vs. DP

Optimal Exercise: American Call, NDP – Example

- $S_0 = 50$, K = 40 -> call option is ITM by \$10, T = 1 month, r = 11%
- Should you exercise the American call?
- Two cases:
  a. You believe the stock is fairly priced or underpriced now and you, thus, would keep the stock upon exercising and for another month (i.e. until T)
     - ⇨ Within the next month, S0 will stay or go up
     - ⇨ You don't think S0 will go lower than \$50.
     - ⇨ No early exercise

  b. You believe the stock is overpriced now and, thus, would sell the stock upon exercising
     - ⇨ No early exercise, as can earn more from selling the option immediately rather than exercising

Case A):

- If you do not exercise today and S_T > 40, you pay \$40 one month later
  - The option is ITM
  - If you believe, it is fairly priced – meaning stay at \$50 => get \$10 out of exercising.
  - But, the better strategy is to keep the option and exercise it at the end of the month.
    - ⇨ You can invest \$40 for a month.
    - ⇨ And, if the saving is large enough => should not exercise

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⇨ Because the stock pays no dividends, no income from the stock is sacrificed.

- If you do not exercise today and $S_T < 40$, you will be glad the decision to exercise early was not taken
  - There is no advantage to exercise the call early if you plan to hold the stock for the rest of the life of the option
  - This is the case, the expectation is wrong, but waiting is better than early exercise.

Case B) – believe overpriced

- What if you sold the option rather than exercising it?
  - The investor is better of selling the option than exercising it.
  - The option will be bought by another investor who wants to hold the stock (at $50). Such an investor must exist otherwise the stock wouldn't be trading at $50
  - The price obtained for the option will be ≥ than the lower bound of the corresponding European call,

$$C\left(S_0, T, K\right) \geq S_0 - Ke^{-rT} \quad = \$10.36$$

    ⇨ Even this is for European, but American will always more.
    ⇨ Given r > 0, it follows C > S0 – K when T > 0 => American call price C is always greater than the option's intrinsic value prior to maturity.
  - You get $10 if you exercise the call
  - You get no less than $10.36 if you sell the call
    ⇨ You are better off selling the option than exercising
  - You could also short the stock and keep the option
    ⇨ Short the stock as you believe is overpriced, but you can keep the option (but not exercising)

You should never exercise **EARLY American calls on NDP assets**!

⇨ It is not only about generate a positive cash flow, but also you can generate more from e.g. selling it.

American Call, NDP: Conceptual Reasons for Not Exercising Early

- Strike price is paid later
  - This relates to time value of money.
  - From the perspective of the option holder, the later the strike price is paid out, the better.

- Insurance against price drop is retained
  - A call option, when held instead of the stock itself, in effect insures the holder against the stock price falling below the strike price.
  - Once the option has been exercised and the strike price has been exchanged for the stock price, this insurance vanishes.

- No income (dividend) is sacrificed
  - Because the NDP case, the underlying does not pay dividends, nothing is giving up (as with the dividends, no exercise = no chance to get the dividend, but in this case, there is no dividend paying anyway)

Save the stroke & retain the insurance & nothing is given up

Optimal Exercise: American Call, NDP

- Since $C\left(S_0, T, K\right) \geq c\left(S_0, T, K\right)$ and $c\left(S_0, T, K\right) \geq S_0 - Ke^{-rT}$

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$$C\left(S_0, T, K\right) \geq S_0 - Ke^{-rT}$$

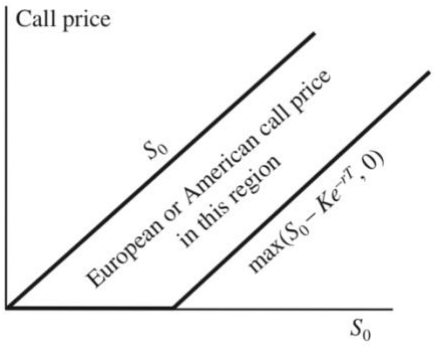
- But, $S_0 - Ke^{-rT} \geq S_0 - K$

$$\boldsymbol{C\left(S_0, T, K\right) \geq S_0 - K}$$

  - i.e., Call value is greater than intrinsic value

American Call on NDP should never be exercised early

- This also implies $C\left(S_0, T, K\right) = c\left(S_0, T, K\right)$, since the American is actually like a European => because never exercise early
- It also implies that the lower bound for $\boldsymbol{C}\left(S_0, T, K\right) \geq Max\left(0, S_0 - Ke^{-rT}\right)$
  - same as for European calls



Optimal Exercise: American Puts, NDP

- For an American put option it can be optimal to exercise early, a put option should always be exercised early if it is sufficiently deep ITM.
- An (extreme) example: K = 10, S = 0
  - If you exercise today, get $10 now (K – S)
  - If you exercise 3 months later, get $10 after three months
    - ⇨ As put option, the maximum you can get is $10 because S cannot go below 0.
    - ⇨ American call save the strike from NOT exercise early, but American put does not save anything (as you receive the strike)
  - S can't fall farther -> collecting K now vs. at time T is worth $Ke^{rT} - K$ extra dollars at time T.
    - ⇨ Early Exercise is optimal in this case

Optimal Exercise: American Puts, NDP

- More generally, it **may be** optimal to exercise an American put on an NDP asset if S is **sufficiently low**
- Like a call option, a put option can be viewed as providing insurance.
  - A put option, when held in conjunction with the stock, insures the holder against the stock price falling below a certain level.
  - However, a put option is different from a call option in that it may be optimal for an investor to forgo this insurance and exercise early in order to realize the strike price immediately.
  - In general, the early exercise of a put option becomes more attractive as *S0 decreases, as r **increases**, and as the volatility decreases*.
  - The early exercise of an American put is attractive when the interest earned on the strike price is greater than the insurance element lost. When interest rates increase, the value of the interest

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earned on the strike price increases making early exercise more attractive. When volatility decreases, the insurance element is less valuable. Again, this makes early exercise more attractive.

- ***Bounds***
  - *European put option* when there are no dividends – the upper and lower bound

$$\max(Ke^{-rT} - S_0, 0) \leq p \leq Ke^{-rT}$$

  - *American put option* when there are no dividends

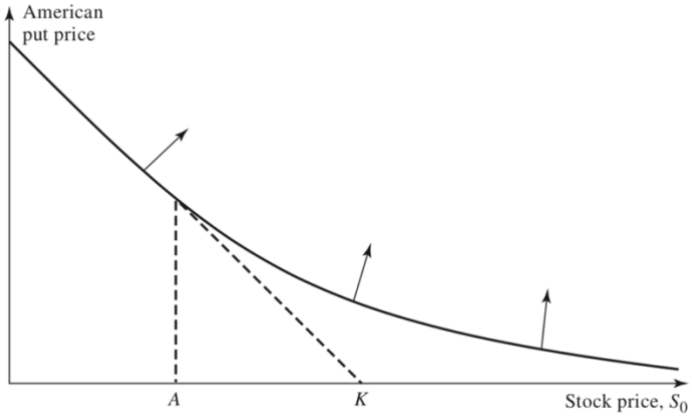
$$P \geq max(K - S_0)$$

⇨ Must apply, because the option can be exercised at any time.

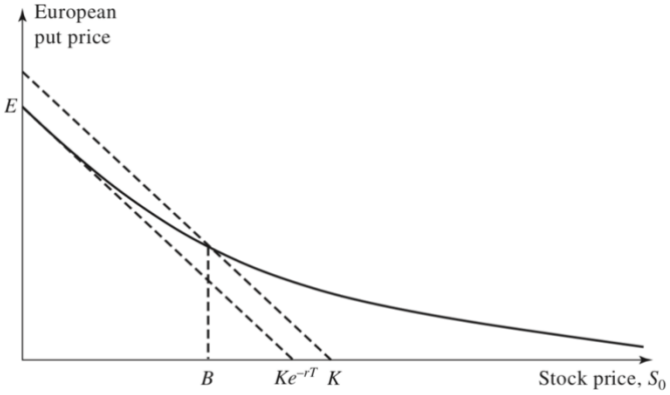
  - Thus, bounds for an American put option on a NDP are:

$$\boldsymbol{max(K - S_0) \leq P \leq K}$$

$$\boldsymbol{P \leq k \quad \& \quad P \geq max\,(K - S0)}$$



- It is always to optimal to exercise an American put early when the stock price is sufficiently low.
- When early exercise is optimal, the value of the option is K - S_0
- The curve represents the value of the put therefore merges into the put's intrinsic value K – S_0 for a sufficiently small value of S_0.
- The line relating the put price to the stock price moves in the direction indicated by the arrows when *r decreases, volatility increases, T increases*.



- The curve represents the relationship between the put price and the stock price for a European option must below the corresponding curve for an American option.
- **Point B**, the price of the option = intrinsic value, must represent a higher value of the stock than point A.

⇨ Because the European curve is below American curve and point E in European is where S0 = 0 and the European put price is Ke^{-rT}

Optimal Exercise: American Puts, DP

- It follows that the same argument can be made for American Puts DP.
  - We do not have any "hard and fast" rules for optimal American puts exercise.

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- If it is dividend paying, by exercising early (sell the option) miss the chance to receive dividend, thus it may be optional to exercise early if S is sufficient low to compensate the loss on dividend.
- But, it is a complicated mathematical problem in terms of quantify the probability of getting more if wait.
  ⇨ There is no rule for American put.

No-Arbitrage Lower Bounds: European Calls & Puts on DP assets

- Denote with D the PV of dividends paid during remaining life of option
- Calls: $c_t \geq Max\{0, S_t - D - Ke^{-rT}\}$
- Puts: $p_t \geq Max\{0, D + Ke^{-rT} - S_t\}$

Optimal Exercise: American Calls, DP

- Assume option is ITM (if not -> no exercise)
- If option is exercised prior to ex-dividend date investor gets $S_t - K$
- If option is not exercised and dividend = D is paid -> stock price drops to $S_t - D$
- Lower bound: $C_t \geq S_t - D - PV(K)$
  - If I sell it, I will get no lower than the lower bound

- If $S_t - K < C_t$ -> should not exercise early
  - Ct is what I get from selling the option is **greater** than exercising the option => should not exercise it.

- So, if $S_t - K \leq S_t - D - PV(K)$: i.e. $D \leq K - PV(K)$ -> should not exercise early
  - If the dividend is not large enough, when it is lower than the interest you gain K – PV(K) from investing the PV of strike => should NOT exercise

Optimal Exercise: American Calls, DP

- Intuitively, If the PV of the dividends to be paid during the remaining life of the call is larger than the PV of the interest that can be earned on the (saved) strike price during the remaining life of the call, then it may be optimal to exercise early (see later in the term for examples)

Put-Call Parity for American Options on NDP Stocks

$$P(S_0, T, K) + S_0 > C(S_0, T, K) + Ke^{-rT}$$

- American option is not an equality

- Proof: since $P(S_0,T,K) > p(S_0,T,K)$ and
  $C(S_0,T,K) = c(S_0,T,K)$ and
  $c(S_0,T,K) + Ke^{-rT} = p(S_0,T,K) + S_0$
  → $p(S_0,T,K) = C(S_0,T,K) + Ke^{-rT} - S_0$

Thus, $P(S_0,T,K) > p(S_0,T,K)$
$= C(S_0,T,K) + Ke^{-rT} - S_0$
So, $P(S_0,T,K) + S_0 > C(S_0,T,K) + Ke^{-rT}$

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Extension of P-C Parity to Include Dividends

- For European options, let D be the present value of future dividends over the life of the contracts. Then, it can be shown that:

$$c\ (S_0, T, K) + Ke^{-rT} + D = p\ (S_0, T, K) + S_0$$

- For American options

$$S - K - D < C - P < S - Ke^{-rT}$$

American calls NDP: never optional – 3 reasons

American put NDP: may be optimal, S is sufficiently low

American puts NP: no rule

American call NP: if $D > K - PV(K)$

Week 10: Binomial Pricing Model

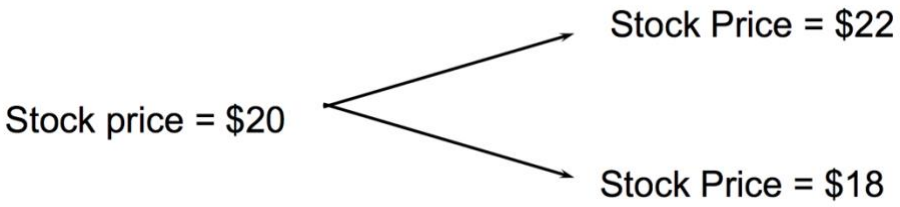
Chapter 12: 12.1-12.8

Binomial Trees: An Initial Disclaimer

- The only assumption needed is that: arbitrage opportunities do not exist.
  - We set a PF of the stock and the option in such a way that there is no uncertainty about the value of the PF at the end of the holding.
  - Because the PF has no risk, the return it earns = risk-free interest rate.
    ⇨ Enable us to work out the cost of setting up the PF and therefore the option's price.

Binomial Model for Stock Price Dynamics

- In a one-period binomial model, the value of an option is the PV of a **probability-weighted average** of two possible values after one period, during which its value is assumed to move either up or down by a chosen size
- The underlying asset price follows a random walk.
- A stock price is currently $20
- In three months, it will be either $22 or $18



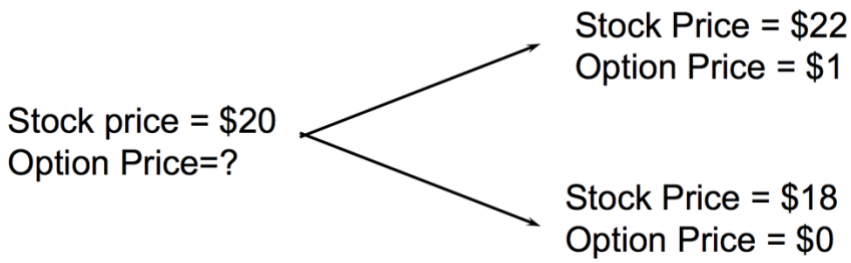
2nd step: Call Option

- A derivative contract depends on the value of the underlying – depends on the stock price will go to $22 or $18, the option will have different values.
- A 3-month call option on the stock has a strike price of 21. Risk-Free Rate is 12%

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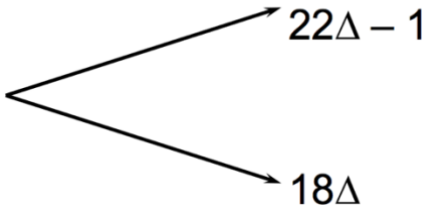
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- The 2nd step in the binomial is: given the possible stock prices, then determine the possible option prices



3rd step: <u>Setting Up a Riskless Portfolio</u>

- Consider the Portfolio: long Δ shares (underlying) & short 1 call option



- We calculate the value of Δ that makes the PF riskless.
- Portfolio is riskless when 22Δ – 1 = 18Δ => Δ = 0.25
  - Long ¼ of shares
  - Short: 1 option
    - ⇨ Whether the underlying goes up or down, will end up with the same amount of money
    - ⇨ Meaning, Δ is the number of shares necessary to hedge a short position in one position.
  - By using Δ in the equation=> portfolio becomes riskless, as eliminate the uncertainty
  - Combine option and underlying, the two are both risky, and make a riskless portfolio => synthetical replication

    - Assume the underlying S either goes up or down
    - Depending on the two values => get knowing the option value
    - Make up a portfolio that has opposite positions (long underlying & short option)
    - Because the PF is riskless => payoff will be certain => price of today is also known with certainty
      - The difference between the price of the overall PF and price of stock => able to get the price of option portion

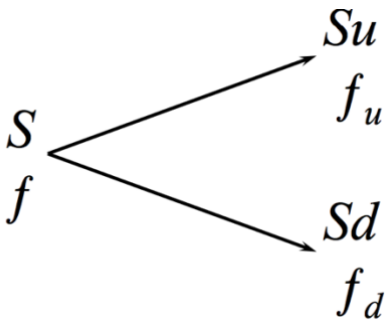
Valuing the Portfolio and the Option

- The riskless portfolio is long 0.25 shares, short 1 call option
- The value of the portfolio in 3 months is: 22 × 0.25 – 1 = 4.50
- The value of the portfolio today is: $4.5e^{-0.12 \times 0.25} = 4.367$ (rf: 0.12, 3 months: 0.25)
- The value of the shares is: 5.000 (= 0.25 × $20)
- By no arbitrage value of the option must be: 5.000 – 4.367 = 0.633 = f
  - -0.633 cents: short the option
  - $5: long the stock
  - Portfolio is worth at $4.367
  - It shows that, in the absence of arbitrage opportunities, the current value of the option must be 0.633.
    - ⇨ If the value of the option > 0.633, the PF would cost less than 4.367 to set up and would earn more than the risk-free rate.
    - ⇨ If the value of the option < 0.633, shorting the PF would provide a way of borrowing money at less than the risk-free rate.

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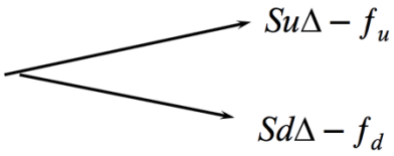
Generalization

- A derivative whose value is denoted by f lasts for time T and is dependent on an underlying asset with spot price S



S: underlying
F: it is generic value -> the market price or value of a derivative asset (it will be either C or P)
U: gross percentage of stock price increase (e.g. in the previous case, price is going up by 10% => u = 1.1.
D: gross percentage of stock price decrease d = 0.9

- Stock gross return per period can only take two values:
  - u (stock price increases) or d (stock price decreases)
  - An up (down) move occurs with probability p (1-p)
    - ⇨ Go up by p
    - ⇨ Go down by 1-p
- Consider the portfolio that is long Δ shares and short 1 derivative
- The portfolio is riskless when $Su\Delta - f_u = Sd\Delta - f_d$ or



$$\Delta = \frac{f_u - f_d}{S_u - S_d}$$

- Value of the portfolio at time T is

$$S_u\Delta - f_u$$

- Value of the portfolio *today* is

$$(S_u\Delta - f_u)e^{-rT}$$

- Cost of Portfolio today is

S Δ (the value of the shares) – f

  - Buying Δ shares of stock, minus f (reduce the cost as shorting the call, this is the money you earn).

- Hence, to rule out arbitrage,

$$f = S\Delta - (Su\Delta - f_u)e^{-rT}$$

- Substituting for Δ we obtain

$$\boldsymbol{f = [p\,f_u + (1-p)f_d]\,e^{-rT}}$$

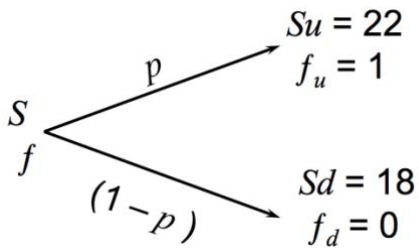
  - It does not involve the probabilities of the stock price moving up or down.
  - And the key reason is that we are not valuing the option in absolute terms. We can calculating its value in terms of the price of the underlying stock.
    - ⇨ The probabilities of future up or down movements are already incorporate into the price of the stock.
    - ⇨ Thus, we do not need to take them into account again when valuing the option in terms of the stock price.

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$$p = \frac{e^{rT} - d}{u - d}$$

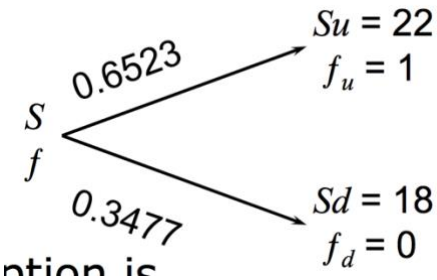
- F is the value of the option => and in bracket, is the expected payoff (p is the probability)

Original Example Revisited



$$p = \frac{e^{rT} - d}{u - d} = \frac{e^{0.12*0.25} - 0.9}{1.1 - 0.9} = 0.6523$$

Original Example Revisited: Valuing the Option



- The value of the option is

$$f = [p\, f_u + (1-p) f_d]e^{-rT}$$

$$= e^{-0.12\times0.25}\,[0.6523\times1 + 0.3477\times0]$$
$$= 0.633$$

- This second procedure for pricing the option is called **risk-neutral valuation**
  - If we discount at the risk-free rate, meaning either asset is riskless or investors do not care about risk.

Risk-Neutral Valuation

- Risk-neutral valuation: when valuing a derivative, we can make assumption that investors are risk-neutral.
  - Investors do not increase the expected return they require from an investment to compensate for increases risk.
- Risk-neutral world has 2 features that simplify the pricing of derivatives
  - The expected return on a stock is risk-free rate
  - The discount rate used for the expected payoff on an option is the risk-free rate.

- The variables p and (1 – p) are the probabilities of up and down movements
- Consider a 1-step binomial tree: the expected stock price is

$$E(S_T) = p\, S_0\, u + (1-p)\, S_0\, d$$

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- Substituting in the above the expression for p, one gets

$$E\,(S_T) = S_0 e^{rT}$$

  - The expected stock price in the future = current price compounded at risk-free rate
  - Shows: stock price grows, on average, at the risk-free rate when p is the probability of an up movement.

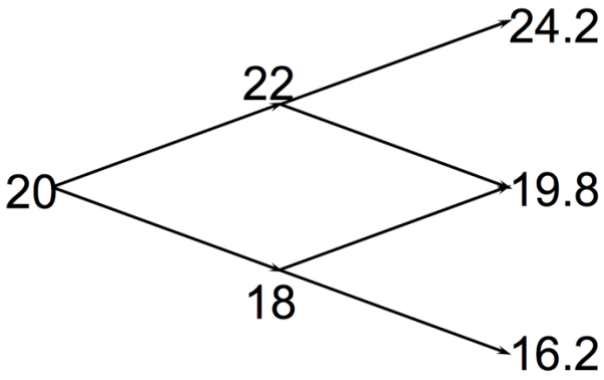
- So, the expected return on a (risky) asset equals the risk-free rate
  - p and 1-p are defined as the risk-neutral probabilities

- Risk neutral means that investors do not seek compensation for risk and are, e.g., satisfied with earning the risk-free rate even on risky assets
- So, risk-neutral probabilities are not the real-world probabilities

Risk-Neutral World and Real World

- The value of a derivative is its expected payoff in a risk-neutral world discounted at the risk-free rate
- But if the valuation formula is based on these non-real world probabilities, how can we possibly hope that it prices correctly derivatives in the real world?
  - Calculation in real-world and imaginary world will be same.
- It is not easy to know the correct discount rate to apply to the expected payoff in the real world.
- The return the market requires on the stock is 16% and this is the discount rate that would be used for the expected cash flows from an investment in the stock.
  - A position in a call option is riskier than a position in the stock.
  - As a result, the discount rate to be applied to the payoff from a call option is greater than 16%, but we do not know how much greater than 16% it should be.
- Using risk-neutral valuation is convenient because we know that in a risk-neutral world the expected return on all assets (and therefore the discount rate to use for all expected payoffs) is the risk-free rate.

A Two-Step Example



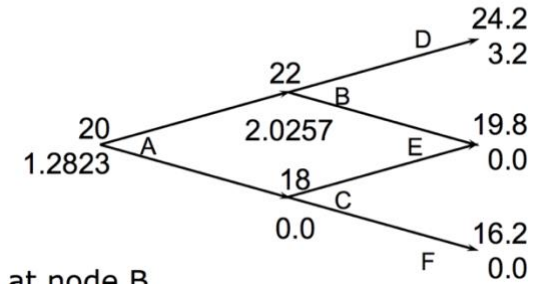
From $22 & $18 go up and down by 10%, at each node, the price can change to another 2 values and by keep doing that, can get many possible values.

- Each time step is 3 months and the maturity is 6 month => 2 step
- K=21, r=12%

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Valuing a Call Option



at node B

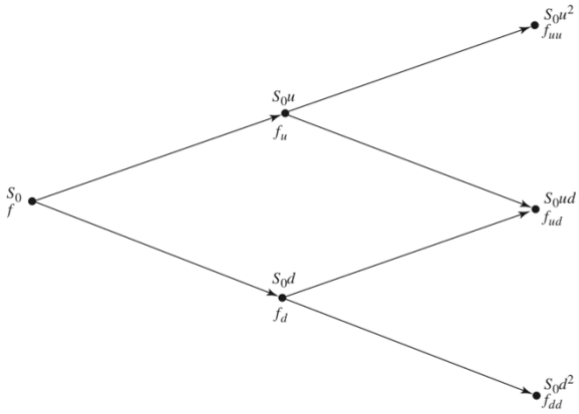
Backward induction

- The strike is $21, and the underlying is $24.2, and the value (payoff fu) is $3.2.

$$f = [p\, f_u + (1-p) f_d]\, e^{-rT}$$

- Value at node B
  $= e^{-0.12\times0.25}(0.6523\times3.2 + 0.3477\times0) = 2.0257$
- Value at node A
  $= e^{-0.12\times0.25}(0.6523\times2.0257 + 0.3477\times0) = 1.2823$

$$f = e^{-2r\Delta t}[p^2 f_{uu} + 2p(1-p) f_{ud} + (1-p)^2 f_{dd}]$$



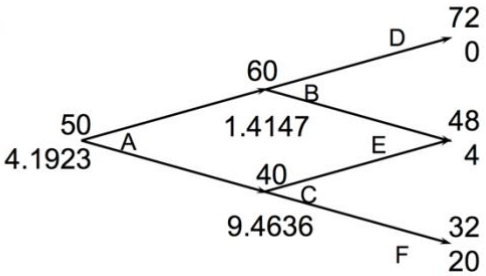
- The option price is always equal to its expected payoff in a risk-neutral world, discounted at the risk-free interest rate.

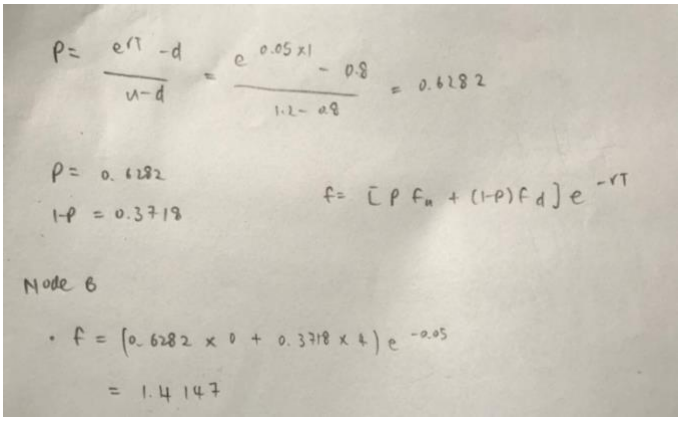
- U = 1.2; d = 0.8 (20%)

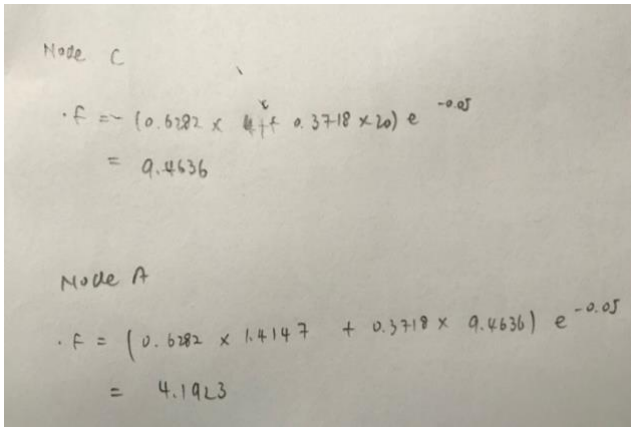
A Put Option Example

$K = 52,\ \Delta t = 1\text{yr}$

$r = 5\%$







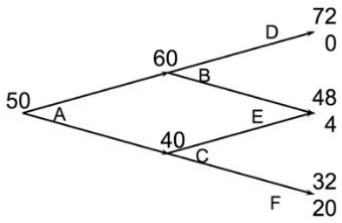
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American Options

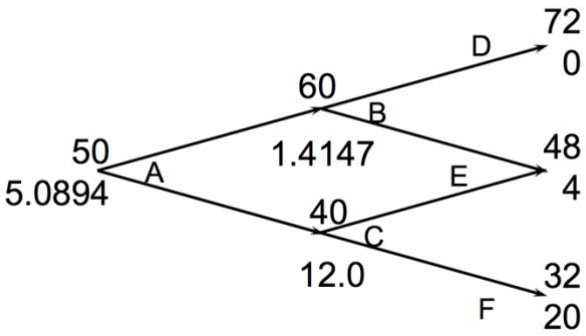
- American options can also be handled in the binomial tree model
- To value an American option, work backwards through the tree as before, BUT test each node to see whether early exercise is optimal.
- The value of the option at the final nodes is the same as a European.
- At earlier nodes the value of an option is the greater of:
  1. its value if it is kept alive (not exercised),
  2. or its payoff from early exercise.
- Always remember what we said about American calls NDP => treat as European

American Put: Example

- S0 = 50, K = 52, T=2 years, Δt = 1yr, r = 5%, u=1.2, d=0.8
  - p = (exp (0.05)-0.8) / (1.2 – 0.8) = 0.6282
  - Δt is the length of each step on the tree
- Find American Put prices with a twostep binomial tree



- Value of (early) exercise at B: K -S => 52-60 = -8 -> do not exercise as OTM
  - Put value at B: (0 x 6282 + 4 x 3728) x exp(-0.05) = 1.4147
- Value of (early) exercise at C: 52 - 40 = 12 -> exercise,
  - As it is > 9.4636 (see 自己的计算), which is the value if not exercised
  - Put value at C is 12
- Value of (early) exercise at A: 52-50 = 2
  - Put Value at A: exp (- 0.05) x (0.6282 x 1.4147+0.3728 x 12) = 5.0894
  - Do not exercise, as 5.0894 >2



Choosing u and d

- The choice of u and d will be made so that the model generates a price behaviour for the underlying that is somehow relate to reality.
- A reasonable objective would be to choose u and d so that the resulting dynamics of S (in the model, i.e., on the tree) matches important features of the dynamics of prices and returns in the real world, such as the expected return and the expected volatility of returns over the life of the option we are pricing
- But to avoid the estimation of expected returns, we can use risk-neutral valuation set the risk-free rate as the expected return on the underlying
  - ⇨ No need to concern expected return as it is risk-natural -> it is the risk-free rate
- We still want to match the volatility of the underlying asset returns

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A Note on T and Δ t

- T is the maturity of the option, expressed as fraction of the year. E.g. 3 months -> T=3/12
- Δt is the time step chosen on the binomial tree, also expressed as a fraction of the year. E.g. if T= 3/12 and one chooses 10 time steps to implement the Binomial tree -> Δt = 3/120
- When using trees with more than **1 step, Δt is needed** to discount back from each node to the preceding one and to compute p

$$p = \frac{e^{r\Delta T} - d}{u - d}$$

Choosing u and d

- One way of obtaining the match is to set

$$u = e^{\sigma\sqrt{\Delta t}}$$

$$d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}}$$

$$p = \frac{a - d}{u - d} \quad a = e^{r\Delta T}$$

  - ⇨ where σ is the annualized return volatility and Δt is the length of the time step (expressed as fraction of one year).
- Later in the term we'll see how to estimate σ

Extending the Tree

- As the number of time steps is increased Δt becomes smaller, the binomial model makes the same assumptions about stock price behaviour as the BSM.

- The extension to more than two steps is purely mechanical but necessary to achieve a realistic underlying asset dynamics.

  - ⇨ The more steps we do, the more the possible values for the underlying asset were allowed, which makes it more realistic.
  - ⇨ To make it more realistic, more than 2 steps is necessary
- Fortunately, plenty of software (such as Derivagem) can do it for us
- Typically, between 30 and a few hundred steps are used, depending in the complexity of the derivative (accuracy vs. speed)

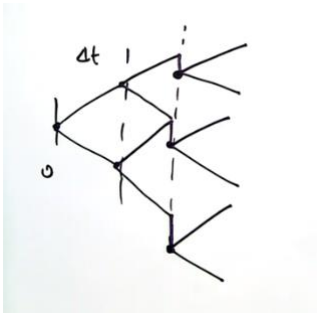
Summary & Generalizations

- The binomial tree method is flexible enough to be used for European as well as American options
- B.T. method can be extended to include dividends ($ dividend or yields) paid by the underlying asset (see next)
- B.T method can be extended to index options, currency options, commodity options and futures options, European or American. Procedure is the same, although the formula for p changes (see next)

As long as you know the price of the underlying asset, then BT can be done. Many kinds of options can be priced with binomial.

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Trees and Dividends



- The main point to remember is that the underlying asset price drops by an amount equal to the dividend payment on the ex-dividend date
    - Need to include those price drops in the binomial tree
    - But might create more nodes than without dividends as the tree might not re-combine.
        - ➔ It is important to eliminate the amount of calculations (nodes) as for practical purposes -> to many nodes, pricing take too long.
    - In practice, a few math "tricks" are used so that the evolution of the underlying asset price reflects the drops and, at the same time the tree recombines (otherwise, "too many" nodes, see fig. 18.6)
        - Want to avoid too many nodes problem

- Tricks:
    - ⇨ Draw the tree for the asset price less the present value of the dividends, as if it was an NDP asset
    - ⇨ Create a new tree by adding the present value of the dividends at each node
- This ensures that the tree recombines

Trees and Dividend *Yields*

- Dividends paid at the *rate q* cause underlying asset price to *grow less* (than if dividends were not paid)
    - ⇨ The payment of a dividend makes the price drop -> the yield makes the price grow less
    - ⇨ Need to include this lower growth of S in the binomial tree
- A convenient way of doing so is lowering the probability of an upward move, p.

    - For non-dividend paying assets

$$p = \frac{e^{r\Delta t} - d}{u - d}$$

    - because expected return on S (in risk neutral world) is r

    - For assets paying a yield, q, expected return on S (in risk neutral world) is r-q
    - For dividend paying assets

$$p = \frac{e^{(r-q)\Delta t} - d}{u - d}$$

- and everything else is the same as in the case of non-dividend paying assets => ***adjust p***
    - ⇨ To account for dividend yield, all one has to do is computing p as in (2) above, and proceed "as usual" with the tree
- But, then, I can use the Binomial Model (with the modified p above) to price options on stock *indices, currencies, investment commodities*
    - ⇨ Remember what we said/did with futures: r_f, u, …

Risk-Neutral World and Real World

- Consider a 1-step binomial tree
- In the risk-neutral world the expected stock price is

$$E(S_T) = p\,S_0 u + (1 - p)\,S_0 d$$

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Substitute

$$p = \frac{e^{rt} - d}{u - d}$$

⇨ $E\ (S_T) = S_0 \exp(rT)$

- So, in the risk-neutral world the expected return on a (risky) asset equals the risk-free rate
- If the model is based on these non-real-world probabilities and expected return, how can we possibly hope that it prices correctly derivatives in the real world?

- Risk-neutral valuation: when valuing a derivative, we can make assumption that investors are risk-neutral.
  - Investors do not increase the expected return they require from an investment to compensate for increases risk.
- Risk-neutral world has 2 features that simplify the pricing of derivatives
  - The expected return on a stock is risk-free rate
  - The discount rate used for the expected payoff on an option is the risk-free rate.

- The variables p and (1 – p) are the probabilities of up and down movements
- Consider a 1-step binomial tree: the expected stock price is

$$E\ (S_T) = \ p\ S_0\ u + (1 - p)\ S_0\ d$$

- Substituting in the above the expression for p, one gets

$$E\ (S_T) = \ S_0 e^{rT}$$

  - The expected stock price in the future = current price compounded at risk-free rate
  - Shows: stock price grows, on average, at the risk-free rate when p is the probability of an up movement.

- So, the expected return on a (risky) asset equals the risk-free rate
  - p and 1-p are defined as the risk-neutral probabilities

Risk-Neutral World and Real World

- First, note that p is not directly necessary to work on the binomial tree
- In fact, we first saw (see last lecture) that the value of a derivative can be computed as

$$f = S\Delta - (Su\Delta - \ f_u)e^{-rT}$$

- Assuming, because the portfolio is risk-free, it has to earn the risk-free, otherwise arbitrage opportunity as people take advantage of arbitrage opportunities, regardless of their preferences
  - Here, we are not discounting something risky, we discount risk-less assets
  - It is based on no-arbitrage principal

- Then, we defined

$$p = \frac{e^{rt} - d}{u - d}$$

⇨ so that we could write the above as

$$f = [p\ f_u + (1 - p)f_d]e^{-rT}$$

⇨ And, only then, we interpreted p as the risk-neutral probability. But in the pricing by arbitrage there is no consideration for risk-neutrality (or for any other attitude towards risk for that matter)

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➔ The expected value of the option discounting back at the risk-free rate
➔ Here, we discount risky assets at risk-free rate because we are discounting the expected option payoff (which is risky)

Overall, the model can be interpreted in two ways

$$f = S\Delta - (Su\Delta - f_u)e^{-rT}$$ – price by no-arbitrage

$$f = [p\, f_u + (1-p) - f_d]e^{-rT}$$ – risky-natural pricing

- So, we know that the pricing by arbitrage "makes sense"
  - ⇨ because it does not assume people are risk-natural, it only assumes people take the arbitrage opportunity, which is reasonable.
- If one can show that the risk-neutral pricing leads to the same price as the arbitrage pricing, then risk-neutral pricing must also "make sense".
- This is pretty easy to show for any of the examples we saw – redo the tree by no-arbitrage and by risk-neutral valuation

- Although complex math can show why, we can get some intuition.
- The option is priced relatively to the underlying
  - ⇨ It prices the whole profile of options and underlying together
- But, through delta-hedging, we know that the option and the underlying can be packaged to yield a riskless portfolio
  - ⇨ Risk-preferences don't matter to price such portfolio
  - ⇨ Because we are pricing a portfolio that is risk-free, thus risk tolerance does not matter.
    - Even more heuristically, we price by

$$\frac{p\, f_u + (1-p) f_d}{e^{rT}}$$

- Both numerator and denominator are "not realistic": it so happens that the two cancel out to yield a "realistic" and convenient pricing framework
  - Numerator: the probability is not the real-world probability
  - Denominator: expected return is the risk-free rate
  - ⇨ Together, make sense in real-world.

$$\boldsymbol{f = [p\, f_u + (1-p) f_d]\, e^{-rT}}$$

- *It does not involve the probabilities of the stock price moving up or down.*
- *And the key reason is that we are not valuing the option in absolute terms. We can calculating its value in terms of the price of the underlying stock.*
  - ⇨ *The probabilities of future up or down movements are already incorporate into the price of the stock.*
  - ⇨ *Thus, we do not need to take them into account again when valuing the option in terms of the stock price.*

$$p = \frac{e^{rT} - d}{u - d}$$

- *F is the value of the option => and in bracket, is the expected payoff (p is the probability)*

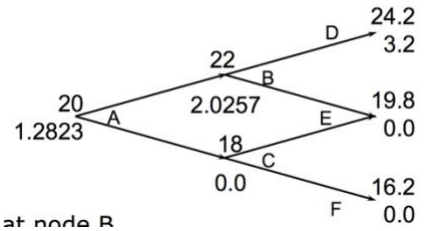
Delta

- The delta Δ of a stock option is the ratio of the *change in the price of the stock option* to the *change in the price of the underlying asset*.

$$\frac{1-0}{22-18} = 0.25$$

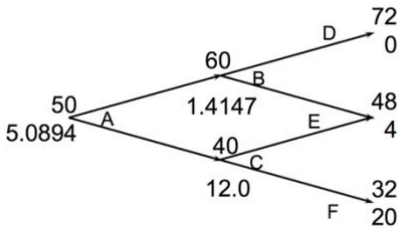
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- It is the number of units of the stock we should hold for each option shorted in order to create a riskless portfolio.
    - To construct a riskless portfolio is sometimes referred to as delta hedging.
    - Delta of a call option is positive; delta for a put option is negative.



at node B

- Delta to stock price movement over the first-time step is: $\frac{2.0257-0}{22-18} = 0.5064$
- Delta to stock price movement over the second-time step is: $\frac{3.2-0}{24.1-19.8} = 0.7273$ for an upward movement



From Figure 12.7, delta is

$$\frac{1.4147 - 9.4636}{60 - 40} = -0.4024$$

at the end of the first time step and either

$$\frac{0 - 4}{72 - 48} = -0.1667 \quad \text{or} \quad \frac{4 - 20}{48 - 32} = -1.0000$$

at the end of the second time step.

Black-Scholes-Merton (BSM) Model

Chapter 13: 13.1 – 13.10 Chapter 15: 15.1, 15.3-6

BSM Model: A BIG DEAL

- Black and Scholes (1973) and Merton (1973) won the Nobel Prize in Economics for their work on option pricing
- To this day, the Black-Scholes model/formula is the most common formula used to price options.
- In fact, this is probably the formula from economics/finance academia that has ever found a universally practical application.

Option Valuation: a Basic Intuition

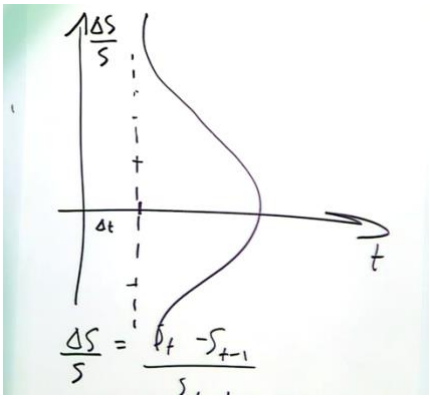
- Ex: European Call Option on NAB stock, K = 40, 6-month maturity; NAB stock currently trading at 30
- How much are you willing to pay for this option?
- If you don't think that NAB stock will climb above 40, you will not want to pay much (or anything at all) for the call
- More generally, the value of the option depends on the probability distribution of the underlying asset price (over the life of the option and/or at maturity)
    - The more you think there is a chance to go up at least to 40, the more you willing to pay
    - The assessment of the value of the option, is related to the probability that option ends up ITM.
        - ⇨ Depend on that, give more or less value to the option.
        - ⇨ Every pricing model will assume something about the probability distribution

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Assumptions of BSM

- The market is frictionless: no transaction costs, no taxes
- No dividends are paid prior to expiration
- The risk-free interest rate is constant
- No riskless arbitrage opportunities exist/persist
- One can borrow/lend at the risk free rate
- Stock prices (and, thus, returns) follow a very specific random process, the random walk
  - The stock price *variance is constant*
  - Stock prices move continuously
    - ⇨ No big jumps in prices (to simply reality)
    - ⇨ Price has a continuous path, if s1, price = $2; the next second, price cannot be $2.5, it must be $2.01 or $1.99
- The stock price behaviour corresponds to the lognormal model with μ and σ constant.

The BSM Random Walk Assumption



- Consider a stock whose price is S
- The BSM considers a non-dividend-paying stock and assumes that the return on the stock in a short period of time is normally distributed.

$$\frac{\Delta S}{S} \sim \phi\ (\mu, \Delta t, \sigma^2, \Delta t)$$

- In a short period of time of length Δt the **discretely compounded** return on the stock $\frac{\Delta S}{S}$ is assumed to be normal with mean $\mu\Delta$ t and standard deviation $\sigma\sqrt{\Delta t}$
  - $\frac{\Delta S}{S}$ is the change in the stock price in time Δ t
  - When Δ t is small, the above equation shows that the percentage change in the stock price in time Δ t is normal with SD $\sigma\sqrt{\Delta t}$

⇨ Suppose that, σ = 0.3, or 30% per annum, and the current stock price is $50. The standard deviation of the percentage change in the stock price in one week (= 1/52 years) is, to a good approximation,

$$0.3 \text{ x } \sqrt{\frac{1}{52}} = 4.16\%$$

⇨ One SD move in the stock price in one week is therefore $50 x 0.0416 = $2.08

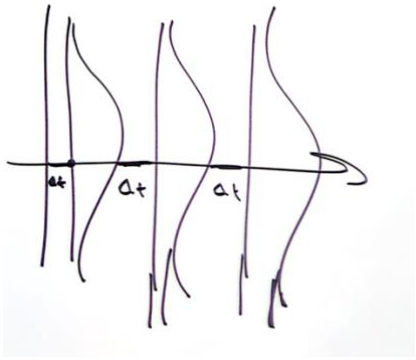
$$\textbf{discretely compounded return: } \frac{\Delta S}{S} = \frac{S_t - S_{t-1}}{S_{t-1}}$$

- It is a normal distribution and depend on mean and SD.

- S0, $\mu\Delta t$ is expected return and $\sigma\sqrt{\Delta t}$ is volatility of returns (where returns are discretely compounded)
- Typically, μ and σ are annual (or, annualized) numbers
- Such distribution is the same for each time interval Δ t

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- In addition, what happens over one time interval does not affect what happens over future time intervals



- **Stock Returns are independent and identically distributed (i.i.d) over time**
  - ⇨ No memories in return => so cannot use past returns to predict future returns
- *In every time interval, the return comes from the same probability distribution.*
  - ⇨ *The model assumes: the return comes from a normal distribution and is the same distribution.*
    - ✧ *What happens in one-time interval, does not affect what happens later => independent random variables*
  - ⇨ $\mu\Delta t$ is expected return and $\sigma\sqrt{\Delta t}$ as SD.

The Lognormal Property

- The stock price at any future time has a lognormal distribution.
- These assumptions imply that ln(ST) (where T is some future time such as the maturity of an option) is normally distributed with mean:

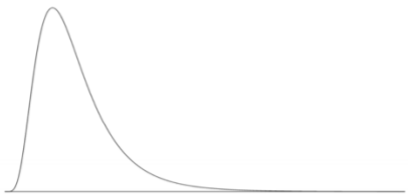
$$In\ S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T$$

- Standard deviation:

$$\sigma\sqrt{T}$$

- Because the ln (ST) is normal, ST is lognormally distributed (it's a theorem from statistics)
- Notice that volatility scales with the square root of time and it is NOT linear in time

The Lognormal Distribution – distribution of prices



Continuous compounded returns:

$$In\ (S_t) - In\ (S_o) = In\ (\frac{S_t}{S_0})$$

$$E(S_T) = S_0\, e^{\mu T}$$

$$\text{var}(S_T) = S_0^{\,2} e^{2\mu T}(e^{\sigma^2 T} - 1)$$

The expected value of St is E(St) and the variance is above.

The Lognormal Property

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$$\ln S_T \sim N\left[\ln S_0 + (\propto - s^2/2)T, s^2 T\right]$$

- - >

$$\ln \frac{S_T}{S_0} \sim N\left[(\propto - s^2/2)T, s^2 T\right]$$

- where N [*m,v*] is a normal distribution with mean *m* and variance *v* → *continuously compounded returns follow a normal distribution*
- Lognormal cannot be negative, also it is right skewed

$\frac{\Delta S}{S_T}$: it is rate of return of stock price, the change in stock price in time ΔT normally distributed with discrete compounding

S_t: lognormal distributed – no compounding about stock
$In\left(\frac{\Delta S}{S}\right)$: normally distributed, with continuous compounding
$In\ (S_t)$: normally distributed, with No compounding about price

Key Concepts Underlying BSM: Replication and Riskless Hedge

- The option price and the stock price depend on the same underlying source of uncertainty
- We can form a portfolio consisting of the stock and the option which eliminates this source of uncertainty – the underlying asset price
  - The reason a riskless portfolio can be set up is that the stock price and the option price are both affected by the same underlying source of uncertainty: stock price movements.
- In any short period of time, the price of a call option is perfectly positively correlated with the price of the underlying stock; the price of a put option is perfectly negatively correlated with the price of the underlying stock
- In both cases, when an appropriate portfolio of the stock and the option is set up, the gain or loss from the stock position always offsets the gain or loss from the option position so that the overall value of the portfolio at the end of the short period of time is known with certainty.

- The portfolio is instantaneously riskless and must instantaneously earn the risk-free rate
- Yes, same basic concepts as in the Binomial Model

BSM: "Derivation" using the riskless hedge – *the no-arbitrage argument*

- Consider a portfolio consisting of a long position in one European call combined with a short position of Δ shares of the underlying stock.
- Today this portfolio is worth:

$$c - \Delta S$$

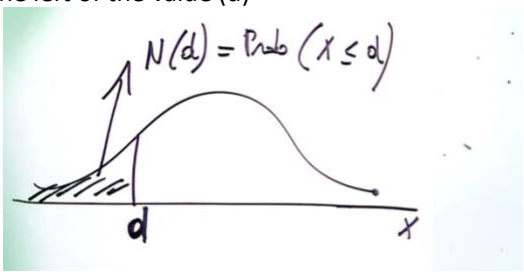
- We know in the binomial model that a careful choice of Δ makes this combination *riskless*— it becomes a synthetic riskless bond.
  - The stock does not only take 2 value, it takes infinite amount as it from a normal distribution
- By using a decades-old mathematical result from the Japanese researcher Ito, Black and Scholes could derive the correct value of Δ in their framework (where the price follows the random walk)

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- One complication is that Δ needs to be adjusted (i.e., the portfolio needs to be rebalanced) continuously to remain riskless
- The math of doing this is beyond our scope. Suffice it to say that it leads you to a *partial differential equation* (PDE)
- Solving such PDE is "easy" for mathematicians and leads directly to the Black-Scholes-Merton formula
- Once you have Δ, then you can find c as you know S.

Variable Definitions

- c = Current European call option value
- p = Current European put option value
- S 0 = Current stock price
- N(d) = Probability of a random draw from a standard normal distribution will be less than d (area under normal curve up to d, which is the cumulative probability function for a standardized normal variable)
  - The area to the left of the value (d)



- K = Exercise price
- r = Risk-free rate, annualized continuously compounded
- T = Time remaining to maturity, in years
- ln = Natural logarithm function
- σ = Volatility (standard deviation) annualized of continuously compounded rate of return of the stock during the life of the option

The Black-Scholes-Merton Formulas: European Calls and Puts, NDP

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

$$p = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$$

$$d_1 = \frac{In\left(\frac{S_0}{K}\right) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{In\left(\frac{S_0}{K}\right) + (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

Risk-Neutral Valuation

- The variable μ, expected stock return, does not appear in the BSM equations
- The equation is independent of investors' risk preference
- Risk-neutral valuation does not state that investors are risk neutral.

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- What it does state is that derivatives such as options can be valued on the assumption that investors are risk neutral.
- It means that investors' risk preferences have no effect on the value of a stock option when it is expressed as a function of the price of the underlying stock.

  - BSM relies on the risk-neutral valuation principle, like the Binomial Tree model
    - The expected return from all investment assets is the rf interest rate.
    - The risk-free interest rate is the appropriate discount rate to apply to any expected future cash flow.

  - *Risk-neutral valuation: when valuing a derivative, we can make assumption that investors are risk-neutral.*
    - *Investors do not increase the expected return they require from an investment to compensate for increases risk.*
  - *Risk-neutral world has 2 features that simplify the pricing of derivatives*
    - *The expected return on a stock is risk-free rate*
    - *The discount rate used for the expected payoff on an option is the risk-free rate.*

  - *The variables p and (1 – p) are the probabilities of up and down movements*
  - *Consider a 1-step binomial tree: the expected stock price is*

$$E(S_T) = p\,S_0\,u + (1-p)\,S_0\,d$$

  - Substituting in the above the expression for p, one gets

$$E(S_T) = S_0 e^{rT}$$

    - The expected stock price in the future = current price compounded at risk-free rate
    - Shows: stock price grows, on average, at the risk-free rate when p is the probability of an up movement.

  - So, the expected return on a (risky) asset equals the risk-free rate
    - p and 1-p are defined as the risk-neutral probabilities

Risk-Neutral World and Real World

- First, note that p is not directly necessary to work on the binomial tree
- In fact, we first saw (see last lecture) that the value of a derivative can be computed as

$$f = S\Delta - (Su\Delta - f_u)e^{-rT}$$

- Assuming, because the portfolio is risk-free, it has to earn the risk-free, otherwise arbitrage opportunity as people take advantage of arbitrage opportunities, regardless of their preferences
  - Here, we are not discounting something risky, we discount risk-less assets
  - It is based on no-arbitrage principal

- Then, we defined

$$p = \frac{e^{rt} - d}{u - d}$$

⇨ so that we could write the above as

$$f = [p\,f_u + (1-p)f_d]e^{-rT}$$

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⇨ And, only then, we interpreted p as the risk-neutral probability. But in the pricing by arbitrage there is no consideration for risk-neutrality (or for any other attitude towards risk for that matter)

➔ The expected value of the option discounting back at the risk-free rate

➔ Here, we discount risky assets at risk-free rate because we are discounting the expected option payoff (which is risky)

Overall, the model can be interpreted in two ways

$$f = S\Delta - (Su\Delta - f_u)e^{-rT}$$ – price by no-arbitrage

$$f = [p\, f_u + (1-p) - f_d]e^{-rT}$$ – risky-natural pricing

- So, we know that the pricing by arbitrage "makes sense"

⇨ because it does not assume people are risk-natural, it only assumes people take the arbitrage opportunity, which is reasonable.

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- This is pretty easy to show for any of the examples we saw – redo the tree by no-arbitrage and by risk-neutral valuation

- Although complex math can show why, we can get some intuition.
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⇨ It prices the whole profile of options and underlying together

- But, through delta-hedging, we know that the option and the underlying can be packaged to yield a riskless portfolio

⇨ Risk-preferences don't matter to price such portfolio

⇨ Because we are pricing a portfolio that is risk-free, thus risk tolerance does not matter.

- Even more heuristically, we price by

$$\frac{p\, f_u + (1-p) f_d}{e^{rT}}$$

- Both numerator and denominator are "not realistic": it so happens that the two cancel out to yield a "realistic" and convenient pricing framework
  - Numerator: the probability is not the real-world probability
  - Denominator: expected return is the risk-free rate

⇨ Together, make sense in real-world.

$$\boldsymbol{f = [p\, f_u + (1-p) f_d]\, e^{-rT}}$$

- *It does not involve the probabilities of the stock price moving up or down.*
- *And the key reason is that we are not valuing the option in absolute terms. We can calculating its value in terms of the price of the underlying stock.*

⇨ *The probabilities of future up or down movements are already incorporate into the price of the stock.*

⇨ *Thus, we do not need to take them into account again when valuing the option in terms of the stock price.*

$$p = \frac{e^{rT} - d}{u - d}$$

- *F is the value of the option => and in bracket, is the expected payoff (p is the probability)*

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- *$\mu\Delta t$ is close to the arithmetic mean of the $\Delta Si / Si$.*
  - *However, the expected return over the whole period covered by the data, expressed with a compounding period of Δt, is close to the geometric mean of the $\Delta Si / Si$.*
  - *This is $\mu - \frac{\sigma^2}{2}$ not μ.*

Implementing BSM

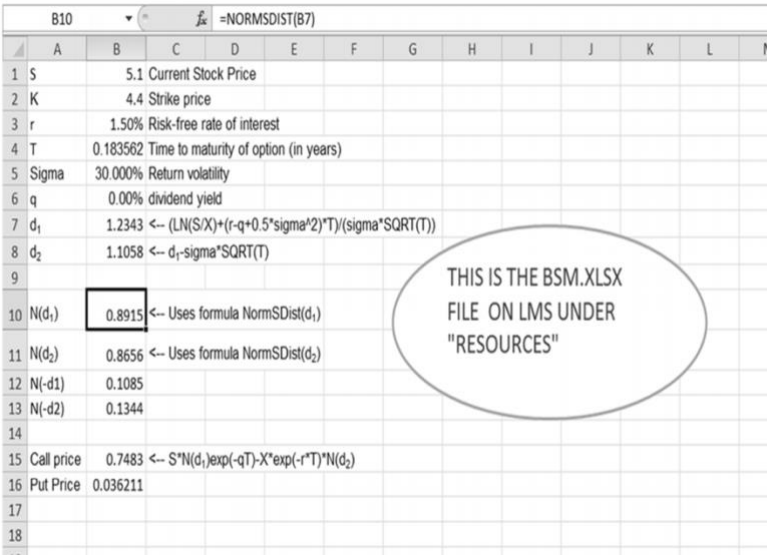
- Assume Telstra (TLS) stock is currently trading at *S = 5.1*.
- The *expected (annualized) volatility* of *continuously compounded* TLS returns is *σ= 0.30*.
- A TLS call option has an *exercise price K = 4.4* and expires on *November 24 (T=67/365)*.
- TLS pays no dividends until March 2017. The current risk-free rate in Australia is 1.5% p.a.
- What should the current price of the 4.4 TLS calls be according to the BSM model?

- First, find d 1 and d 2:

$$d_1 = \frac{In\left(\frac{S_0}{K}\right) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} = 1.2343$$

$$d_2 = d_1 - \sigma\sqrt{T} = 1.1058$$

- Using tables (see textbook p.590-91) for normal distribution probabilities or Excel (function is NORMSDIST, see BSM.xlsx):
  - N(1.2343) = 0.8915
  - N(1.1058) = 0.8656



- Using the BSM formula: $c = S_0\, N(d_1) - Ke^{-rT}\, N(d_2)$
- c = 5.1*0.8915 – 4.4*exp(- 0.015*(67/365))*0.8656
- c= $0.7483

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Volatility

- A record of stock price movements can be used to estimate volatility.
- The volatility σ is the annualized standard deviation of the continuously compounded (c.c) rate of return on the underlying asset over the life of the option
- The standard deviation (or, volatility) of the c.c. return over a time interval Δt is $\sigma \sqrt{\Delta t}$
- If a stock price is $50 and its volatility is 25% per year what is the volatility of the price change in one day?
  - It is 0.25*sqrt(1/252) = 1.58%
  - Note: 252 is used conventionally as the number of trading days in a calendar year

Price does not move when the market is closed => number of trading days is used

Estimating Volatility (σ)

- Given the key role of the underlying asset volatility in determining the option value, volatility forecasting is a HUGE deal in the options world.
- The volatility of a stock σ, is a measure of our uncertainty about the returns provided by the stock.
- A variety of methods have been proposed in the industry. We'll look at the two basic ones:
  - Historical Volatility: simple to implement
  - Implied Volatility: most common used one
- Many others have been proposed
- Empirically, the issue of what model provides the best volatility forecast is still open

Historical Volatility – backward looking

1. Take observations S0, S1, . . . , Sn at intervals of τ years
   - Note: τ =1/12 for monthly data, 1/52 for weekly, 1/252 for daily (trading vs. calendar days)
   - τ is the length of time interval in years.
   - Between now and the expiration of the future, will be similar as the recent past
     ⇨ Using what happened in recent history to predict what will happen in near future

2. Define the continuously compounded return as:

$$u_i = In\left(\frac{S_i}{S_{i-1}}\right)$$

   ⇨ Because sigma is defined as the SD of continuously compounded return

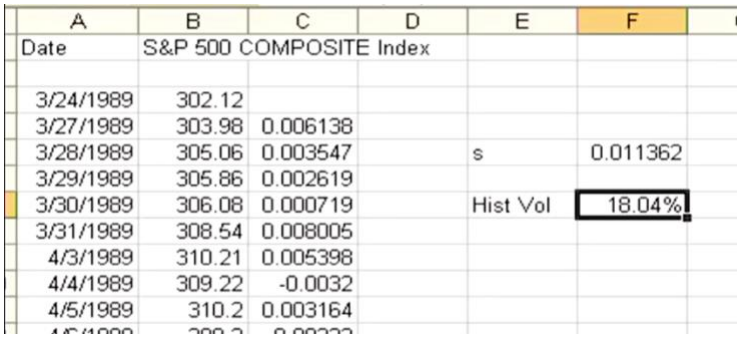
3. Calculate the standard deviation, $\underline{s}$, of the $\underline{u_i}$´$\underline{s}$
4. The historical annualized volatility estimate is, then

$$\hat{\sigma} = \frac{s}{\sqrt{\tau}}$$

   ⇨ The sigma I am looking for is the annualized SD => so need to annualize the SD

   ⇨ E.g. if I am doing daily returns, compute $\underline{s}$ as a daily SD and then annualize it.

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	A	B	C	D	E	F
	Date	S&P 500 COMPOSITE Index				
	3/24/1989	302.12				
	3/27/1989	303.98	0.006138			
	3/28/1989	305.06	0.003547		s	0.011362
	3/29/1989	305.86	0.002619			
	3/30/1989	306.08	0.000719		Hist Vol	18.04%
	3/31/1989	308.54	0.008005			
	4/3/1989	310.21	0.005398			
	4/4/1989	309.22	-0.0032			
	4/5/1989	310.2	0.003164			

- 's' here is the daily SD, and daily return on average in very small => you expect on average per day, with 60% / 70% probability, either 1.1% move up or down
- Historical vol is the annualized SD.

Historical Vol

- Advantages: simple and intuitive
- Drawbacks: it does not reveal the volatility over the remaining life of the option.
  1. Needs to choose data window and frequency
  2. gives the same weight to all past observations
  3. Assumes that future closely reflects past
- In practice, 2-3 years of daily data are used for standard traded equity and FX options
- Also, research has found that higher frequency data (daily or intra-daily) yield more accurate volatility assessment than lower frequency data (weekly, monthly…)
  - ⇨ The higher the frequency, the better

Implied Volatility – forward looking

- The sigma: is the volatility expected over the life of the option
- Prices aggerate information and opinion so the volatility is implied in the price that people are trading with.
- Given an option pricing model and observed option market prices, what is the volatility value that *makes the model price = market price*?
  - ⇨ Implied volatility: it is the volatility implied by an option price observed in the market.
  - ⇨ It is the expected annualized standard deviation of underlying asset continuously compounded returns over the remaining life of the option.
  - ⇨ For a given market price, there is only one value of sigma that matches this market price
- The is a one-to-one correspondence between prices and implied volatilities
  - ⇨ It should be a good reflection of what people expect
  - ⇨ C is an increasing function of σ

- Traders often quote the implied volatility of an option rather than its price.
  - ⇨ This is convenient because the implied volatility tends to be less variable than the option price.
  - ⇨ The implied volatility of an option does depend on its strike price and time to maturity.

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- Implied vols. are inherently forward looking: should provide market's best estimate of future volatility
- Should be able to anticipate abrupt changes in vols. better than backward looking methods (such as Historical vol)
    - Especially useful in times of stress
    - Especially when the market condition changes dramatically, implied vol should be a much better predictor than historical as the historical assumes there is no change.

- Drawbacks:
    1. If option is not traded.. as need an observed market price
    2. Assumes that adopted pricing model is the one used by (majority of) market participants

- Calculations: it's a minimization problem (find parameter that minimizes deviation between, say, B-S price and market price)

The VIX "Index

- Weighted Average of (annualized) implied vols from near-term S&P500 index options (SPX)

Weighted average of volatility, of options with fairly short maturities – options maturities up to 30 days

The very short-term options have less weight and the most heavily traded have more weight.

- Implied vols used to be from BSM model, now from more complex approach
- Considered as a measure of market expectations of near-term equity market (i.e., S&P500) volatility
    - ⇨ The future value of options on the S&P 500 depend on both the level of the S&P 500 and its volatility.
    - ⇨ By contrast, a future contract on the VIX is a bet only on volatility.
- CBOE promotes it as 'considered by many to be the world's premier barometer of investor sentiment and market volatility'
- Often referred to as the fear index
    - ⇨ Volatility going up means uncertainty going up => uncertainty going up

- e.g. VIX: 11.93: the close of that trading day, the average implied volatility for short-term S&P 500 options was 11.93%
- When S&P goes up, VIX goes down => fear kicks in

The VIX

- VIX and S&P500 tend to move in opposite directions
    - negative correlation = -0.72 on daily data 1990-2016
- But the correlation changes a lot over time

By supplying a script for replicating volatility exposure with a portfolio of SPX options, this new methodology transformed VIX from an abstract concept into a practical standard for trading and hedging volatility.

- Volatility can be shorted or longed

Volatility and VIX

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- The original VIX measures aggregate US equity volatility
- CBOE (and others) by now provides volatility indicators for a **variety** of assets and asset classes
    - ⇨ Not only aggregate US equity volatility
- VIX itself has become an asset underlying futures and options
    - it is, hence, possible to get pure volatility exposures for several assets
    - Volatility itself has become an important asset class
        - ⇨ It is an asset that can be part of your portfolio

Including Dividends in BSM

- The BSM formula we have seen and used thus far applies to European calls and puts on non-dividend paying (NDP) assets
- The main points to remember when the underlying pays dividends are that
    a. the underlying asset price drops by an amount (approximately) equal to the dividend payment on the ex-dividend date
    b. the option holder does not receive the dividend as only the holder of the underlying asset can receive the dividends
    c. dividends decrease value of European calls and increase value of European puts, all other things equal

Including Dividends in BSM

- European options DP are valued by substituting into the BSM formula the **stock price *less the present value of dividends*, computed at the *risk-free rate*** (公式中需要自己减去 pv of dividends)
- European options can be analysed by assuming that the stock price is the sum of two components:
    - ⇨ a riskless component that will be used to pay the known dividends during the life of the option – it is the *PV of all the dividends during* the life of the option discounted from the ex-dividend dates to the present at the rf rate.
    - ⇨ a risky component – S0
    - ⇨ *σ in the BSM should be the volatility of the risky component of the stock price, not the volatility of the stock price itself.*
- Only dividends with ex-dividend dates during life of option should be included
- For short maturity options, dividends are typically known with certainty
- For longer maturity options dividend payments *may need to be estimated*

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Example 13.6 Using Black–Scholes–Merton when there are dividends

Consider a European call option on a stock with ex-dividend dates in two months and five months. The dividend on each ex-dividend date is expected to be \$0.50. The current share price is \$40, the exercise price is \$40, the stock price volatility is 30% per annum, the risk-free rate of interest is 9% per annum, and the time to maturity is six months. The present value of the dividends is

$$0.5e^{-0.09\times 2/12} + 0.5e^{-0.09\times 5/12} = 0.9741$$

The option price can therefore be calculated from the Black–Scholes–Merton formula with $S_0 = 40 - 0.9741 = 39.0259$, $K = 40$, $r = 0.09$, $\sigma = 0.3$, and $T = 0.5$:

$$d_1 = \frac{\ln(39.0259/40) + (0.09 + 0.3^2/2) \times 0.5}{0.3\sqrt{0.5}} = 0.2020$$

$$d_2 = \frac{\ln(39.0259/40) + (0.09 - 0.3^2/2) \times 0.5}{0.3\sqrt{0.5}} = -0.01012$$

Using the NORMDIST function in Excel gives

$$N(d_1) = 0.5800 \quad \text{and} \quad N(d_2) = 0.4959$$

and from equation (13.5) the call price is

$$39.0259 \times 0.5800 - 40e^{-0.09\times 0.5} \times 0.4959 = 3.67$$

or \$3.67.

Valuing European Options on Stocks Paying Dividend Yields

- Often (e.g., for long maturity options, for stock indices) dividends are looked at as yields
  - ⇨ To make it grow less, the yield will cause the price of the option be less.
- Similarly to the case of dollar dividends, dividends paid at the (continuous) rate q cause stock price to grow less
- E.g. if with Dividend Yield = q, S grows to St, then without dividends S0 grows $S_T\, e^{(qT)}$
- Equivalently, without dividend $S_0\, e^{(-qT)}$ grows to ST (i.e., still by q% per period for T periods)

Valuing European Options on Stocks Paying Dividend Yields

- The probability distribution for the stock price at time T is the same in the following two cases:
  1. The stock starts at price S0 and provides a dividend yield = q
  2. The stock starts at price $S_0\, e^{(-qT)}$ and provides no dividend income => this one starts with a lower price but grows faster because there is no dividend and the two will end up as the same number.

- We can value European options with BSM by adjusting (i.e., reducing) the stock price to $S_0\, e^{(-qT)}$

- The lower bound for the European option price, is:

$$c \geq \max(S_0 e^{-qt} - Ke^{-rT}, 0)$$

  - ⇨ One European call option + an amount of cash equal to Ke^{-rT}
  - ⇨ e^{-qT} shares with dividends being reinvested in additional share

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$$p \geq \max(Ke^{-rT} - S_0 e^{-qt}, 0)$$

⇨ One European put option + e^{-qT} shares with dividends on the shares being reinvested in additional shares

⇨ An amount of cash equal to Ke^{-rT}

Extensions of BSM to include Dividend Yield

$$c = S_0 e^{(-qT)} N(d_1) - Ke^{-rT} N(d_2)$$

$$p = Ke^{-rT} N(-d_2) - S_0 e^{(-qT)} N(-d_1)$$

$$d_1 = \frac{In\left(\frac{S_0}{K}\right) + (r - q + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{In\left(\frac{S_0}{K}\right) + (r - q - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$$

⇨ If the dividend yield rate is known but not constant during the life of the option, equation of *c and p* are still true, with q equal to the average annualized dividend yield during the option's life.

⇨ The calculation of q should include only dividends for which the ex-dividend dates occur during the life of the option.

Valuing European Currency Options with BSM

- As we saw previously, a foreign currency is an asset that provides a continuous "dividend yield" equal to rf
- We can use the BSM formula for an option on a stock paying a continuous dividend yield:
  - Set S0 = current exchange rate (units of d*omestic currency for 1 unit of foreign currency*)
  - Set q = rf

BSM Formulas for European Currency Options

$$c \geq \max(S_0^{-r_f T} - Ke^{-rT}, 0)$$

$$c = S_0 e^{(-r_f T)} N(d_1) - Ke^{-rT} N(d_2)$$

$$p = Ke^{-rT} N(-d_2) - S_0 e^{(-r_f T)} N(-d_1)$$

$$d_1 = \frac{In\left(\frac{S_0}{K}\right) + (r - r_f + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{In\left(\frac{S_0}{K}\right) + (r - r_f - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$$

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- In some circumstances, it is optimal to exercise American currency and index options prior to maturity.
- Thus, American currency and index options are worth more than their European counterparts.
- In general, call options on high-interest currencies and put options on low-interest currencies are the most likely to be exercised prior to maturity.
  - ⇨ The reason is that a high-interest currency is expected to depreciate
  - ⇨ low-interest currency is expected to appreciate
- Also, call options on indices with high dividend yields and put options on indices with low dividend yields are most likely to be exercised early.

American Options and Dividends

- American Call NDP = European option as it should never exercise early
- American Put DP, NDP, American Call DP
  - BSM is not well suited. We need alternative tools: binomial trees, Monte Carlo simulation

Binomial vs. Black & Scholes

If we..

1. Price a European option with a binomial tree as if it was European
2. Repeat for larger and larger number of steps on the tree
3. Price the same European option with Black and Scholes

The price we get as we increase the number of steps will change and as we increase the steps, at some point, price will converge.

- Binomial tree price for European puts and calls converges to Black and Scholes price as the number of time periods increases to infinity and the length of each time period is infinitesimally short
- Broadly speaking, this is because the binomial distribution converges to the normal as the number of trials approaches infinity

Stock Index Options (Ch. 15)

- Stock index option is an option whose underlying asset is index
- The most popular underlying indices in the U.S. are
  - The S&P 100 Index (OEX American and XEO European)
  - The S&P 500 Index (SPX European)
  - The Dow Jones Index (DJX European)
  - The Nasdaq 100 Index (NDX European)
    - ➔ **Contracts are on $100 times index value** (0.01 times DJX) ; they are settled in cash
- In Australia:
  - ASX 200 (XJO, European, cash settled, $25 x index value)

Stock Index Option: Example

- Consider a *call option* on an index with a strike price of 560
- Suppose 1 contract is exercised when the index level is 580
- Index multiplier is $100
- What happens on exercise date?
- Holder gets (S-K)*$100 = 20*$100=$2000 per contract in cash

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Valuing European Stock Index Options with BSM

- We can use the formula for an option on a stock paying a continuous dividend yield
- Set S0 = current index level
- Set *q = average (across index components) dividend yield* expected during the life of the option
- Instead of forecast the dividend yield by myself to ask people their expectations, or I can look some prices that embedded dividend yield

Using Forward or Futures prices in the BSM formula

- Often times, traders and market makers prefer to use the forward and/or the futures price rather than the spot price (in the BSM) of the underlying when pricing options because those embedded dividend yields
- To this end, they keep track of forward and futures curves
- They, then, use forward or futures prices in the BSM model rather than the spot
- This practice bypasses the needs to input dividend yields in the model (don't need to predict the dividend)

BSM Formulas with Futures/Forward Price

- Using the futures (or, forward) price, F0, at time 0 for delivery at T

$$c = e^{-rT}[F_0 N(d_1) - KN(d_2)]$$

$$p = e^{-rT}[KN(-d_2) - F_0 N(-d_1)]$$

⇨ There is no s and there is no q, but a F (which is **a future price**) and use the future price that expires as close as possible to the expiration of the option.

⇨ Option and Forward/Futures maturity must be the same

$$d_1 = \frac{In\left(\frac{F_0}{K}\right) + \frac{\sigma^2 T}{2}}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Stock Index Options for Portfolio Insurance

- This is a protective put applied to a stock index, and referred as portfolio insurance either to prevent losses or lock in gains
- Options are routinely used to help manage the risk and return of a portfolio.
- Portfolio insurance is a strategy of either preventing a large negative return shock to a portfolio or locking in the gains that have already accumulated in a portfolio without liquidating the underlying portfolio
- Portfolio insurance puts a floor on portfolio value
- It's a Protective Put strategy (can be done with individual assets as well)
- In what follows we will ignore the cost of the put (i.e., we will focus on the equity portion of the overall position)

Stock Index Options for Port folio Insurance: Example 1

- Your portfolio is worth $12,000,000 and *perfectly replicates* the S&P500 (-> β =1)

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- S&P500 is currently at 800 -> your portfolio is 12,000,000 / 800 = 15,000 times the index
- You want your portfolio value not to fall below $10,800,000 over the next 6 months (i.e., maximum loss = 10%)
  - ⇨ This determines how much insurance going to buy
- As the index drops in value, the payoff of a put option on the index will increase 1 for 1.
  - ⇨ For every drop of 1 in the index, a put option will payoff 1 extra dollar.
- Maximum tolerable loss is exceeded when S&P500 drops below 720 (800 x 0.9 = 720)
  - ➔ Buy puts on the index with strike = 720
- How many contracts?
  - ➔ Buy 12,000,000 / (100*800) = 150 contracts

  The LOSS of the PF: $\frac{12m}{800}\ (800 - 680) = 1800K$

  The Value of the PF: 12M – 1800 K = 1020K
- If, e.g., index drops to 680 -> portfolio value drops to $10,200.000, but puts pay off

  150 contract x (720-680) x 100 multiplier = $600,000
  - ➔ Total Portfolio Value = $10,800,000
  - ➔ Index goes down, option pay off

Using Index Options for Portfolio Insurance: Generalization

- Suppose the $ value of a portfolio is P and the spot value of the index is S0
- If a portfolio has a β of 1.0, portfolio insurance is obtained by buying 1 put option contract on the index for each 100 S0 dollars held, i.e., P / (100 S0) contracts
- If the β is not 1.0, the portfolio manager buys *β put options* for *each 100 S0 dollars* held
  - ⇨ The higher the beta, the riskier the portfolio is and the more insurance going to buy.
  - ⇨ E.g. $\beta \times \frac{P\ (value\ of\ PF)}{100 \times S_O(S\&P)}$
- In both cases, a strike for the put, K, is chosen to yield the desired insurance level
  - ⇨ K depends on maximum tolerable loss.
  - ⇨ E.g. 720 was the K because I don't want to lose more than 10%.

The Greeks I – Ch. 17

Chapter 17: 17.1, 17.2, 17.4

Example

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- Bank has sold European calls (writer of European call) on 100,000 share of a non-dividend paying stock for $300K
    - Stock price = $49
    - Strike price = $50
    - Risk free rate = 5% per annum
    - Volatility of stock price = 20% p.a.
    - Time to maturity is 20 weeks
    - Black Scholes price: ~$240,000
        - ⇨ Option sold for $60K more than model price
        - ⇨ What model says not necessarily reflect the market place

Hedging the Option

- Naked position
    - ⇨ Do nothing…but can be risky
        - ➔ If stock price falls, get $300K profit
        - ➔ If stock price rises above $53, losses accumulate
- Covered
    - ⇨ Buy stock to cover the option
        - ➔ But if stock price falls can have big losses in spot position, only partially offset by income from option writing
- You are the writer of the call, so you don't want to see a price increase in the option
    - ⇨ Meaning, the price of the underlying is not your only concern because the price of the underlying is not the only thing that can make the price of the option go up.

Hedging the Option

- Options positions are exposed to several sources of risk: time, volatility, risk-free rate, etc, not only to changes in the price of the underlying
- A writer of options is concerned about managing these risks
- Goal: manage all of the risks so that overall risk level is acceptable

Hedging with the Greeks

- More sophisticated than naked or covered positions
- Involves calculating **sensitivities** of option price to changes in factors affecting value (time, volatility, risk-free rate, etc)
- These sensitivities are delta, gamma, theta, vega, and rho

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- ⇨ Each of these measures captures a specific sensitivity (i.e., w.r.t. one of the factors)
- ⇨ Each *Greek letter measures a different dimension of the risk of an option*
- ⇨ In calculus terms, the Greeks are the partial derivatives of the option price with respect to the risk factors

Speculating with the Greeks

- Speculators care Greeks as well because it tells you how much you can alter the risk profile.
- As always, depending on one's goals/perspective, risk measures can also be used to speculate
- In general, the Greeks are the building blocks for creating portfolios with specific risk characteristics.

Greeks: Local measures

- These risk measures are all local: they measure an option's price sensitivity to a small change in the relevant risk factors.

Delta – the price of the underlying asset

- Defined as the change in an option price w.r.t. the price of the underlying asset
  - ⇨ It measure how sensitive you option price to the price of the underlying, which is how much the option price changes if the underlying asset changes by little bit.
  - ⇨ It is the slope of the curve that relates the option price to the underlying asset price.
- Similar to a slope coefficient
- Already seen when building riskless portfolio in binomial model and BSM model
  - ⇨ E,g: Delta = 0.25 is # of units of underlying to short for every call we are long
    Delta 0.25 represents if you long 1 call option, you have to go short (0.25) units of underlying asset to create a riskless hedge.
  - ⇨ But that is because when underlying price moves by 1, *call price moves by 0.25 (25%)*, so that overall portfolio value is unchanged, it tells how much option moves when the underlying moves by 1 (by little bit)

- **Delta (Δ) is the slope of the tangency line**
  - ⇨ The curve is the value of the option as a function of stock price
  - ⇨ Compute the option price by varying s (for different values of s compute c)

    The relationship between s and c is a non-linear function.

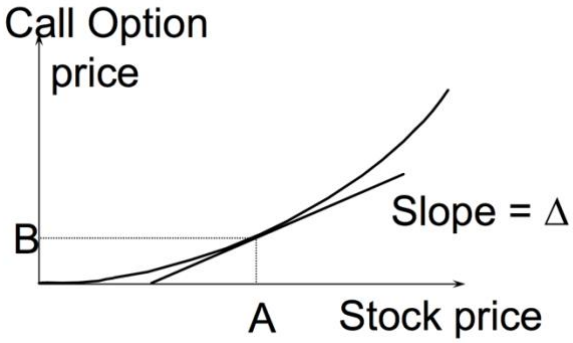
    In general, the delta of a call equals $\Delta c / \Delta s$

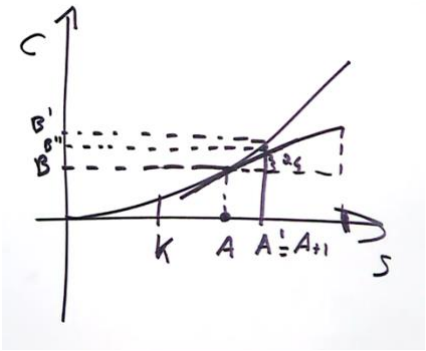
    ➔ Δs is a small change in the stock price

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➔ Δc the resulting change in the call price.

⇨ The slope coefficient of the line approximates what happens to the option price when underlying moves a bit as delta assumes there is a linear relationship.





⇨ E.g. the slope coefficient is 0.4 => when the underlying goes from A to A', C goes up approximately by 0.4.

⇨ The smaller the change in s, the better the approximation. The approximation error becomes much bigger as the change in price becomes much bigger.

⇨ ***Suggesting, using the slope coefficient to approximate what happens in the option price is fairly accurate as long as change in stock price is small***.

⇨ If S goes up, the slope of the tangency line becomes stepper and stepper, close to 45-degree line => the option in deep ITM

⇨ If S goes down, below K, the slope of the tangent line gets flatter and flatter and the minimum of the slope line can be 0 (which is a flat line).

- For call option, delta is the number between 0 – 1, and will be close to 1 for deep ITM option and close to 0 for deep OTM option.
  - ⇨ Delta can be the probability of exercising, but not accurate enough (e.g. delta = 0.7, 70% chance of exercising the option)
    - Delta is the partial derivate of call option price with respect to the underlying asset price.
    - Or it is the N(d1) from BSM is the delta of the option.

For puts:

- Low asset price compares to K, deep ITM puts
- High asset price comparing to K, deep OTM, delta closes to 0.

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- Because it is a put, it has a negative sign => delta of a put is a negative number and because of that, delta will be -1 to 0.
- Delta is negative, which means that a long position in a put option should be hedged with a long position in the underlying stock, and a short position in a put option should be hedged with a short position in the underlying stock
- Long call & short put: positive delta -> go short the underlying
- Similar to duration measure for bond pricing: delta as a measure of option sensitivity is only reasonable for very small changes in the stock price
- But, then, for relatively 'large' changes in the stock price delta is only an approximation to the change in the option price

Delta and Underlying Asset Price

- As S grows larger than K then delta is close to one
- As S goes further below K then delta is close to zero
- Delta approximates probability that the option will be exercised

Delta in the BSM Model

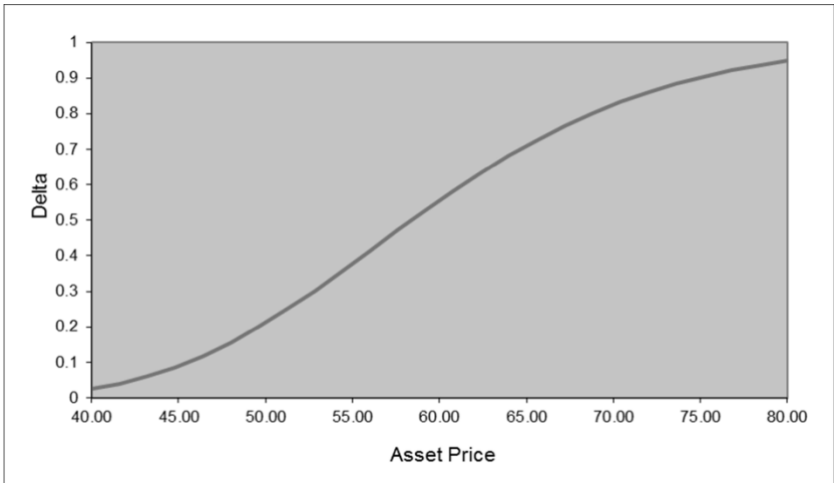
- The delta of a European call on a non-dividend-paying stock in the BSM model is

$$\Delta = \frac{\partial c}{\partial S} = N\left(d_1\right)$$

- The delta of a European put on the stock in the BSM model is

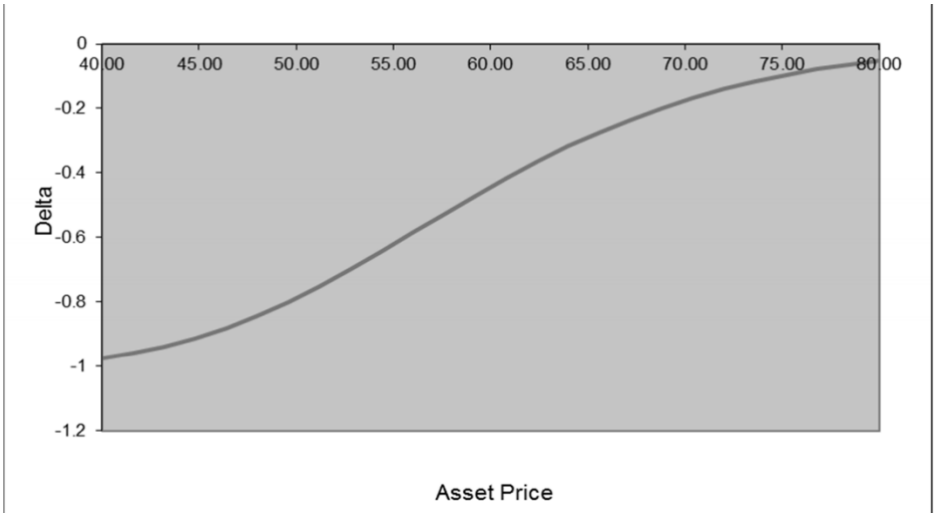
$$N\left(d_1\right) - 1$$

(BSM) Delta of European Call K=60, vol=30, T=5/12



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(BSM) Delta of European Put K=60, vol=30, T=5/12



Delta Hedging (and its limitations)

- Delta: manage the risk of the option position, delta tells how much the option value is affected by changes in the underlying asset
- Involves creating/maintaining a delta neutral portfolio (i.e., a portfolio with Delta=0)
  - ⇨ Option has its own delta, combine the option with the underlying, and this combination can be made delta neutral
- But since delta is only an approximation to the change in the option price, a delta hedged portfolio is not necessarily riskless!
  - ⇨ Even if the portfolio has a delta = 0, it is not exactly riskless, it is only approximate riskless.
    1. It tells us what happens to the option price only the underlying changes by a little bit – delta is an approximation
- Even if the price of the underlying experiences a small change, other factors (volatility, interest rates,…) affecting the option price may change
  - ⇨ delta neutral portfolio is not necessarily riskless!
  - ⇨ It is only **approximate** riskless.
  - And you can't delta hedge and forget…
  - The overall position should be delta natural – this is an obligation

<u>Dynamic</u> Delta Hedging

- As the underlying asset price moves Delta moves as well!
- The position must be (frequently…) rebalanced in order to keep it delta neutral

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- As we will see in the continuation of our example, delta hedging a written option involves a "buy high, sell low" trading rule
    - How is the above a reasonable strategy?

⇨ As the tables illustrate, delta hedging a short position generally involves selling stock just after the price has gone down and buying stock just after the price has gone up. It might be termed a buy-high, sell-low trading strategy! The present value of the expected cost is the option price, or $240,000. This comes from the average difference between the price at which stock is purchased and the price at which it is sold.

⇨ Buy high sell low is not losing money, rather is paying insurance => this is the cost of hedging.

⇨ It is reasonable because in exchange of reducing risk, you pay a cost.

Example (same…)

- Bank has sold European calls on 100,000 share of a non-dividend paying stock: $300K
- Stock price = $49, Strike price = $50, Risk free rate = 5% per annum, Volatility of stock price = 20% p.a., Time to maturity is 20 weeks
- Black Scholes price: ~$240,000
    - ⇨ Option sold for $60K more than model price
- Bank wants to keep a Delta Neutral ($\Delta = 0$) Position by buying/selling stock

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Table 17.2 Simulation of Dynamic delta hedging.

Week	Stock price	Delta	Shares purchased	Cost of shares purchased ($000)	Cummulative cost ($000)	Interest cost ($000)
0	49.00	0.522	52,200	2,557.8	2,557.8	2.5
1	48.12	0.458	(6,400)	(308.0)	2,252.3	2.2
2	47.37	0.400	(5,800)	(274.7)	1,979.8	1.9
3	50.25	0.596	19,600	984.9	2,966.6	2.9
4	51.75	0.693	9,700	502.0	3,471.5	3.3
5	53.12	0.774	8,100	430.3	3,905.1	3.8
6	53.00	0.771	(300)	(15.9)	3,893.0	3.7
7	51.87	0.706	(6,500)	(337.2)	3,559.5	3.4
8	51.38	0.674	(3,200)	(164.4)	3,398.5	3.3
9	53.00	0.787	11,300	598.9	4,000.7	3.8
10	49.88	0.550	(23,700)	(1,182.2)	2,822.3	2.7
11	48.50	0.413	(13,700)	(664.4)	2,160.6	2.1
12	49.88	0.542	12,900	643.5	2,806.2	2.7
13	50.37	0.591	4,900	246.8	3,055.7	2.9
14	52.13	0.768	17,700	922.7	3,981.3	3.8
15	51.88	0.759	(900)	(46.7)	3,938.4	3.8
16	52.87	0.865	10,600	560.4	4,502.6	4.3
17	54.87	0.978	11,300	620.0	5,126.9	4.9
18	54.62	0.990	1,200	65.5	5,197.3	5.0
19	55.87	1.000	1,000	55.9	5,258.2	5.1
20	57.25	1.000	0	0.0	5,263.3	

- Delta: 0.522: means because I am ***shorting these options*** to have a delta hedge, I need to go ***long the underlying*** because the delta is 0.522 I need to long 0.522 shares for every option that I am shorting.

- That is the delta of the option, if you want to consider the delta of the position, then you need to multiple 0.52 by -100,000 which is the position in the option.

  -

  ⇨ Share purchased = 100k x 0.522 = 52,200

  ⇨ Cost of the shares 52,200 x 49 = $2557800

  ⇨ Interest costs: 2557.8K x $e^{0.05\frac{7}{365}} - 2557.8K = 2453.86$

- Delta: 0.458: means I only need 45800 shares, means I need to sell some shares (6400 shares).
  ⇨ By selling 6400 shares at $48.12 I receive $308000
- Delta is 1 at maturity because delta gives a rough approximation with the probability of exercise, if on the expiration day, the option is ITM, it will be exercised for sure, so delta = 1.
- The cost to purchase the shares is $5263.3K, but as you can the call option Writer, you have 100,000 shares each with a strike of $50 (so you will receive 5M).
  ⇨ The net cost for you is therefore $263.3k to hedge your option position, to keep it as delta natural.
- In this case, rebalancing is every week, but between rebalancing, my portfolio is NOT delta hedge.
  ⇨ The more frequently you balance, the more the delta is hedged, the more you pay for the transaction costs.

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⇨ The trade-off is between transaction costs and the accuracy of the hedge.

- This is the case where the option is OTM, the holder will not exercise.

Table 17.3 Simulation of dynamic Delta hedging

Week	Stock price	Delta	Shares purchased	Cost of shares purchased ($000)	Cumulative cost ($000)	Interest cost ($000)
0	49,00	0.522	52,200	2,557.8	557.8	2.5
1	49.75	0.568	4,600	228.9	2,789.2	2.7
2	52.00	0.705	13,700	712.4	3,504.3	3.4
3	50.00	0.579	(12,600)	(630.0)	2,877.7	2.8
4	48.38	0.459	(12,000)	(580.6)	2,299.9	2.2
5	48.25	0.443	(1,600)	(77.2)	2,224.9	2.1
6	48.75	0.475	3,200	156.0	2,383.0	2.3
7	49.63	0.540	6,500	322.6	2,707.9	2.6
8	48.25	0.420	(12,000)	(579.0)	2,131.5	2.1
9	48.25	0.410	(1,000)	(48.2)	2,085.4	2.0
10	51.12	0.658	24,800	1,267.8	3,355.2	3.2
11	51.50	0.692	3,400	175.1	3,533.5	3.4
12	49.88	0.542	(15,000)	(748.2)	2,788.7	2.7
13	49.88	0.538	(400)	(20.0)	2,771.4	2.7
14	48.75	0.400	(13,800)	(672.7)	2,101.4	2.0
15	47.50	0.236	(16,400)	(779.0)	1,324.4	1.3
16	48.00	0.261	2,500	120.0	1,445.7	1.4
17	46.25	0.062	(19,900)	(920.4)	526.7	0.5
18	48.13	0.183	12,100	582.4	1,109.6	1.1
19	46.63	0.007	(17,600)	(820.7)	290.0	0.3
20	48.12	0.000	(700)	(33.7)	256.6	

- OTM option at the expiration will have a delta = 0, ITM delta = 1.
  - ⇨ For puts, OTM delta = 0; ITM = -1

Dynamic Delta Hedging

- In the previous Tables (17.2, 17.3) the total discounted cost of hedging is roughly the Black & Scholes price of $240,000.
  - ⇨ The cost of hedging (the cost for keeping the portfolio neutral) is always roughly the same for both ITM and OTM option.
  - ⇨ If the hedge worked perfectly, the cost of hedging would, after discounting, be exactly equal to the BSM price for every simulated stock price path.
- There is no perfect match with the Black & Scholes price because rebalancing occurs only once a week (not continuously)
- More frequent rebalancing would bring the cost of hedging closer to the Black & Scholes value (see Table 17.4)
  - ⇨ The more I create a risk-free PF, the closer I get to earn the rf rate.
  - ⇨ The better you are hedging, the closer to the rf rate.
  - ⇨ The more frequent balance, the better hedge my position is, which is closer as to hold a risk-free asset

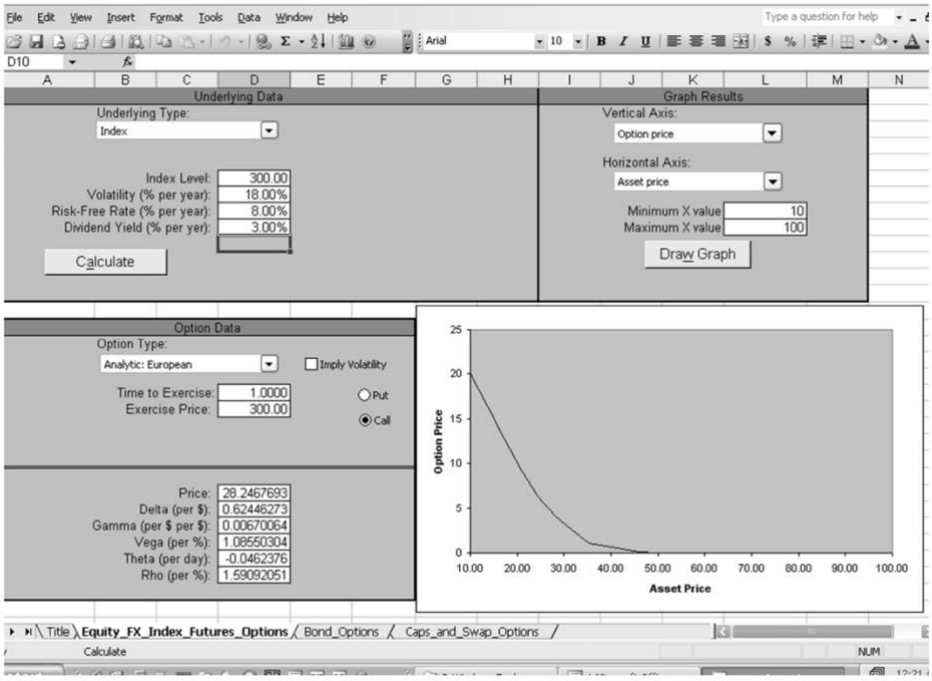
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➔ The max I can earn is the risk-free rate.

- ***Delta hedging aims to keep the value of the financial institution's position as close to unchanged as possible***.

Delta Neutrality: Example (new) and Limitations

- Sacamano Bank has **sold (written)** 100 European call stock option contracts (i.e., it is short 10,000 calls)
- S = $300; K = $300; T = 1 year; σ = 18%; r = 8%; q = 3%.
- Denote with n0 the # of call options (in this case n 0=10,000)
- How should the bank hedge its option position?
- To make position delta neutral, bank needs to buy (delta*10,000) shares of underlying stock



- Loss in the option, makes money in the underlying because you are short long – means loss money when the stock price increases

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DELTA-NEUTRAL HEDGE

$n_0 = -10{,}000 \rightarrow$ Long $n_S = 6{,}245$ shares

Case A1: **S increases from \$300 to \$301.**

Portfolio	Initial Value	New value	Change
-10000Calls	- \$282,500	- \$288,800	- \$6,300
6,245S	\$1,873,500	\$1,879,745	\$6,245

→ Hedging Error: - \$55

Case A2: **S decreases from \$300 to \$299.**

Portfolio	Initial value	New value	Change
-10000Calls	- \$282,500	- \$276,200	+ \$6,300
6,245S	\$1,873,500	\$1,867,255	- \$6,245

→ Hedging Error: + \$55

- For a small change in the underlying, assuming other factors remain unchanged, the value of the PF is very little affected by the change.
- The hedging error is small.

Case B: **S increases from \$300 to \$310.**

Portfolio	Initial Value	New value	Change
-10000Calls	- \$282,500	- \$348,100	- \$65,600
6,245S	\$1,873,500	\$1,935,950	\$62,450

→ Hedging Error: - \$3,150

The point here is that Delta has changed significantly and .6245 no longer applies.

S =	\$300	\$301	\$310
Δ =	.6245	.6311	.6879.

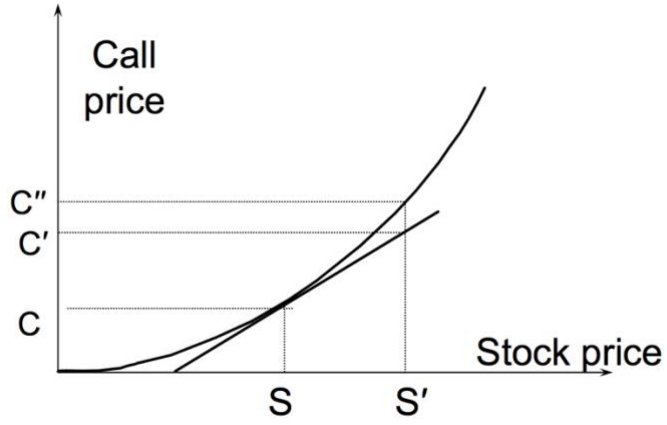
We conclude that the delta-neutral portfolio must be adjusted for "large" changes of the underlying asset price.

→ Need Gamma....

- gramma: how quickly delta can change, how much it may change

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Gamma Addresses Delta Hedging Errors Caused by Curvature



- delta hedge assumes that the option price moves from C to C', when in fact moves from C to C''.
- the difference C and C'' leads to a hedging error.
- The size of the hedging error depends on the curvature of the relationship between the option price and the stock price.
- Gamma measures this curvature.
    - So, the stepper the curve is, the bigger the space between the curve and tangent line => the gap is the hedging error.
        ⇨ Hedging error will be bigger, the stepper the curve is.

Gamma (Γ)

- Delta: change in an option price w.r.t. the price of the underlying asset
- The smaller the change in s, the better the approximation. The approximation error becomes much bigger as the change in price becomes much bigger.
    - **Delta (Δ) is the slope of the tangency line**
- ***Suggesting, using the slope coefficient to approximate what happens in the option price is fairly accurate as long as change in stock price is small***.

- Gamma: change in the (portfolio's) delta due to a "small" change in the underlying asset price.
- It is the rate of change of the PF's delta with respect to the price of the underlying asset.
    - Measures the curvature in the relation between the option price and stock price (second partial derivative of option price w.r.t. stock price)
    - Similar to convexity for bonds

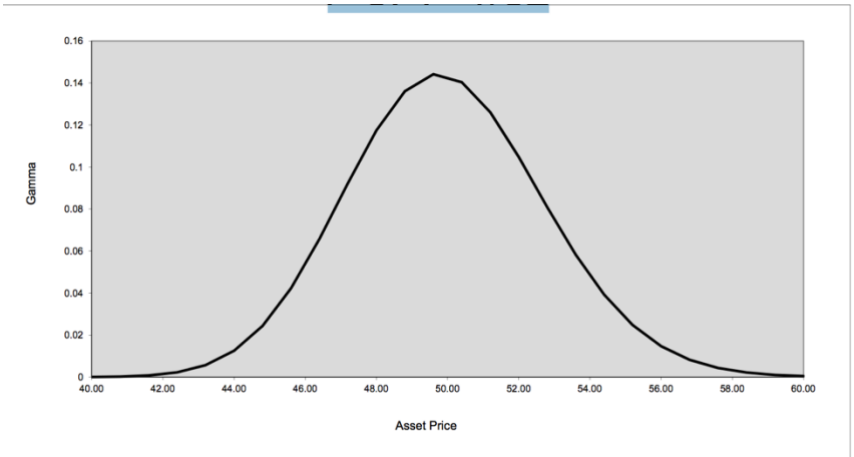
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- If gamma is small, delta changes slowly, and adjustments to keep a portfolio delta neutral need to be made only relatively infrequent. As gamma increases, so does the sensitivity of the portfolio to a change in the underlying price
    - Small gammas mean slow changes in delta -> infrequent changes to hedg
    - Small gamma = delta not move much
- If gamma is highly negative or positive, delta is very sensitive to the price of the underlying asset
    - ⇨ Harder to maintain a delta neutral position
- Traders often refer to delta as the "speed" of the option's position and to gamma as its "acceleration"
    - ⇨ Delta: tells how much the option moves for a given change in S – speed.
    - ⇨ Gamma: tells how quickly it moves from Point A to B – acceleration.

- But a lot of acceleration is a double-edged sword: if the price of the underlying moves against you
    - ⇨ Too much acceleration: lose control on your position

- Gamma is positive for long option positions (calls and puts), negative for short option positions (calls and puts)
    - ⇨ Because long: delta will go up, when the underlying goes up, which means gamma is positive as **gamma is the change the delta**.
        - ➔ Delta goes up, means a positive change for gamma

    - **Gamma can be very high for ATM options near expiration**
        - ⇨ EX: Gamma vs. S European Call, S=50, K=50, vol=40, r=5, T=1/52



        - ⇨ ATM option, close to maturity, lots of acceleration

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➔ Because e.g. if the current price is $49 and the strike is $50, now the delta is = 0 (because currently the option is OTM); but it only requires move up a little for the option to be ITM, where delta = 1.

➔ This is the biggest move the delta can make => lots of acceleration

➔ Thus, ATM option, especially close to maturity have lots of acceleration. Delta can change a lot, even S changes a little bit.

➔ Short-life at-the-money options have very high gammas, which means that the value of the option holder's position is highly sensitive to jumps in the stock price.

Gamma Neutrality and Hedging

- To be better hedged, want delta not to change => to create a gamma natural PF, gamma = 0: delta not change as gamma is the changing delta.
- Delta neutrality provides protection vs. small changes in the underlying's price between rebalancing
- Gamma neutrality provides protection vs. larger changes in the underlying's price between rebalancing
  - ⇨ *Delta of an option* tells me how much the option price changes if the underlying changes
  - ⇨ *Delta of underlying* is how much the underlying change if underlying changes.
    - ➔ Delta of underlying = 1
    - ➔ Gamma of underlying, and gamma is the changing delta => thus gamma of underlying = 0.
- Option has a non-zero gamma (long: +; short: -) and underlying has a gamma = 0 and cannot be used to change the gamma of a PF to realize gamma neutrality.

Gamma Neutrality and Hedging

- Gamma neutrality cannot be achieved by taking position in the underlying **because its value is linear in its price** (and so its gamma is zero!)
- To make a portfolio gamma neutral we need to take positions in additional derivatives (which have a non-zero gamma)

Gamma Neutrality

- How to make a portfolio gamma neutral?
- See Sacamano Bank example
- We need another derivative.

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- Assume the following call option on the same stock is considered:
- K = $305 T = 90 days σ= 18% r = 8% q = 3% Compute Δ= .4952 Γ= .0148

DELTA-GAMMA-NEUTRAL PORTFOILO

Δ(portfolio) = 0: $n_S + n_0(.6245) + n_1(.4952) = 0$

Γ(portfolio)= 0: $n_0(.0067) + n_1(.0148) = 0$

Solution:

$n_0 = -10{,}000$

$n_1 = -(.0067)(-10{,}000)/.0148 = 4{,}527$

$n_S = -(.6245)(-10{,}000) – (4{,}527)(.49520 = 4{,}003$

Short the initial call : $n_0 = -10{,}000$

Long call #1 $n_1 = 4{,}527$

Long 4,003 shares $n_S = 4{,}003$

- The delta of each of the three positions by the number of units in the position.

⇨ The underlying Ns has a delta = 1

⇨ Initial position n0 has a delta of 0.6245

⇨ Additional position n1

- Gamma: the gamma of each position times the number of units in the position

⇨ The gamma of the underlying is 0.

- By treating two more things (underlying and other option, make delta and gamma both to 0 (natural)
- No speed and no acceleration

DELTA-GAMMA-NEUTRAL PORTFOLIO

Case B1: **S increases from $300 to $301?**

Portfolio	Initial value	New value	Change
0) -10,000	- $282,500	- $288,800	-$6,300
1) 4,527	$45,360	$47,657	$2,297
S) 4,003	$1,200,900	$1,204,903	$4,003
		Error:	0

Case B2: **S increases from $300 to $310.**

Portfolio	Initial value	New value	Change
0) -10,000	- $282,500	- $348,100	- $65,600
1) 4,527	$45,360	$70,930	$25,570
S) 4,003	$1,200,900	$1,240,930	$40,030
		Error:	0

- When the underlying moves by little bit, no changes in the PF at all.
- In this case, delta natural is enough

- Here, if price moves a lot in the underlying, your PF is still protected
- Because what you lose on the initial calls is perfectly offset by the profits made on the other call and on the underlying.
- But it required additional cost as to purchase an additional option

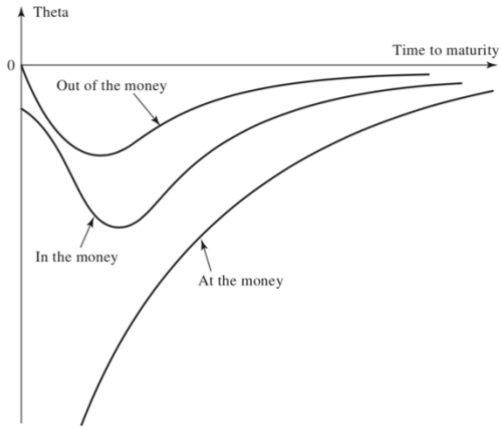
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Caveat (again…)

- The hedging errors in the delta-gamma neutral case still assume that changes in the option values are induced only by changes in the underlying asset price
  - ⇨ There are interest and volatility that can still affect the performance of the PF.
- But we know that other factors affect the option value
  - ⇨ A gamma-neutral PF is risk-free when interest rate and volatility remains unchanged.
- Restated, even the delta-gamma neutral strategy assume "all else equal"
  - ⇨ Need to measure effect of other factors on option value

Theta (Θ)

- Rate of change of the value of the option with respect to the passage of time
- Referred to as the "time decay" of an option (portfolio)
  - ⇨ For most options, as maturity shrinks, everything else stay the same, the option value decrease.
- Usually negative for options: as time to maturity decreases, options become less valuable
  - ⇨ For a call option, when the stock price is very low (S-K), theta is close to 0.
  - ⇨ For an ATM option, theta is large and negative.
- But as time passage is certain, no need to hedge against theta
  - ⇨ There is uncertainty about the future stock price, but there is no uncertainty about the passage of time as the passage of time is NOT a risk factor.
  - ⇨ It makes sense to hedge against changes in the price of the underlying asset, but it does not make any sense to hedge against the passage of time.
  - ⇨ Thus, many traders regard theta as a useful descriptive statistic for a portfolio to measure it and record how much your option position will change day by day if others remain unchanged.
- Theta is given in terms is $/1 day.
- See Formula sheet (NOT textbook section 17.12)
- Ex: Θ(c) = - $0.0118 means that if time to expiration decreases by one calendar day, the call price (option value) will decreases by 1.18 cents per calendar day
- This is the information we want to have, but not a factor we want to hedge.



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Vega (ν) or Kappa (κ)

- While Black-Scholes assume that volatility of the underlying is constant, we know in practice that volatilities change over time. Therefore, option values are likely to change because of movements in volatility
- Vega is the rate of change of the value of the PF with respect to the volatility of the underlying asset.
- Vega is always positive (negative) for long (short) option positions
    - ⇨ If vega is highly positive or negative, the PF's value is very sensitive to small changes in volatility.
    - ⇨ If close to 0, volatility changes have relatively little impact on the value of the PF.
    - ⇨ The option price is always a positive function of volatility.
    - ⇨ Thus, Vega positive when long.

- *As gamma increases, so does the sensitivity of the portfolio to a change in the underlying price*
    - *Small gammas mean slow changes in delta -> infrequent changes to hedg*
    - *Small gamma = delta not move much*
- *If gamma is highly negative or positive, delta is very sensitive to the price of the underlying asset*
    - ⇨ *Harder to maintain a delta neutral position*

- Gamma neutral position is typically not vega neutral
- To obtain gamma and vega neutrality simultaneously one needs two additional traded contracts
- Vega is in terms of $/1% change in vol.
    - ⇨ EX: vega = 12.1 means that a 1% increase (from 20% to 21%, 100 basis points) in volatility will increase the **option value** by 0.01*12.1 = $0.121

Rho (ρ)

- Rate of change of the value of the option WRT the **interest rate**
    - ⇨ Sensitivity of option value to changes in the risk free interest rate
- In the case of currency options, two ρ's exist corresponding to the two interest rates.
- Rho is in terms of $/1% change of r
- EX: ρ (c) = 8.91 means that a 1% increase (increase 100 basis points) (from 5% to 6%) in r will increase the option value by 8.91*0.01 = 0.0891

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Managing Delta, Gamma, Vega and Rho: Summary

- Δ can be changed by taking a position in the underlying asset
- To adjust Γ (gamma – acceleration: changing delta) , ν (Vega – volatility), and ρ (rho - interest rate) it is necessary to take a position in 3 additional derivatives
- How?
- Let's go back to the Sacamano bank example
- Identify two more call contracts

Delta-Gamma-Vega-Rho-neutral portfolio

CALL	0	1	2	3
K	300	305	295	300
T(days)	365	90	90	180
Volatility	18%	18%	18%	18%
r	8%	8%	8%	8%
Dividends	3%	3%	3%	3%
c	$28.25	$10.02	$15.29	$18.59

Delta-Gamma-Vega-Rho-neutral portfolio

CALL	Δ	Γ	υ	ρ
0	.6245	.0067	.0109	.0159
1	.4954	.0148	.0059	.0034
2	.6398	.0138	.0055	.0044
3	.5931	.0100	.0080	.0079
S	1.0	0.0	0.0	0.0

Delta = Δ = 0

- nS+n0(.6245)+n1(.4954)+n2(.6398)+n3(.5931)=0

Gamma = Γ = 0

- n0(.0067)+n1(.0148)+n2(.0138)+n3(.0100)=0

Vega = υ = 0

- n0(.0109)+n1(.0059)+n2(.0055)+n3(.0080) =0

Rho = ρ = 0

- n0(.0159)+n1(.0034)+n2(.0044)+n3(.0079) =0

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The "final" solution is:

	Exact n
Short 100 calls #0;	-10,000
Short 339 calls #1;	-33,927
Long 265 calls #2;	26,534
Long 204 calls #3;	20,420
Short 6,234 shares.	-6,234

THE DELTA-GAMMA-VEGA-RHO- NEUTRAL PORTFOLIO

S increases from $300 to $310
r increases from 8% to 9%
σ increases from 18% to 24%

Portfolio	Initial Value	New value
0) - 10,000	- $282,468	- $428,071
S) - 6,234	- $1,870,200	- $1,932,540
1) - 33,927	- $340,023	- $664,552
2) 26,534	$405,668	$694,062
3) 20,420	$379,677	$622,240
TOTAL	- $1,707,356	$1,708,861

Hedging Error:$1,505 or .00088%.

Hedging in Practice

- Dynamic hedging of the Greeks can quickly become cost prohibitive and often the Greeks work against one another
    - ⇨ Keep treading will be expensive
- In practice, portfolios are often rebalanced daily to maintain delta neutrality while vega (kappa) and gamma are monitored frequently
    - ⇨ Almost no one hedge all of these at once
    - ⇨ Delta hedge is necessary and obligated
- Read Business Snapshot 17.1

Note on Greek Formulas

- In Chapter 17 (sections 17.4-17.9), textbook reports series of formulas for the Greeks
- Let's not forget that those expressions are all from Black-Scholes and refer to NDP assets
- Then, textbook (section 17.12) reports modified formulas for DP assets
- No need to memorize any of those
- They are on the Formula Sheet
- Greeks from different models generate different values

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